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Learning objectives

1. Get practice understanding *abstract* definitions and notation.
    a. Working through exercises with numbers instead of variables can help us gain intuition for general results.
    b. We practice using notation on simple exercises to get comfortable with it before more advanced problems.
2. Get practice with problem solving.
    a. If a problem initially appears too difficult, how can you approach it or try to reduce its complexity?
    b. How can you use previous results to solve parts of a problem?
3. Learn properties of the summation operator.
    a. We will use the summation operator a lot for probability and statistics.

Notation (1/3): x and x_i

- If we write x, this refers to an unspecified number.
- If we write x_i, this refers to the i'th number of an unspecified sequence of numbers.
  - e.g., x_1, x_2, x_3, x_4, x_5 could be the incomes of 5 people.
- We often don't know the values in the sequence.
- Here is one example with values (units are \$1,000's):

$$x_1 = 60, \quad x_2 = 32, \quad x_3 = 80, \quad x_4 = 50, \quad x_5 = 28.$$

  - Just like we often don't assign specific numbers to the x_i, we let the letter n represent the (often unknown) length of the sequence.
  - x_n refers to the last number of the sequence. In this example $n = 5$, so $x_n = x_5 = 28$).
  - x_{n-1} refers to the 2nd-to-last number in the sequence. In this example $x_{n-1} = x_{5-1} = x_4 = 50$.

Notation (2/3): "..."

- We write a sequence of n unspecified numbers as:

$$x_1, x_2, \ldots, x_n.$$

- The "..." means "continue the pattern". For example,

$$x_1, x_2, \ldots, x_9 = x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9.$$

- Similarly,

$$x_1, x_3, \ldots, x_9 = x_1, x_3, x_5, x_7, x_9.$$

  The pattern of the first 2 elements tells how to fill the dots.

- If a second element is not specified, it is assumed the subscript increases by 1 (i.e., the step size is 1):

$$x_1, \ldots, x_9 = x_1, x_2, \ldots, x_9 = x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9.$$

Notation (3/3): "$\sum$", "$:=$", and "$+\cdots+$"

Definition 1 (summation operator). For a function f and integer n, define $\sum$ such that

$$\sum_{i=1}^{n} f(i) := f(1) + f(2) + \cdots + f(n).$$

- The symbol "$:=$" means "is defined to be".
- "$+\cdots+$" means to continue adding following the pattern until the last index, n, is reached.
- "i" is called the index of summation.
- 1 is called the lower bound of summation.
- n is called the upper bound of summation.

Notation (3/3): "$\sum$", "$:=$", and "$+\cdots+$"

Definition 1 (summation operator). For a function f and integer n, define $\sum$ such that

$$\sum_{i=1}^{n} f(i) := f(1) + f(2) + \cdots + f(n).$$

- We read the expression as "The sum of 'f' from i equals 1 to n" or as "The sum from i equals 1 to n of f".
- The following three notations are equivalent:

$$\sum_{i=1}^{n} f(i) = \sum\nolimits_{i=1}^{n} f(i) = \sum_{i=1}^{i=n} f(i).$$

Example 1. Evaluate $\sum_{i=1}^{3} i$.

Here, the function f is defined by simply $f(i) = i$.

So applying Def. 1, we get:

$$\sum_{i=1}^{3} i = f(1) + f(2) + f(3) = 1 + 2 + 3 = 6.$$

Exercise 1. Evaluate $\sum_{i=1}^{3} i^2$.

Here, the f is defined by $f(i) = i^2$. Applying Def. 1, we get:

$$\sum_{i=1}^{3} i^2 = f(1) + f(2) + f(3) = 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14.$$

Remark 1. $\left(\sum_{i=1}^{3} i\right)^2 = 6^2 = 36 \neq 14 = \sum_{i=1}^{3} i^2$.

Intuition: for numbers a, b: $(a + b)^2 \neq a^2 + b^2$.

Definition 2. A constant is a variable that does not depend on the index of summation.

Example 2. Let a be a variable. Simplify $\sum_{i=1}^{n} a$.

- Are you stuck? If so, think about how we can solve an easier problem as an intermediate step.
- Whenever you see a problem that seems too difficult or general, plug in a number for one of the variables.

Solving an easier problem

Exercise 2. Let a be a variable. Simplify $\sum_{i=1}^{3} a$.

a does not change when i changes. That is, $f(i) = a$. We get:

$$\sum_{i=1}^{3} a = f(1) + f(2) + f(3) = a + a + a = 3a.$$

Remark 2. It will be helpful later to observe the following:

$$\sum_{i=1}^{3} a = a + a + a = a(1 + 1 + 1) = a \sum_{i=1}^{3} 1,$$

i.e., we can "pull out" a from inside the summation to outside of it.

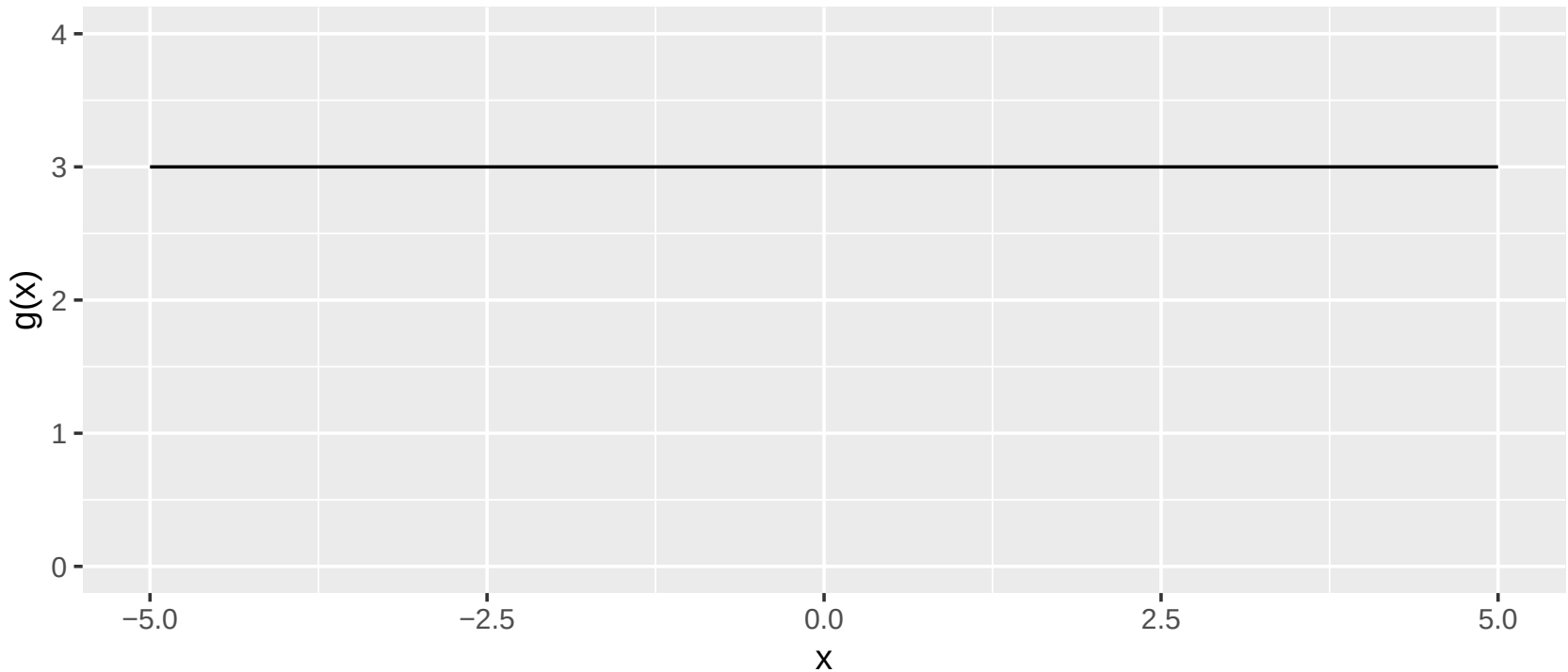
How can $f(i)$ not depend on i? (1/2)

- For the summation $\sum_{i=1}^{3} a$, to apply Def. 1 we specified $f(i) = a$.
- It feels awkward that f is a function of i but there is no i on the RHS.
- Let's step away from summation for a brief moment and consider the function g defined by $g(x) = 3$.
  - Although g is a function of x there is no x on the RHS.
  - Why no x on the RHS? Because the value g takes does not depend on x.

How can $f(i)$ not depend on i? (2/2)

Recall that g is defined by $g(x) = 3$.

Let's graph g over the domain $(-5, 5)$ (these are the values x can take in the graph):



Back to the more general problem

From Exercise 2, we found that $\sum_{i=1}^{3} a = 3a$.

Exercise 3. Let a be a variable. Simplify $\sum_{i=1}^{n} a$. First, we apply Def. 1, using the same f as in Exercise 2:

$$\sum_{i=1}^{n} a = f(1) + f(2) + \cdots + f(n) \tag{1}$$

$$= \underbrace{a + a + \cdots + a}_{n \text{ terms}} \tag{2}$$

$$= na. \tag{3}$$

- Similar to Exercise 2, observe that $\sum_{i=1}^{n} a = a \sum_{i=1}^{n} 1 = an$.

Exercise 4. Let $x_1 = 4, x_2 = -10, x_3 = 1$. Evaluate $\sum_{i=1}^{3} x_i$.

Here, f is defined by $f(i) = x_i$. For example, $f(2) = x_2$. Using Def. 1, we get the following:

$$\sum_{i=1}^{3} x_i = f(1) + f(2) + f(3) \tag{1}$$

$$= x_1 + x_2 + x_3 \tag{2}$$

$$= 4 + (-10) + 1 = -5. \tag{3}$$

Remark 3. There's nothing special about using "i" for the index. For example, the following equalities hold:

$$\sum_{i=1}^{3} x_i = \sum_{q=1}^{3} x_q = \sum_{\square=1}^{3} x_\square = -5.$$

Exercise 5. Simplify $\sum_{i=1}^{n} a x_i z^c$.

The first step is to ask yourself "What are the constants?"

Here, f is defined by $f(i) = a x_i z^c$. Applying Def. 1, we get:

$$\sum_{i=1}^{n} a x_i z^c = a x_1 z^c + a x_2 z^c + a x_3 z^c + \cdots + a x_n z^c.$$

In each term, only x_i is changing. a, z, and c do not change as i changes, so they are the constants. We factor them out:

$$= a z^c \left(x_1 + x_2 + x_3 + \cdots + x_n \right).$$

Finally, we recognize that we can use the summation operator to simplify the terms inside the parentheses:

$$= a z^c \sum_{i=1}^{n} x_i.$$

Example 3. Is the following equality necessarily true?

$$\sum_{i=1}^{n}(ax_i + by_i) \stackrel{?}{=} a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} y_i$$

Exercise 6. Yes, it is true. But what's the intuition for it?

- Whenever you're stuck proving a general statement, it's often useful to try a special case. For example, what if n ***equals 2***?

Then the question reduces to the following:

$$(ax_1 + by_1) + (ax_2 + by_2) \stackrel{?}{=} a(x_1 + x_2) + b(y_1 + y_2) \quad (1)$$

$$\iff \quad (2)$$

$$(ax_1 + ax_2) + (by_1 + by_2) \stackrel{?}{=} a(x_1 + x_2) + b(y_1 + y_2) \quad (3)$$

- We're summing the same numbers, just in a different order!

Distributive property of the summation operator

Theorem 1 (distributive property). For any sequence $x_1, \ldots, x_n$ and any sequence $y_1, \ldots, y_n$, and any constants a, b,

$$\sum_{i=1}^{n} (ax_i + by_i) = a \sum_{i=1}^{n} x_i + b \sum_{i=1}^{n} y_i.$$

Exercise 7. Is the following equality true for any x_i and y_i? Why?

$$\sum_{i=1}^{n} (5x_i - 2y_i) \stackrel{?}{=} 5 \sum_{i=1}^{n} x_i - 2 \sum_{i=1}^{n} y_i.$$

Yes. If we substitute $a = 5$ and $b = -2$ into Theorem 1 we get the equation in Exercise 7. i.e., $\sum_{i=1}^{n} (5x_i - 2y_i) = \sum_{i=1}^{n} (5x_i + (-2)y_i)$.

The index of summation can start at an index other than 1.

Example 4. Let π_y represent profit in a company in year y. We can represent profit between 2000 and 2019 as $\sum_{y=2000}^{2019} \pi_y$.

Example 5. The sum of all integers from -2 to 3 is $\sum_{i=-2}^{3} i$.

- It can be useful to break sums up: $\sum_{i=-2}^{3} i = \sum_{i=-2}^{0} i + \sum_{i=1}^{3} i$.
- Let's expand each of the above 3 summations:

$$(-2) + (-1) + (0) + (1) + (2) + (3) = (-2) + (-1) + (0) + \\ + \ (1) + (2) + (3).$$

- e.g., we can break up total profit by decades:
$$\sum_{y=2000}^{2019} \pi_y = \sum_{y=2000}^{2009} \pi_y + \sum_{y=2010}^{2019} \pi_y.$$

The mean: definition and exercise

Definition 3. Given a sequence of numbers $x_1, \ldots, x_n$, we define the mean (or average) as

$$\bar{x} := \frac{1}{n} \sum_{i=1}^{n} x_i.$$

Exercise 8. Suppose the mean amount of euros in the pockets of people in a room is 10€. And there are 35 people in the room. What is the total number of euros in the room? How can we use the above equation to answer this question?

Solution to Exercise 8 (recall $\bar{x} := \frac{1}{n}\sum_{i=1}^{n} x_i$)

1. The hard part is just setting up the problem: Let x_i be the amount of € that student i has in their pocket. So x_4 is the amount of € that the 4th student has in their pocket.
2. Then, we want to solve for the total amount of € in the room, which in our notation is $\sum_{i=1}^{35} x_i$.
3. From the definition of $\bar{x}$ we have the following (I switched the LHS and RHS for following steps): $\frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$.
4. Let's plug in the numbers from the problem: $\frac{1}{35}\sum_{i=1}^{35} x_i = 10$.
5. To solve for $\sum_{i=1}^{35} x_i$, we multiply both sides of the equation by 35 and get the following: $\sum_{i=1}^{35} x_i = 10 \times 35 = 350$.
6. The following relationship is useful more generally (and again follows directly from Def. 3): $\sum_{i=1}^{n} x_i = n\bar{x}$.

Sum of deviations from the mean

Example 6. Simplify $\sum_{i=1}^{n}(x_i - \overline{x})$.
We start by using the distributive property,

$$\sum_{i=1}^{n}(x_i - \overline{x}) = \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} \overline{x} \tag{1}$$

$$= n\overline{x} - \sum_{i=1}^{n} \overline{x} \quad \text{(by Exercise 8)} \tag{2}$$

$$= n\overline{x} - \overline{x}\sum_{i=1}^{n} 1 \quad \text{(see Remark 2)} \tag{3}$$

$$= n\overline{x} - \overline{x}n \quad \text{(by Exercise 3)} \tag{4}$$

$$= 0. \tag{5}$$

Why is $\overline{x}$ a constant? (1/2)

Why could we take out $\overline{x}$ from the summation in step (3)?

- Because $\overline{x}$ does not depend on i.
- Let's do an exercise with numbers to make this point clear.

Exercise 9. Let $x_1 = 1,\ x_2 = 6,\ x_3 = 5$. Evaluate $\overline{x}$.

$$\begin{aligned}
\bar{x} &= \frac{1}{3}\sum_{i=1}^{3} x_i && (1)\\
&= \frac{1}{3}\left(x_1 + x_2 + x_3\right) && (2)\\
&= \frac{1}{3}\left(1 + 6 + 5\right) && (3)\\
&= \frac{1}{3}\cdot 12 && (4)\\
&= 4. && (5)
\end{aligned}$$

Why is $\overline{x}$ a constant? (2/2)

Exercise 10. And what does $\sum_{i=1}^{n} \overline{x}$ equal for this specific case? (recall that $x_1 = 1,\ x_2 = 6,\ x_3 = 5$)

n is 3, $\overline{x}$ is 4, so by Def. 1 we have:

$$\sum_{i=1}^{n} \overline{x} = \sum_{i=1}^{3} 4 = 4 + 4 + 4.$$

You should see that 4 does not depend on i, so we could have pulled it out of the summation:

$$\begin{aligned} \sum_{i=1}^{3} 4 &= 4 \sum_{i=1}^{3} 1. \\ &= 4 \cdot (?) \\ &= 4 \cdot 3 = 12. \end{aligned}$$

Confirming a special case (recall: $x_1 = 1,\ x_2 = 6,\ x_3 = 5$)

Let's check that the general result of Exercise 6, $\sum_{i=1}^{n} (x_i - \overline{x}) = 0$, indeed holds for this specific sequence x_1, x_2, x_3.

Let's start with a very simple step, plugging in for n:

$$\sum_{i=1}^{n} (x_i - \overline{x}) = \sum_{i=1}^{3} (x_i - \overline{x}) .$$

Now, let's apply Def. 1. What is $f(i)$ in this case?
$f(i) = x_i - \bar{x}$. What is the next step then to applying Def. 1?

$$\begin{aligned} \sum_{i=1}^{3} (x_i - \overline{x}) &= (x_1 - \overline{x}) + (x_2 - \overline{x}) + (x_3 - \overline{x}) \\ &= (1 - 4) + (6 - 4) + (5 - 4) \\ &= -3 + 2 + 1 = 0. \end{aligned}$$

Order of operations

$\sum$ has a higher "order of operations" than the minus sign, which is why we use parentheses when we write $\sum_{i=1}^{n} (x_i - \overline{x})$.

$$
\begin{aligned}
0 = \sum_{i=1}^{n} (x_i - \overline{x}) &\neq \sum_{i=1}^{n} x_i - \overline{x} && (1)\\
&= \left(\sum_{i=1}^{n} x_i \right) - \bar{x} && (2)\\
&= n\bar{x} - \bar{x} && (3)\\
&= \bar{x}(n-1). && (4)
\end{aligned}
$$

The $\neq$ means "not necessarily equal for *every* sequence x_i".

Indicator function

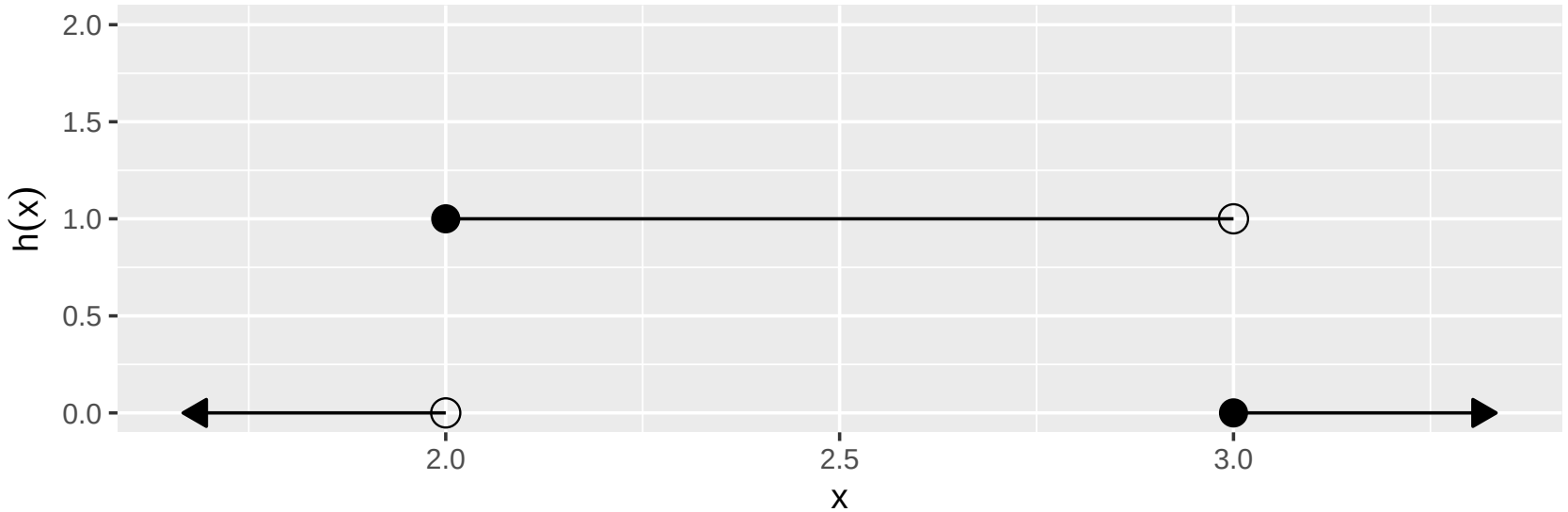
- Indicator functions are a class of functions that "indicate" whether a specific condition is true.
- If the condition is true, the function outputs 1; otherwise it outputs 0.
- That is, an indicator function gives either 0 or 1 as output.

Example 7. Consider the function h defined by $h(x) = \mathrm{I}\{2 \leq x < 3\}$. The I in the definition of h is the notation for the indicator function. The part "$2 \leq x < 3$" is the condition, which will be true for some values of x and not for others.

Exercise 11. Evaluate h at the values -7, 2, 2.5, 3, 4:

- $h(-7) = 0$, $h(2) = 1$, $h(2.5) = 1$, $h(3) = 0$, $h(4) = 0$.

Indicator functions (cont.), recall $h(x) = \mathrm{I}\{2 \leq x < 3\}$



- An equivalent way to define h is to define it "piecewise":

$$h(x) = \begin{cases} 0 & x < 2 \\ 1 & 2 \leq x < 3 \\ 0 & x \geq 3 \end{cases}$$

- The advantage to using an indicator function is that it is more concise, easier to read, and you can use it in expressions.

- Let g_y be the GDP growth (%) in the US in year y.
- Let "low growth" be defined as 2% or lower.

Exercise 12. Express the number of years of low growth in the US between 2016 and 2021. (hint: use $\sum$ and $\mathrm{I}\{\cdot\}$) $\sum_{y=2016}^{2021} \mathrm{I}\{g_y \leq 2\}$.

Example 8. The table below gives g_y for 2016—2021 in the US:

year	2016	2017	2018	2019	2020	2021
GDP growth (%)	1.7	2.3	2.9	2.3	-3.4	5.7

$$\begin{aligned}\sum_{y=2016}^{2021} \mathrm{I}\{g_y \leq 2\} &= \mathrm{I}\{g_{2016} \leq 2\} + \mathrm{I}\{g_{2017} \leq 2\} + \mathrm{I}\{g_{2018} \leq 2\} + \\ &\quad + \mathrm{I}\{g_{2019} \leq 2\} + \mathrm{I}\{g_{2020} \leq 2\} + \mathrm{I}\{g_{2021} \leq 2\} \\ &= \mathrm{I}\{1.7 \leq 2\} + \mathrm{I}\{2.3 \leq 2\} + \mathrm{I}\{2.9 \leq 2\} + \\ &\quad + \mathrm{I}\{2.3 \leq 2\} + \mathrm{I}\{-3.4 \leq 2\} + \mathrm{I}\{5.7 \leq 2\} \\ &= 1 + 0 + 0 + 0 + 1 + 0 = 2.\end{aligned}$$

Dropping outliers

Exercise 13. Suppose the incomes in a population of size n are ordered from x_1 (the richest) to x_n (the poorest). We might want to know the mean income, but dropping the richest person (e.g., Bill Gates). How would we write that mean?

Hint: write it out where the population is of size 3, and use simple numbers. Then generalize to a population of size n.

a. $\frac{1}{n}\sum_{i=1}^{n} x_i$

b. $\frac{1}{n-1}\sum_{i=2}^{n} x_i$ (correct) x_1 is not used, and divide by $n-1$.

c. $\frac{1}{n-1}\sum_{i=1}^{n} x_i$

d. $\frac{1}{n-1}\sum_{i=1}^{n-1} x_i$

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What is a distribution

Definition 4. The distribution of a rv is the list of probabilities that the rv is in each interval of $\mathbb{R}$.

Definition 5. The distribution of a population is the list of proportions of the population that are in each interval of $\mathbb{R}$.

- Note there is no randomness in Definition 5.

Example 9. Suppose a rv W is drawn *at random* from the population of incomes in Catalunya. Then, the distribution of W equals the distribution of incomes in Catalunya.

- e.g., $\mathrm{P}[W \in (20, 30)]$ equals exactly the proportion of incomes in Catalunya that are in the interval $(20, 30)$.
- Since there is no randomness in the second interpretation, we should prefer it. It's important to practice communicating in the least technical way possible.

Same distribution

Definition 6. Two rvs X and Z have the same distribution if $\mathrm{P}[X \in A] = \mathrm{P}[Z \in A]$ for all intervals A.

Exercise 14. Let $X \sim N(0,1)$ and $Y \sim N(3,2)$. Do they have the same distribution?

No. They both have a normal distribution, not the normal distribution. Normal distributions are a family of (different) distributions.

Exercise 15. Prove that X and Y do not have the same distribution. (hint: use Definition 6)

Proof: dist of $X \sim N(0,1)$ doesn't = dist of $Y \sim N(3,2)$

- We need to find an interval for which they have different probs.
- Let's take $A := (-\infty, 0)$. Note that for any rv Z, $\mathrm{P}[Z \in (-\infty, 0)] = \mathrm{P}[Z < 0]$.
- $\mathrm{P}(X < 0) = \frac{1}{2}$ and $\mathrm{P}(Y < 3) = \frac{1}{2}$ because normal distributions are symmetric around their means.
- $\mathrm{P}(Y < 3) = \mathrm{P}(Y < 0) + \mathrm{P}(0 \leq Y < 3)$. What can we say about $\mathrm{P}(0 \leq Y < 3)$? $\mathrm{P}(0 \leq Y < 3) > 0$ because normal rvs have *strictly* positive density for all $x \in \mathbb{R}$.

$$\begin{aligned} \text{Thus,} \quad \mathrm{P}(X < 0) = \frac{1}{2} &= \mathrm{P}(Y < 3) && (1) \\ &= \mathrm{P}(Y < 0) + \mathrm{P}(0 \leq Y < 3) && (2) \\ &> \mathrm{P}(Y < 0). && (3) \end{aligned}$$

- Thus we found an interval $A = (-\infty, 0)$ for which $\mathrm{P}[X \in A] \neq \mathrm{P}[Y \in A]$ so X and Y do not have the same distribution.

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Outline and learning objectives

1. Sum of ***squared*** deviations
   - This will give us more practice with the summation operator and we will prove a useful result.
2. Introduction to sets
   - Sets are an important mathematical tool used in probability/statistics.
3. Random variables
   - These are the core building blocks of probability and help us discuss uncertainty.
   - Uncertainty is typically ignored in the introductory classes to economics... and yet, in real life, whether you work for a company, the government, or in almost any job, there is considerable uncertainty about the future that affect decisions you need to make.

Sum of squared deviations from the mean

- We saw that the sum of deviations is not interesting (it's always 0). The sum of *squared* deviations is more interesting:

$$\sum_{i=1}^{n}(x_i-\overline{x})^2$$

- The exponent is a higher-order operation than summation:

$$\sum_{i=1}^{n}(x_i-\overline{x})^2=\sum_{i=1}^{n}\left\{(x_i-\overline{x})^2\right\}\neq\left\{\sum_{i=1}^{n}(x_i-\overline{x})\right\}^2=\{0\}^2=0.$$

- We often use an equivalent formula:

$$\sum_{i=1}^{n}(x_i-\overline{x})^2=\sum_{i=1}^{n}x_i^2-n\overline{x}^2.$$

- In general, how do we prove that an equality is true? We either start with the LHS and use algebra to end up with the RHS... or we start with the RHS and transform it into the LHS.

Proving $\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$ (1/3)

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n}\left(x_i^2 - 2x_i\bar{x} + \bar{x}^2\right) \quad (1)$$

$$= \sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n} 2x_i\bar{x} + \sum_{i=1}^{n}\bar{x}^2 \quad (2)$$

- The reason we break up the square in (1) is so that we can distribute the summation operator in (1) $\rightarrow$ (2).
- Let's pause and look at where we want to go. We want to get to the RHS of our original equation. We already have one of the terms! Which one?
  - So on line (2) we don't want to change the first term, $\sum_{i=1}^{n} x_i^2$.
- Can you simplify the second term on line (2)?

Proving $\sum_{i=1}^{n}(x_i - \overline{x})^2 = \sum_{i=1}^{n} x_i^2 - n\overline{x}^2$ (2/3)

$$= \sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n} 2x_i\overline{x} + \sum_{i=1}^{n} \overline{x}^2 \quad (1)$$

$$= \sum_{i=1}^{n} x_i^2 - 2\overline{x}\sum_{i=1}^{n} x_i + \sum_{i=1}^{n} \overline{x}^2 \quad (2)$$

$$= \sum_{i=1}^{n} x_i^2 - 2\overline{x}(n\overline{x}) + \sum_{i=1}^{n} \overline{x}^2 \quad (3)$$

$$= \sum_{i=1}^{n} x_i^2 - 2n\overline{x}^2 + \sum_{i=1}^{n} \overline{x}^2 \quad (4)$$

Now let's simplify the third term.

Proving $\sum_{i=1}^{n}(x_i - \overline{x})^2 = \sum_{i=1}^{n} x_i^2 - n\overline{x}^2$ (3/3)

$$= \sum_{i=1}^{n} x_i^2 - 2n\overline{x}^2 + \sum_{i=1}^{n} \overline{x}^2 \tag{1}$$

$$= \sum_{i=1}^{n} x_i^2 - 2n\overline{x}^2 + \overline{x}^2 \sum_{i=1}^{n} 1 \tag{2}$$

$$= \sum_{i=1}^{n} x_i^2 - 2n\overline{x}^2 + \overline{x}^2 n \tag{3}$$

$$= \sum_{i=1}^{n} x_i^2 + n\overline{x}^2(-2+1) \tag{4}$$

$$= \sum_{i=1}^{n} x_i^2 - n\overline{x}^2. \tag{5}$$

Introduction to sets

A set is a mathematical "container". The items in the container are called elements.

Example 10. Suppose each of us in the classroom has a single favorite color, and that C is the set of all of the favorite colors in the classroom.

- We define a set explicitly (i.e., list all of its elements) using curly braces, and separate elements with commas.
  - e.g., let $C = \{\text{pink}, \text{black}, \text{blue}, \text{green}, \text{yellow}, \text{red}\}$.
- The elements of sets do not need to be numbers.
  - Many variables in economics are not numbers: gender, race, college major.

Set membership, and subsets

Let's continue with $C = \{\text{pink}, \text{black}, \text{blue}, \text{green}, \text{yellow}, \text{red}\}$.

- We use $\in$ ("in") and $\notin$ ("not in") to denote whether a particular value is an element of a set or not.
  - If violet $\notin C$, then no one's favorite color in this classroom is violet.
  - If violet $\in C$, at least one person's favorite color in this classroom is violet.
- A subset is a set whose elements are elements of another set.
  - Suppose B is the set of favorite colors of people sitting in the back row. Then, B is a subset of C: if $x \in B$, then $x \in C$.
  - e.g., maybe $B = \{\text{pink}, \text{black}, \text{red}\}$.

Properties of sets

Two important properties of sets:

1. Sets are not ordered: e.g.,
   $B = \{\text{pink}, \text{black}, \text{red}\} = \{\text{black}, \text{red}, \text{pink}\}$.
   - Analogy: $2 + 7 = 7 + 2$.
2. A set does not keep track of duplicate elements.
   $\{\text{pink}, \text{pink}, \text{pink}, \text{black}, \text{black}\} = \{\text{pink}, \text{black}\}$.

Remark 4. If you want to keep track of order (e.g., least to most popular favorite colors), or duplicates, instead use a sequence, which is ordered.

Examples of sets you have seen

- The interval $(2, 5)$ is the set of all (real) numbers between 2 and 5, excluding 2 and 5.
  - $2.84 \in (2, 5)$, $\pi \in (2, 5)$, $\frac{7}{2} \in (2, 5)$, $7 \notin (2, 5)$.
  - We don't use curly braces because we don't explicitly list the elements of the set.
- $[2, 5]$ is the set of all (real) numbers between 2 and 5, *including* 2 and 5.
  - That is, $5 \in [2, 5]$ but $5 \notin (2, 5)$.
- $\mathbb{R} := (-\infty, \infty)$.
- $\mathbb{R}^+ := [0, \infty)$, $\mathbb{R}^- := (-\infty, 0]$.
- $\mathbb{Z}$ is the set of all integers. i.e., $\mathbb{Z} := \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.

Defining sets in terms of others

We often define subsets based on a membership criteria.

Example 11. $D = \{z \in \mathbb{Z} | z < 10,\ z \text{ is prime}\}$ means "the set of all integers such that they are less than 10 and they are prime."

- In this case, it is easier to write $D = \{2, 3, 5, 7\}$.

Example 12. $(3, 6) = \{x \in \mathbb{R} | 3 < x < 6\}$.

Example 13. Let f be defined by $f(x) = x^2$.
Simplify $\{x \in \mathbb{R} | f(x) < 4\}$.

- It is the set of all numbers whose square is less than 4.
- $\{x \in \mathbb{R} | f(x) < 4\} = \{x \in \mathbb{R} | x^2 < 4\} = (-2, 2)$.

Example 14.
$B = \{x \in C | \text{a person in the back row has favorite color } x\}$.

Experiments and rvs

- Let's take an example of flipping a coin.
- When we flip a coin, there are two potential outcomes of the experiment that can occur: Either the coin can be heads or it can be tails.
- The sample space, Ω, is the set of potential outcomes of the experiment.
  - For the coin experiment, we might write the sample space as $\Omega = \{H, T\}$.
  - We often refer to an element of Ω using ω.

Definition 7 (random variable). A random variable (rv) is a ***function*** $\Omega \mapsto \mathbb{R}$.

- A rv essentially summarizes/"translates" each outcome of a specific experiment into a number.

Example: flipping a coin

- We might define a specific rv, X, as the function defined by:

$$X(H) = 1$$
$$X(T) = 0$$

- An equivalent definition of X is: $X = \mathrm{I}\{\omega = H\}$.
- A third way to define the same rv X is:

$$X = \begin{cases} 1 & \text{if } \omega = H \\ 0 & \text{if } \omega = T \end{cases}$$

- Note that we could have chosen numbers other than 1 and 0.

♡s and 5s: the experiment

- Let's make up a game, and call it "hearts (♡s) and 5s".
- To play the game, we shuffle a 52-card deck, and draw 2 cards without replacement.
  - without replacement means after drawing the 1st card, we do not put it back in the deck before drawing the 2nd card.
- You get \$1 for each of the two cards that is a heart and \$2 for each card that is a 5.
- Ω is quite large. e.g., $(1\spadesuit, 3\clubsuit) \in \Omega$, $(J\heartsuit, 7\diamondsuit) \in \Omega$.
  - Ω does not have any information about the winnings.
- We might want to define a rv that reports our winnings from the game.

♡s and 5s: the earnings rv

- Recall: we get \$1 for each ♡ and \$2 for each 5.
- Since each potential outcome, ω, is two cards, it's helpful to break it down as $\omega = (\omega_1, \omega_2)$, where ω_1 is the first card, and ω_2 is the second card drawn.
- Then, we define our earnings rv $H(\omega) = H((\omega_1, \omega_2))$ as:

$$H(\omega) = \mathrm{I}\{\omega_1 \text{ is a } \heartsuit\} \cdot 1 + \mathrm{I}\{\omega_2 \text{ is a } \heartsuit\} \cdot 1 + \\ + \mathrm{I}\{\omega_1 \text{ is a } 5\} \cdot 2 + \mathrm{I}\{\omega_2 \text{ is a } 5\} \cdot 2.$$

Exercise 16. Evaluate $H((7\diamondsuit, 10\heartsuit))$.

$$\begin{aligned} H((7\diamondsuit, 10\heartsuit)) &= \mathrm{I}\{7\diamondsuit \text{ is a } \heartsuit\} \cdot 1 + \mathrm{I}\{10\heartsuit \text{ is a } \heartsuit\} \cdot 1 + && (1) \\ &\quad + \mathrm{I}\{7\diamondsuit \text{ is a } 5\} \cdot 2 + \mathrm{I}\{10\heartsuit \text{ is a } 5\} \cdot 2 && (2) \\ &= 0 \cdot 1 + 1 \cdot 1 + 0 \cdot 2 + 0 \cdot 2. && (3) \\ &= 1. && (4) \end{aligned}$$

The support of a rv

Definition 8. The support of a rv X is the set of potential outcomes of X and is denoted $\text{supp}(X)$.

- For the coin example, 0 and 1 are the potential outcomes of X, so $\text{supp}(X) = \{0, 1\}$.
- A useful first step when encountering a random variable is to ask yourself, "What values can the rv take (*as output*)?"
  - We often say "X can take values (*as output*) in the set {0,1}."

Exercise 17. What is the support of H? $\text{supp}(H) = \{0, 1, 2, 3, 4, 5\}$.

- If the cards were drawn *with* replacement in the experiment, then what would the support be? {0,1,2,3,4,5,6}.

Observed outcomes of a rv

- We might flip the coin 5 times, and we would say that each of those 5 specific flips produces an observed outcome of X. We could write down the outcomes as follows:

flip #	obs. outcome of Ω	$X(a_i)$ (=obs. outcome of X)
1	$a_1 = T$	$x_1 = X(T) = 0$
2	$a_2 = H$	$x_2 = X(H) = 1$
3	$a_3 = H$	$x_3 = X(H) = 1$
4	$a_4 = H$	$x_4 = X(H) = 1$
5	$a_5 = T$	$x_5 = X(T) = 0$

- We record observed outcomes using lowercase letters.

Introducing events

Before defining events, let's motivate why events are important:

The probability that $\underbrace{_____}_{\textbf{\textit{This}} \text{ is an event!}}$.

Or equivalently, using math notation:

$$\underbrace{\mathrm{P}(\quad)}_{\textbf{\textit{This}} \text{ is an event!}}$$

Anything you can "take the probability of" is an event! Examples:

1. The fire alarm in Matherly goes off in the next hour.
2. Someone in our class falls asleep before this lecture is over.
3. A woman is elected president next election.
4. The University of Florida wins its next football game.

Formal definition of event

Definition 9 (event of a rv). An event of a ***rv*** X is any subset A of $\mathbb{R}$. If $X(\omega) \in A$ then we say the event A happened (for X). If $X(\omega) \notin A$, then the event A did not happen (for X).

Example 15. Examples of events of H:

1. $\{0\}$ (0 winnings)
2. $\{1, 2, 3, 4, 5\}$ (winnings > 0)
3. $\{1, 3, 5\}$ (win an odd amount of \$)
4. $(1, 4)$. For H, prob of this event is the same as $\{2, 3\}$.

Exercise on potential outcomes

Exercise 18. Let the rv D take as value the roll of a 6-sided die.

1. How many distinct outcomes are possible with a roll of the die?
   There are 6 potential outcomes of the experiment.
2. Write out the sample space.
   $\Omega = \{1, 2, 3, 4, 5, 6\}$.
3. Write out the mathematical definition of the function D.
   In this simple case we can define D as $D(\omega) = \omega$.

Events (cont.)

Exercise 19. Give examples of *events* of the rv D.

Here are some examples of events. We assign variables to them so we can discuss them.

1. $A = \{4\}$.
2. $B = \{1, 3, 5\}$.
   - B is the event that the number rolled is odd.
3. $C = \{1, 2, 3, 4\}$.
   - C is the event that the number rolled is less than or equal to 4.

Events (cont.), $A = \{4\}, B = \{1, 3, 5\}, C = \{1, 2, 3, 4\}$

- It's important to understand that we only roll the die ***once***.
- For example, $\{1, 3, 5\}$ does not mean that we roll the die three times. In the *one* roll of a die ω, $D(\omega) \in \{1, 3, 5\}$, i.e., we get a number in that list, then the event happened.
- If the outcome of the roll is 5, we say the event B "happened".
- Two or more events can happen. For example, if we roll a 4, then the events A and C both happened because $4 \in A$ and $4 \in C$. The event B did not happen because $4 \notin B$.

Notation: $X = 1$ versus $X(\omega) = 1$

- Suppose we flip the coin and get heads. We *don't* write $X = 1$ to record an observed outcome.
- X is a function and 1 is a number, so it would be weird to say that a function were equal to a number.
- Really, $X = 1$ is shorthand notation to refer to the *event* $\{1\}$.
- X is a function, so it never actually equals a number. The correct statement for $X(\omega) = 1$ is "The observed outcome of X was 1 for this particular flip".

Exercise: events vs. observed outcomes

We might flip the coin 5 times, and we would say that each of those times results in an observed outcome of X.

Exercise 20. Can we record the outcomes as follows?

flip #	outcome of Ω	$X(\omega)$ (=outcome of X)
1	$a_1 = T$	$X = 0$
2	$a_2 = H$	$X = 1$
3	$a_3 = H$	$X = 1$
4	$a_4 = H$	$X = 1$
5	$a_5 = T$	$X = 0$

- No. $X = 0$ refers to the *event* $\{0\}$. Stating $X = 0$ does not record that the event *happened* for a particular coin flip.

Exercise: events vs. observed outcomes

We might flip the coin 5 times, and we would say that each of those times results in an observed outcome of X.

Exercise 20. Can we record the outcomes as follows?

flip #	outcome of Ω	$X(\omega)$ (=outcome of X)
1	$a_1 = T$	$x_1 = 0$
2	$a_2 = H$	$x_2 = 1$
3	$a_3 = H$	$x_3 = 1$
4	$a_4 = H$	$x_4 = 1$
5	$a_5 = T$	$x_5 = 0$

- To record the *outcome* of X for a specific outcome of the experiment, we would use a lowercase letter.

"$X = 1$" is an event

- "$X = 1$" refers to the event that X is 1. i.e., the event $\{1\}$.
- $\mathrm{P}(X = 1)$ = "the probability (of the *event*) that X is 1". i.e., $\mathrm{P}(X = 1) = \mathrm{P}(X \in \{1\})$.
- In this case, we have that $\mathrm{P}(X = 1) = \mathrm{P}(H)$, i.e., the probability of getting a heads.
- Similarly, $\mathrm{P}(2 \leq D \leq 4)$ = "the probability (of the *event*) that D is in-between 2 and 4." i.e., $\mathrm{P}(2 \leq D \leq 4) = \mathrm{P}(D \in [2, 4])$.

What is probability?

We've been using the term "probability", but how do we *define* it?

Definition 10. The probability of an event is the long-run proportion of times the event happens.

Example 16. Consider the rv D and the event $A := \{1, 4\}$. We know intuitively that $\mathrm{P}(A) = \frac{2}{6}$. But why?

- $\mathrm{P}(A) = \frac{2}{6}$ *because* if we roll the die n times, the proportion of times A will happen is approximately $\frac{2}{6}$ if n is large.
- If we take the limit as n goes to infinity, the "approximately" just above becomes "exactly".

Exercise 21. If we roll a die 1,000 times, the event A will happen approximately how many times? $\frac{2}{6} \cdot 1000$.

- Generalizing, if we roll a die n times, the event A will happen approximately $\frac{2}{6} \cdot n$ times.

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Outline and learning objectives

1. Example with rvs that is more interesting in economics.
   a. Understand the basic intuition of a simple model.
   b. Think about an "experiment" more generally, and what it means for something to be "random".
   c. Get practice with defining some rvs in terms of others.
2. New example using coins.
   a. Evaluate probabilities of events of a rv.
3. Define a probability mass function (pmf) and do examples.

The phrase "is a function of"

We use the phrase is a function of to mean that the value of a variable ***depends on*** the value of another variable.

Exercise 22. The time you get out of bed in the morning is a function of which variables?

1. The time you set your alarm to.
2. The time you went to bed.
3. The time of your first class.
4. The time the other people in your home wake up.
5. The temperature.

Ice cream

- Luca is an ice cream salesperson.
- Every day he wakes up, buys ice cream from the supermarket and bikes an hour to the beach to sell individual ice creams.
- Some days he makes a lot of money and some days less.
- We will create a model to show how the uncertainty of profit affects Luca's decisions.
- We need a ***simple*** mechanism for *why* profit fluctuates.
  - For the ♡s and 5s example, the mechanism was drawing cards.
  - For this ice cream business example, our mechanism will be the weather: people buy more ice cream on sunny days.

Getting a feeling for the model

When studying a model, there are 3 important questions:

1. What are the choice variables?
   - A choice variable is a variable a person can choose in order to maximize something (e.g., profit).
   - In our model, Luca chooses the ***quantity*** of ice cream he buys.
2. What information does the decision-maker (Luca) have ***when he makes his choice***?
   - He will have partial information about the beach's weather:
   - He uses the weather at his house to *guess* the beach's weather.
3. What are the sources of randomness? i.e., why would Luca make more money on some days and less on others?
   - In our model, the weather drives all randomness.

Simplifications

To keep our model easy to understand, we make some simplifications:

1. Let p_S be the price per pint of ice cream that Luca buys from the super market.
   - Suppose that p_S is ***constant***.
     i.e., it does not change depending on the weather.
2. Let p_B be the price per pint that Luca sells at the beach.
   - Suppose that $p_B > p_S$ and is ***constant***. In a more complex model, Luca would *choose* p_B.
3. Luca is the only ice cream salesperson at the beach.
4. There are only 2 states of the weather at a certain location: *cloudy* or *sunny*.

What is the "experiment" here?

- The experiment is observing "day-specific characteristics" that affect Luca's ice cream business.
- With our simplifications, the day-specific characteristics are:
  1. The weather at Luca's house.
  2. The weather at the beach.
- Let $w_L \in \{\textit{cloudy}, \textit{sunny}\}$ be the weather at Luca's house, and let $w_B \in \{\textit{cloudy}, \textit{sunny}\}$ be the weather at the beach.
- An outcome is a pair of weather states: $\omega = (w_L, w_B)$.

Exercise 23. What is the sample space?

$$\Omega = \{(\textit{cloudy}, \textit{cloudy}), (\textit{cloudy}, \textit{sunny}), (\textit{sunny}, \textit{cloudy}), (\textit{sunny}, \textit{sunny})\}$$

Demand

- Demand is how many pints people at the beach ***would like*** to buy (at the fixed price, p_B) for a day.
- Since p_B is fixed, we do not write demand as a function of p_B.
- We model demand as depending entirely on the beach weather.
- Since the beach weather is random, demand is random.
- Demand could be the following, e.g.:

$$Q_D(\omega) = Q_D((w_L, w_B)) \tag{1}$$
$$= \mathrm{I}\{w_B = \textit{cloudy}\} \cdot 2 + \mathrm{I}\{w_B = \textit{sunny}\} \cdot 10. \tag{2}$$

Exercise 24. Does Q_D depend on "all" of $\omega = (w_L, w_B)$?

No, on line (2) w_B appears but w_L does not. So we can write $Q_D(\omega) = Q_D(w_B)$.

Revenue

- Luca's revenue comes from how much ice cream he sells at the beach, which depends on:
  1. The demand (we defined this on the prev. slide).
  2. $Q_S(\omega) :=$ How many pints he bought at the supermarket in the morning, which is Luca's choice variable.
- Thus, revenue is $R(\omega) = p_B \cdot \min(Q_D(\omega), Q_S(\omega))$.
  - $Q_D(\omega) > Q_S(\omega)$ if Luca didn't buy enough ice cream.
  - $Q_D(\omega) < Q_S(\omega)$ if Luca bought too much ice cream.

Exercise 25. What ω would lead to $Q_D(\omega) > Q_S(\omega)$?

If $\omega = (w_L, w_B) = (\textit{cloudy}, \textit{sunny})$, then Luca doesn't buy much ice cream since he thinks it is likely cloudy at the beach, but it ends up being sunny.

Cost

Exercise 26. Is cost a ***random*** variable? Yes.

Definition. A variable is random if it is a function of ω.

- Luca's cost is the cost of the ice cream he buys from the supermarket (i.e., quantity times price).
- Cost is random because each day the cost could change:

$$C(\omega) = C(w_L) = p_S \cdot Q_S(w_L).$$

Profit

- Is profit random? Yes, because revenue and cost are random:

$$\pi(\omega) := R(\omega) - C(\omega).$$

Remark 5. We typically use more concise notation: $\pi := R - C$.

$$\begin{aligned}
\pi(\omega) &= \pi((w_L, w_B)) && (1)\\
&= R((w_L, w_B)) - C(w_L) && (2)\\
&= \underbrace{p_B \cdot \min(Q_D(w_B), Q_S(w_L))}_{\text{revenue}} - \underbrace{p_S \cdot Q_S(w_L)}_{\text{cost}}. && (3)
\end{aligned}$$

Exercise 27. Review: what is Luca's choice variable? Q_S.
Recall that we are assuming p_S is fixed.

Profit maximization

- Luca's choice variable is how much ice cream he buys, Q_S.
- But $Q_S = Q_S(w_L)$ is a *function*. What does it mean to "choose" a function?
- He chooses 2 numbers: $Q_S(w_L = \textit{sunny})$, and $Q_S(w_L = \textit{cloudy})$. What strategy does he use?
- Profit is *random*, and he has to make a choice, based on w_L, before he even knows w_B, which is a determinant of profit.
- Put simply, how can he choose how much ice cream to buy if he doesn't know what demand will be?
- We will come back to this question in a future lecture, after we've built up the necessary tools to answer it.

Back to coins!

Exercise 28. Suppose we define one experiment to be *two* coin flips. Give (1) the sample space, Ω; and (2) define a (*any*) rv on Ω.

(1) The sample space is: $\Omega = \{(H, T), (T, H), (H, H), (T, T)\}$.

(2) One example of a rv defined on Ω is the function Y defined by:

$$Y((H, T)) = 1$$
$$Y((T, H)) = 1$$
$$Y((H, H)) = 2$$
$$Y((T, T)) = 0$$

- Y represents the number of heads of the two coin flips.
- We lose information if we focus on the outcome of Y instead of the outcome of the experiment: If we observe $Y(\omega^*) = 1$ for an outcome, we don't know if $\omega^* = (H, T)$ or $\omega^* = (T, H)$.
- Think about rvs as *summarizing* information we care about.

rvs summarize information

Suppose I offer you the following wager:

> *"Give me \$1 and I'll flip two coins and I'll give you \$1 for each head that comes up."*

- In this case, the rv Y that we defined is an interesting one because it captures ***revenue***.
- If the payout were different for (H, T) than (T, H) we would want to define a different rv (that does not map them to the same number).

rvs based on other rvs

We often define rvs as functions of other rvs.

Exercise 29. With Y defined as before and with the example of the wager, how would you interpret the rv Z defined by $Z := Y - 1$? Recall, this notation means: $Z(\omega) := Y(\omega) - 1$.

It represents your profit for the outcome of the experiment.

Exercise 30. Consider the rv W defined by

$$\begin{aligned} W((H,T)) &= 0 \\ W((T,H)) &= 0 \\ W((H,H)) &= 1 \\ W((T,T)) &= -1 \end{aligned}$$

What can we say about W? $W = Z$.

Simplifying notation

- For notational simplicity, we often write $Y(H, T) = 1$ instead of $Y((H, T))$.
  - But we must remember that Y really only takes one argument (one element of Ω).
- Similarly, in the ice cream example, we often write $\pi(w_L, w_B)$ instead of $\pi((w_L, w_B))$.
  - Technically, π takes one pair of weather states as input.
  - Practically, it's equivalent to think of π as taking two individual weather states as input.

Evaluating probabilities of events of a rv (1/2)

- A useful strategy for evaluating the probability of an event of a rv is to use the probability of outcomes of the experiment.
- Let Y map two *fair* coin flips to the number of heads.
- Recall $\Omega = \{(H, T), (T, H), (H, H), (T, T)\}$ and $Y((H, T)) = 1, Y((T, H)) = 1, Y((H, H)) = 2, Y((T, T)) = 0$.

Exercise 31. What are $\mathrm{P}(Y = 0), \mathrm{P}(Y = 1), \mathrm{P}(Y = 2)$?

Note that $\mathrm{P}((H, H)) = \mathrm{P}((H, T)) = \mathrm{P}((T, H)) = \mathrm{P}((T, T)) = \frac{1}{4}$.

- Then, the desired probabilities of Y follow:
- $\mathrm{P}(Y = 0) = \mathrm{P}((T, T)) = \frac{1}{4}$.
- $\mathrm{P}(Y = 1) = \mathrm{P}(\{(H, T), (T, H)\}) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.
- $\mathrm{P}(Y = 2) = \mathrm{P}((H, H)) = \frac{1}{4}$.

Evaluating probabilities of events of a rv (2/2)

In case evaluating $\mathrm{P}(Y = 1)$ is confusing, it helps to make a table:

ω	$\mathrm{P}(\omega)$	$Y(\omega)$	$Y(\omega) = 1$?
(H, H)	$\frac{1}{4}$	2	$\times$
(H, T)	$\frac{1}{4}$	1	✓
(T, H)	$\frac{1}{4}$	1	✓
(T, T)	$\frac{1}{4}$	0	$\times$

Now, we sum up $\mathrm{P}(\omega)$ in the rows with checkmarks.

$$\mathrm{P}(Y = 1) = \mathrm{P}((H, T)) + \mathrm{P}((T, H)) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Discrete rvs

Definition 11. A rv is a discrete rv if it can take on a finite or countably infinite number of values.

- The examples of rvs we've seen so far have all been discrete.
  - Let X be 1 if H, 0 if T. Then, $\text{supp}(X) = \{0, 1\}$.
  - Let D be the number from rolling a die. Then, $\text{supp}(D) = \{1, 2, 3, 4, 5, 6\}$.
  - These sets are *discrete*. e.g., we can write all the values on the board that they can take. *1,2,3,4,5,6*.
- Don't worry about the term "countably infinite" for our class.



  - If you want an example to satisfy your curiosity, consider an experiment where you flip a coin until you get a tails. Define Q to be the number of coin flips it takes until you get a tails. Then, $\text{supp}(Q) = \{1, 2, 3, \ldots\}$.
  - From now on, in our class we will use "infinite" to mean "***un***countably infinite" (e.g, $[3, 4]$).

Exercise on notation for support

Suppose W is a discrete rv with $\text{supp}(W) = \{w_1, \ldots, w_k\}$.

Exercise 32 (parsing abstract notation). What is k equal to in each of the following? Additionally, give the values of w_j for $j = 1, \ldots, k$.

1. Let $\Omega = \{H, T\}$, $X(H) = 1, X(T) = 0$. $k = 2$.
   $w_1 = 0, w_2 = 1$.
2. Let Y be the number of heads from two coin flips. $k = 3$.
   $w_1 = 0, w_2 = 1, w_3 = 2$.
3. Let D be the number rolled from a six-sided die. $k = 6$.
   $w_j = j$ for $j = 1, \ldots, 6$. $\iff w_1 = 1, w_2 = 2, \ldots, w_6 = 6$.

Probability mass functions (pmfs)

Definition 12 (pmf). Suppose W is a discrete rv. We define the probability mass function (pmf) as the function $f_W : \mathbb{R} \to [0, 1]$ such that $f_W(a) = \mathrm{P}(W = a)$.

- Every function takes an input and gives an output.
- A pmf must be able to take any number as input.
- As output, it gives the probability that the rv takes that value.

Applying definition to construct pmfs

Exercise 33. Write the pmf for each of the rvs in Exercise 32 assuming the coin for the underlying experiment of X and Y is fair and the die underlying the experiment for D is fair.

1. Let $\Omega = \{H, T\}$, $X(H) = 1, X(T) = 0$.

Let f_X be the pmf of X. Then, $f_X(0) = \mathrm{P}(X = 0) = \frac{1}{2}$.

$f_X(a) = \frac{1}{2}$ for $a \in \{0, 1\}$. Is that all we need? Not quite. We also need to specify: $f_X(a) = 0$ for $a \notin \{0, 1\}$.
We can define this function as:

$$f_X(a) = \begin{cases} \frac{1}{2} & a = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

We could also write the above as: $f_X(a) = \frac{1}{2} \cdot \mathrm{I}[a \in \{0, 1\}]$.

Solution to Exercise 33, part (2)

2. Let Y be the number of heads from two fair coin flips.

Define the function f_Y such that

$$
\begin{aligned}
f_Y(0) &= \mathrm{P}(Y = 0) = ? \\
f_Y(1) &= \mathrm{P}(Y = 1) = ? \\
f_Y(2) &= \mathrm{P}(Y = 2) = ? \\
f_Y(x) &= ? \text{ for } x \notin \{0, 1, 2\}
\end{aligned}
$$

Solution to Exercise 33, part (2)

2. Let Y be the number of heads from two fair coin flips.

Listing them all out, we have:

$f_Y(0) = \mathrm{P}(Y = 0) = \mathrm{P}(\omega = (T, T)) = \frac{1}{4}$.

$f_Y(1) = \mathrm{P}(Y = 1) = \mathrm{P}(\omega \text{ is } (H, T) \text{ or } (T, H)) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$.

$f_Y(2) = \mathrm{P}(Y = 2) = \mathrm{P}(\omega = (H, H)) = \frac{1}{4}$.

$f_Y(a) = 0$ for $a \neq 0, 1, 2$.

The function f_Y defined above is the pmf of Y.

Solution to Exercise 33, part (3)

3. Let D be the number rolled from a six-sided die.

Define the function f_D such that

$$f_D(a) = \begin{cases} \frac{1}{6} & a = 1, 2, 3, 4, 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

The function f_D is the pmf of the rv D.

Equivalently, $f_D(a) = \frac{1}{6} \cdot \mathrm{I}\left[a \in \{1, 2, 3, 4, 5, 6\}\right]$.

Properties of pmfs

The following are properties of pmfs that follow from the definition. Let W be a discrete rv with $\text{supp}(W) = \{w_1, \ldots, w_k\}$, and let f_W be the pmf of W.

1. f_W is a function.
   - Every function has an input and gives an output. The input value is a real number, and the output is the probability with which W takes that value.
2. For $j = 1, \ldots, k$, $0 \leq f_W(w_j) \leq 1$.
3. $\sum_{j=1}^{k} f_W(w_j) = 1$.
   - A fun statistic: researchers have found that 97% of teachers at UPF are lacking common sense... luckily for you, I'm in the other 5%.

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5. Continuous random variables, pdfs, and cdfs

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Learning objectives and outline

1. Introduce continuous random variables and probability density functions (pdfs).
2. Learn the important properties of normal pdfs.
3. Introduce cumulative distribution functions (cdfs).
   - These are an alternative to using pmfs and pdfs.
   - And a cdf works for *both* discrete and continuous rvs.
4. Do an example using the cdf of a continuous rv.
   - Show how it compares to using the pdf.

Motivation for *continuous* rvs

- We have only discussed discrete rvs so far. However, many rvs in economics are not discrete.
- There are many variables in economics that do not take on a finite or countable number of values.
- For example, it does not make much sense to model income as a discrete random variable, since from a practical perspective income takes on many different values.
- Thus, in order to talk about these types of rvs, and especially probabilities of *events* of these rvs, we need a different tool.
- Continuous rvs can take on an infinite amount of values.
- In practice, if a discrete rv X does not take any value with high probability, then we usually approximate it as a continuous rv.

Example of a continuous rv: spinning a quarter

- Let Q be the exact (i.e., unrounded) number of seconds until a quarter stops spinning. Let's say $\text{supp}(Q) = (0, \infty)$.
- Recall we use probability to calc. probabilities of events of rvs.

Examples of events of Q:

1. $\{\pi\}$ (the quarter stops at exactly π seconds)
   - $\mathrm{P}(Q \in \{\pi\}) = \mathrm{P}(Q = \pi) = 0$. (i.e., exact equality).
   - What ***if*** $\mathrm{P}(Q = x) = 0.0001$ for $x \in (3.14, 3.15)$? Is it possible? No, because $\infty \cdot 0.0001 = \infty > 1$.
   - Thus, we don't care about events like this of exact equality. Rather, we care about events that are intervals.
2. $[2, 3]$ (the quarter lasts between 2 and 3 seconds inclusive)
3. $(5, \infty)$ (the quarter lasts at least 5 seconds)
4. $\{x \in \mathbb{R} | x \in (0, 3) \text{ or } x \in (5, \infty)\}$
   (either it stops spinning before 3 seconds or after 5 seconds)

How to get prob of events that are uncountable sets?

- For discrete rvs, we calculated the probs. of events using pmfs.
- A pmf is a natural tool for discrete rvs because the probability of an event of a discrete rv can always be expressed as the summation of individual probabilities.
- For a continuous rv, the probability of an event cannot be calculated by the sum of the individual probabilities.
  - e.g., we cannot sum a function $f(x)$ over all $x \in [2, 3]$.
- We need a new tool.

Definition 13. Let X be a rv. Then, X is a continuous rv if there exists a function, $f_X : \mathbb{R} \mapsto [0, \infty)$, called a probability *density* function (pdf), such that for any numbers $c_1 \leq c_2$,

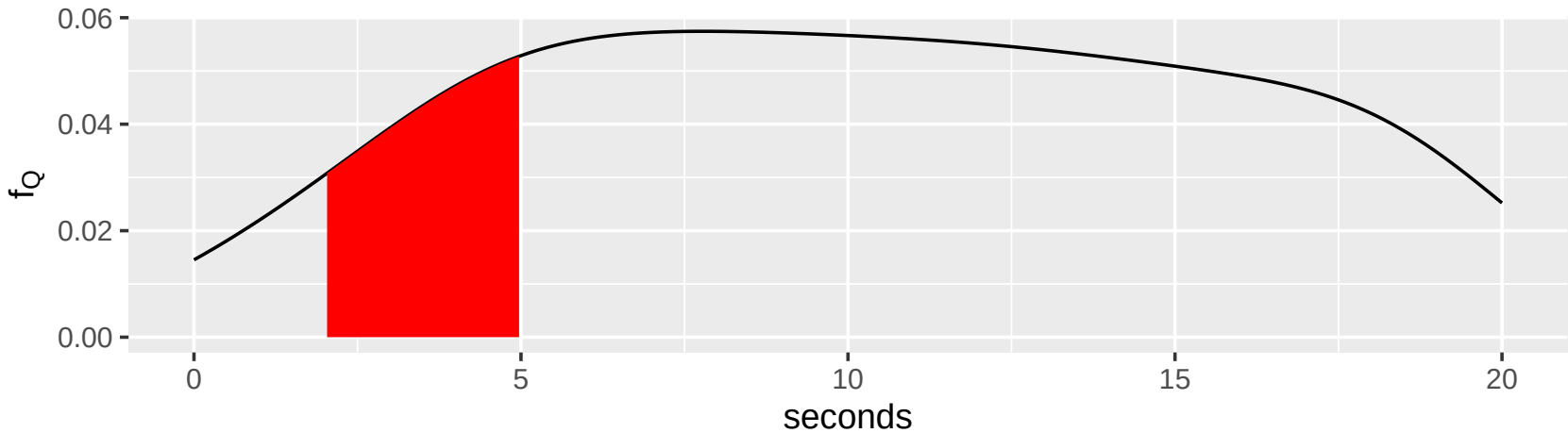
$$\mathrm{P}(X \in [c_1, c_2]) = \mathrm{P}(c_1 \leq X \leq c_2) = \int_{c_1}^{c_2} f_X(x)dx.$$

- f_X is the name of the function. It is common to specify the subscript to remind us that it is the pdf of X. We might have f_X and f_Q if we have rvs X and Q.
- A pdf is a tool, i.e., invention. It doesn't give us probability directly, but rather density. We have to *integrate* density to get probability.

Definition 13. Let X be a rv. Then, X is a continuous rv if there exists a function, $f_X : \mathbb{R} \mapsto [0, \infty)$, called a probability *density* function (pdf), such that for any numbers $c_1 \leq c_2$,

$$\mathrm{P}(X \in [c_1, c_2]) = \mathrm{P}(c_1 \leq X \leq c_2) = \int_{c_1}^{c_2} f_X(x)dx.$$

Let f_Q be the pdf of Q (the seconds until a quarter stops):



We can express the prob of any event of Q in terms of f_Q.

Exercise 34. $\mathrm{P}(Q \in [2, 5]) = \mathrm{P}(2 \leq Q \leq 5) = \int_2^5 f_Q(q)dq$.

Definition 13. Let X be a rv. Then, X is a continuous rv if there exists a function, $f_X : \mathbb{R} \mapsto [0, \infty)$, called a probability *density* function (pdf), such that for any numbers $c_1 \leq c_2$,

$$\mathrm{P}(X \in [c_1, c_2]) = \mathrm{P}(c_1 \leq X \leq c_2) = \int_{c_1}^{c_2} f_X(x)dx.$$

Property 1: Let X be a continuous rv with pdf f_X. Then, $\mathrm{P}(X = z) = 0$ for all $z \in \mathbb{R}$.

Proof.

$$\mathrm{P}(X = z) = \mathrm{P}(z \leq X \leq z) \tag{1}$$

$$= \int_z^z f_X(x)dx \quad \text{(by Def. 1)} \tag{2}$$

$$= 0. \tag{3}$$

Exercise 35. What is $\mathrm{P}(Q = \pi)$? 0, by Property 1.

- Similarly, $\mathrm{P}(Q = 3) = 0$.
- With continuous rvs, events of interest are *intervals*.

Definition 13. Let X be a rv. Then, X is a continuous rv if there exists a function, $f_X : \mathbb{R} \mapsto [0, \infty)$, called a probability *density* function (pdf), such that for any numbers $c_1 \leq c_2$,

$$\mathrm{P}(X \in [c_1, c_2]) = \mathrm{P}(c_1 \leq X \leq c_2) = \int_{c_1}^{c_2} f_X(x)dx.$$

Definition 14. The support of a continuous rv X with pdf f_X is $\mathrm{supp}(X) = \{x | f_X(x) > 0\}$.

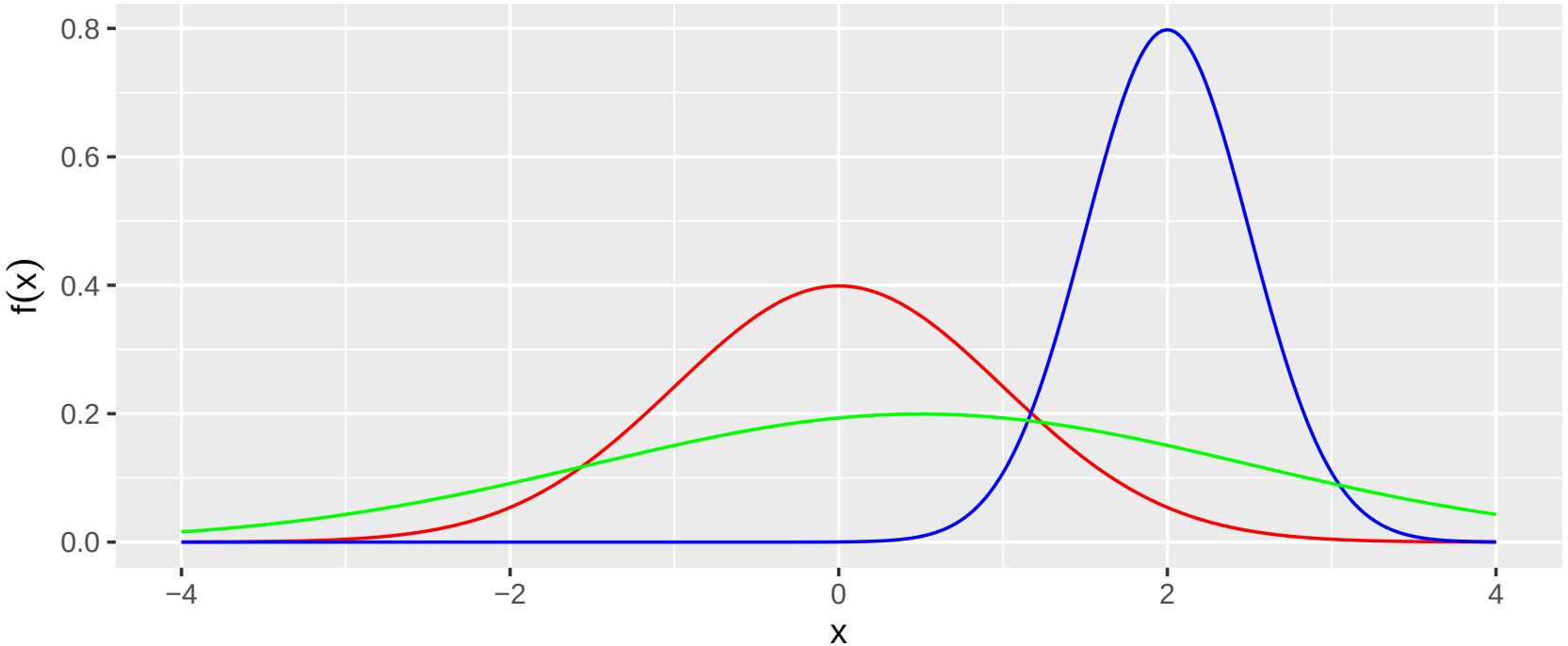
Let X be a continuous rv with pdf f_X. Then, X and f_X satisfy the following properties:

1. $\mathrm{P}(X = z) = 0$ for all $z \in \mathbb{R}$.
2. $f_X(z) \geq 0$, for all $z \in \mathbb{R}$.
   - Density can equal 0 for some z. e.g., $f_Q(z) = 0$ for $z < 0$.
3. $\int_{-\infty}^{\infty} f_X(z)dz = \mathrm{P}\left(X \in (-\infty, \infty)\right) = \mathrm{P}(-\infty < X < \infty) = 1$.

Normal pdfs

- The most common type of pdf of a continuous rv is the "normal pdf", also referred to as the "Gaussian pdf".
- The normal pdf is a critical component of the Central Limit Theorem, which we will rely on heavily when we get to the statistics part of the course.
- The next exercise is only for those of you who have seen a normal pdf before (don't worry if you haven't).

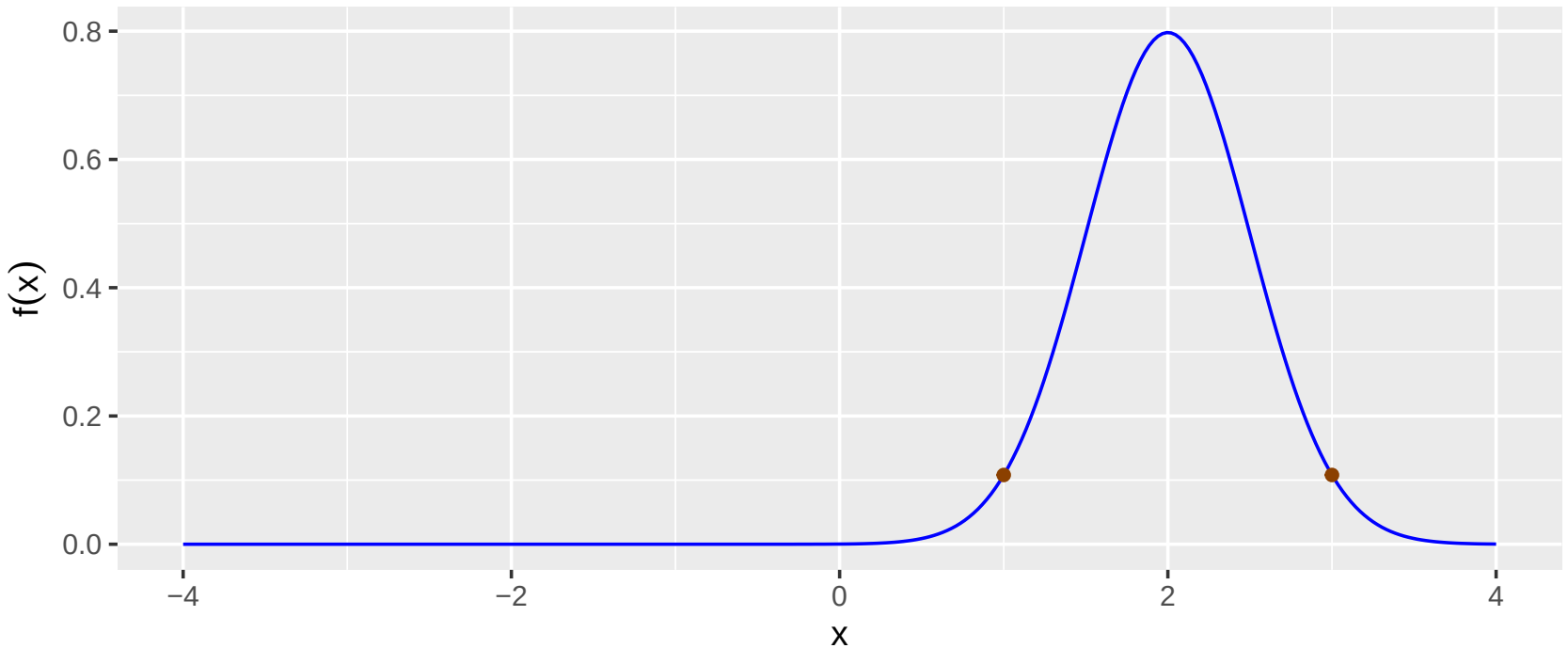
Properties of a normal pdf, f



Exercise 36. Which of the above is the pdf of a normal pdf?

A: All of them! In fact, there is not just one single "normal pdf". It is a family of pdfs.

Properties of a normal pdf, f

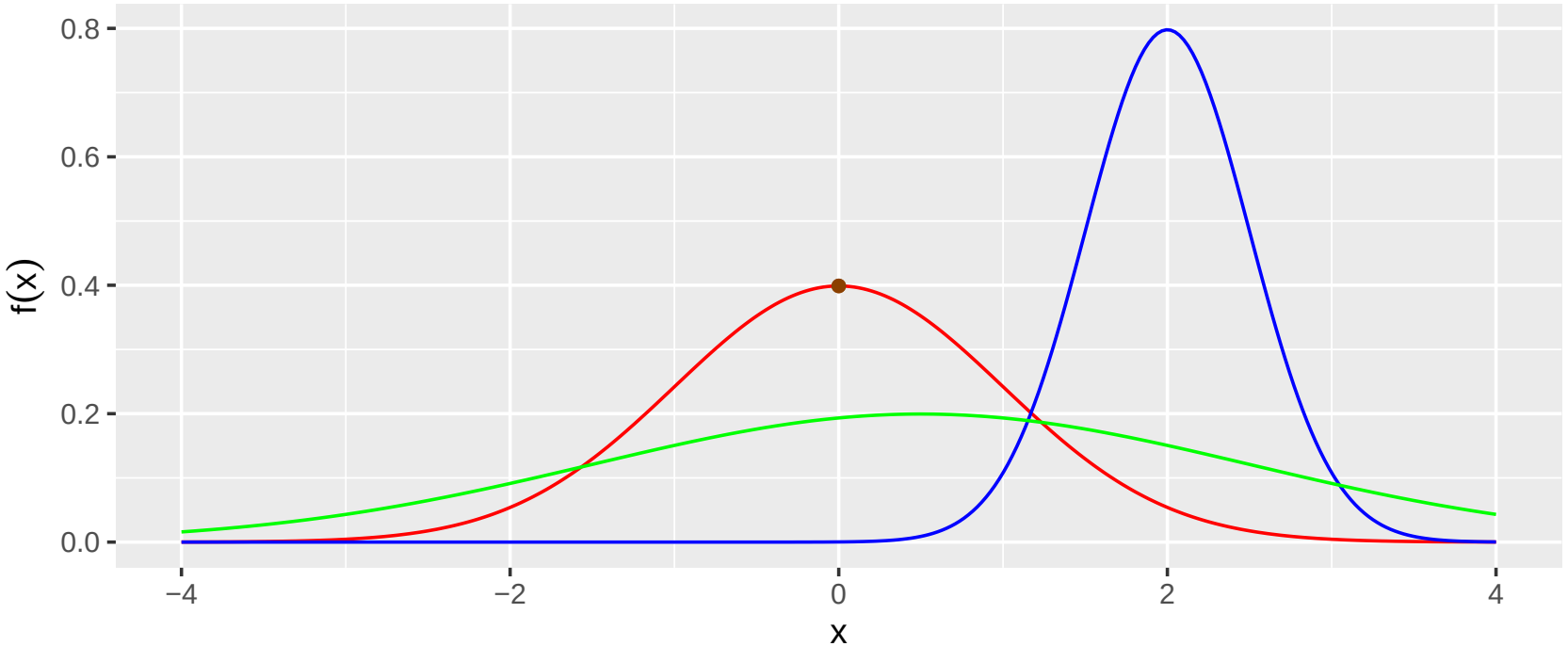


Every normal pdf f has the following properties:

1. They have one mode (peak), denoted μ.
2. $f(x) > 0$ for all $x \in \mathbb{R}$. i.e., the support is $(-\infty, \infty)$.
3. They are symmetric about μ. Formally: for any normal pdf f and any number c, $f(\mu + c) = f(\mu - c)$.

Exercise 37. If $\mu = 2$, then $f(3) = f(2 + 1) = f(2 - 1) = f(1)$.

Properties of a normal pdf, f



- The pdf of a normal rv is: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$.
- μ and σ are parameters of the pdf. e.g., if we plug in $\mu = 2, \sigma = 0.5$, we get the blue pdf: $f(x) = \frac{1}{0.5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-2}{0.5}\right)^2}$.
- If $\mu = 0$, $\sigma = 1$, we use ϕ for f and call it the *standard* normal pdf. e.g., $\phi(0) = \frac{1}{1\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{0-0}{1}\right)^2} = \frac{e^0}{\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi}} \approx 0.40$.

Using the pdf of Q

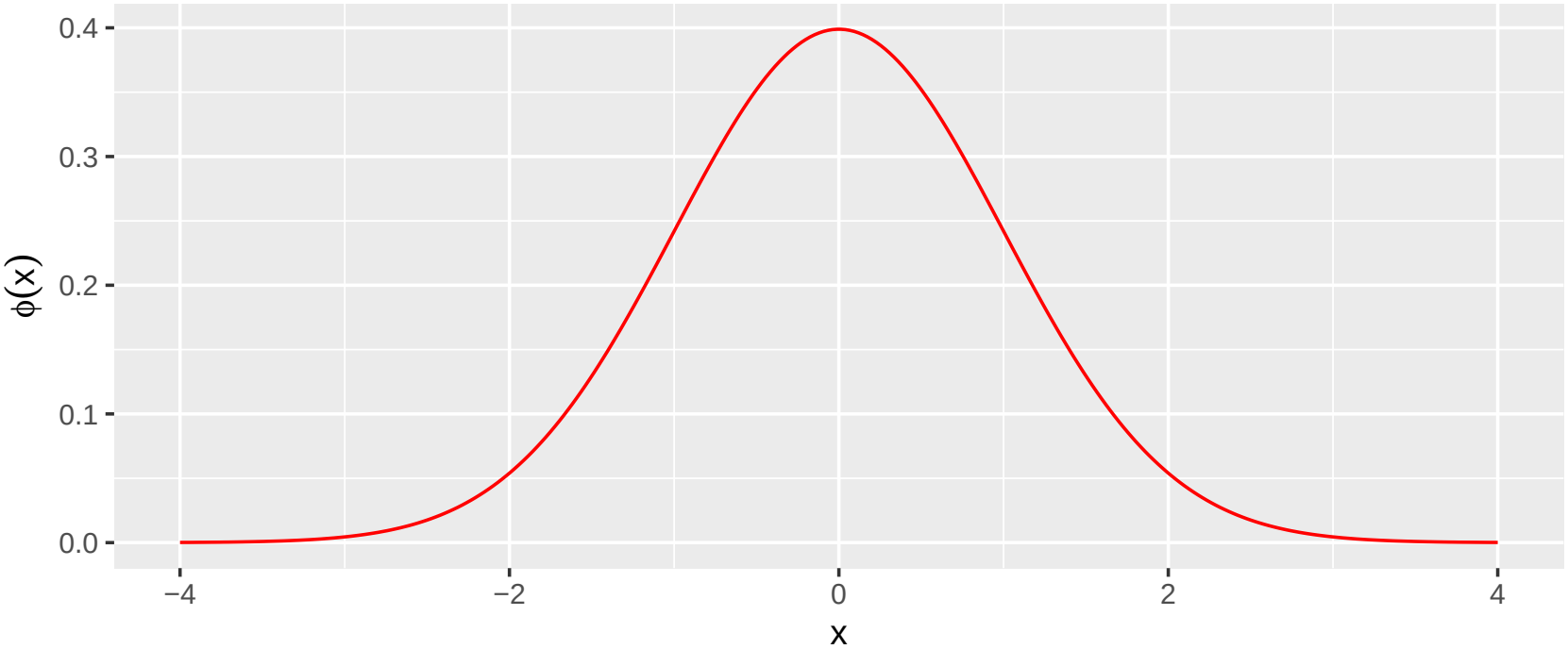
Let f_Q be the pdf of Q.

Exercise 38. What is the prob that the quarter stops spinning between 2 and 3 seconds, in terms of f_Q? $\mathrm{P}(2 < Q < 3) = \int_2^3 f_Q(q)dq$.

Exercise 39. Do you think f_Q is in the normal family of pdfs? Why?

- It can't have an *exact* normal pdf, because $\mathrm{supp}(Q) = (0, \infty)$.
- If f_Q *were* a normal pdf, then $\mathrm{P}(-6 \leq Q \leq -2) = \int_{-6}^{-2} f_Q(x)dx > 0$ by prop. 2.
- It's possible it has an *approximately* normal pdf.
  - e.g., maybe $\mathrm{P}(2 < Q < 3) = \int_2^3 f_Q(q)dq \approx \int_2^3 f(x)dx$ for some normal pdf f.

Density is not probability (1/3)

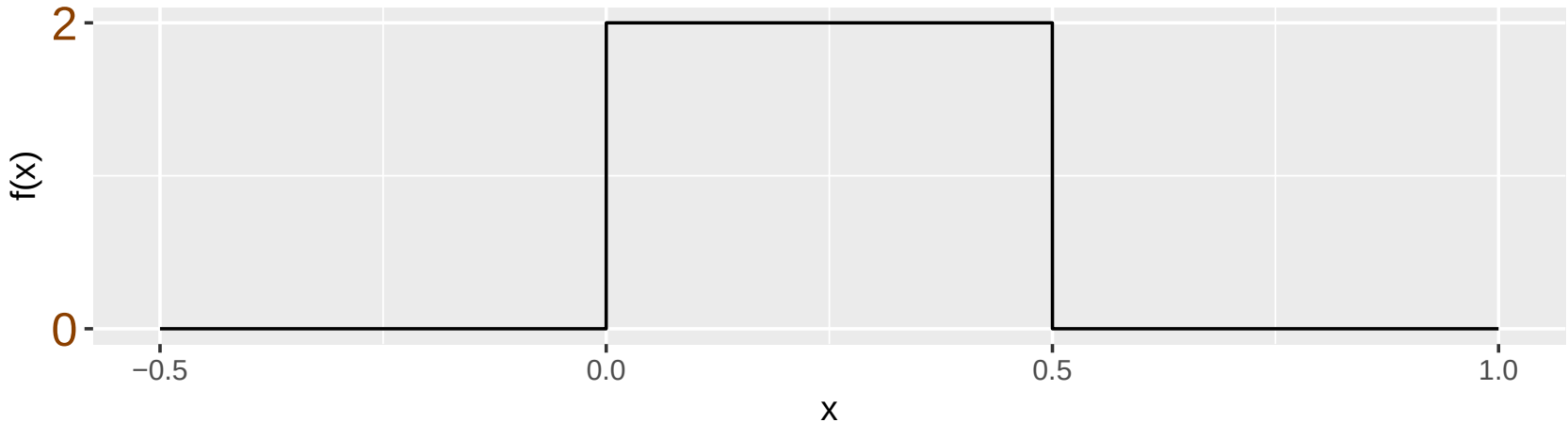


- The above is the standard normal pdf, ϕ.
- The output of a pdf can't be directly interpreted as probability.
- e.g., $\phi(0) \approx 0.4 \neq 0 = \mathrm{P}(X = 0)$. 0.4 is *density*, not prob.
- To obtain probability from a pdf, we must *integrate* the pdf.

Density is not probability (2/3)

For any pdf f, $\int_{-\infty}^{\infty} f(x)dx = \mathrm{P}(-\infty < X < \infty) = 1$.

Exercise 40. Can density of any pdf be above 1? Or would this violate a rule of probability?



- What does b equal? (hint: area of the rectangle must be 1.) Thus, width x height $= 0.5 \cdot b = 1$. So $b = \frac{1}{0.5} = 2$.
- The pdf above is a uniform pdf with parameters 0 and 0.5. $f(x) = 2 \cdot \mathrm{I}\{0 < x < 0.5\}$.

Density is not probability (3/3)

Definition 15. Let X be a continuous rv with pdf f_X. X has a uniform pdf with parameters c and d if f_X is constant over (c, d), and 0 elsewhere.

1. We know that $\int_{-\infty}^{\infty} f_X(x) = 1$ by props. of a pdf.
2. "constant over (c, d) and 0 elsewhere" translates to: $f_X(x) = b \cdot \mathrm{I}\{x \in (c, d)\}$ for some $b \in \mathbb{R}$.

Exercise 41. Find b, in terms of c and d.

We substitute $f_X(x)$ from (2) into (1) to get the following:

$$1 = \int_{-\infty}^{\infty} b \cdot \mathrm{I}\{x \in (c, d)\}\, dx \tag{1}$$

$$= \int_{c}^{d} b dx = b \cdot (d - c) \tag{2}$$

$$b = \frac{1}{d - c}. \tag{3}$$

Cumulative distribution function (cdf)

- We could use pdfs directly and integrate, but it turns out there's often an easier way.
- And in fact, this easier way can be used for *both* **discrete** and **continuous** random variables. So it will be nice to have something that always works.

Definition 16. The cumulative distribution function (cdf) of a rv X is the function $F_X : \mathbb{R} \mapsto [0, 1]$ such that:

$$F_X(a) = \mathrm{P}(X \leq a).$$

- The $\leq$ is important to remember.
- We will discuss a specific case where $\mathrm{P}(X \leq a) = \mathrm{P}(X < a)$, but remember that the definition of a cdf for all cases uses a weak inequality.

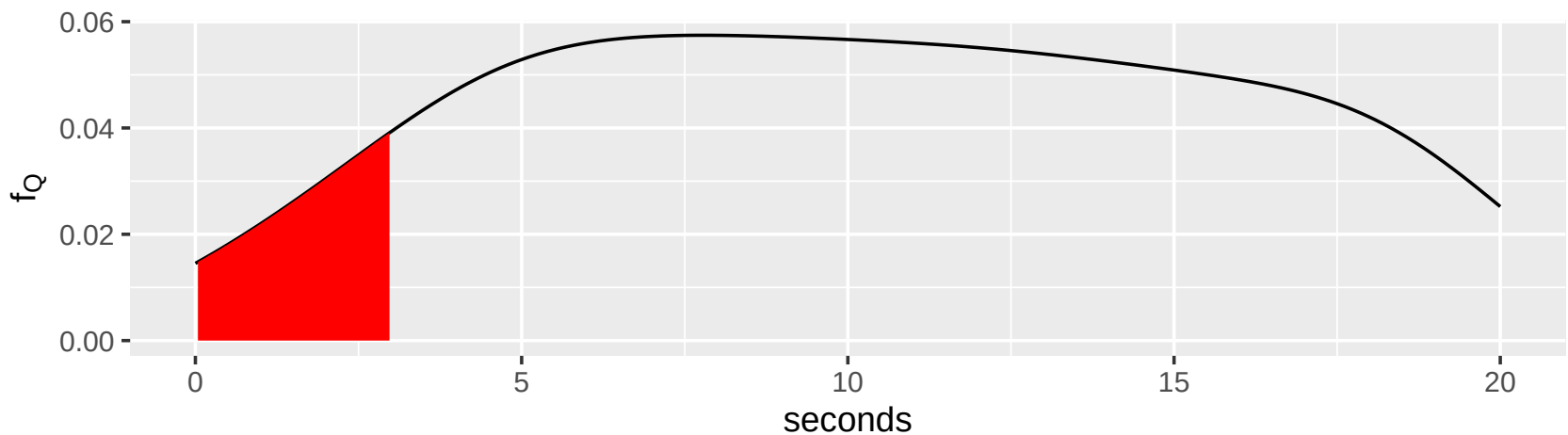
Relationship between pdf and cdf

Exercise 42. What does $F_Q(3)$ equal, in terms of f_Q?
$F_Q(3) = \mathrm{P}(Q \leq 3)$. How do we obtain $\mathrm{P}(Q \leq 3)$ from f_Q?

$$\mathrm{P}(Q \leq 3) = \mathrm{P}(Q \in (-\infty, 3]) \tag{1}$$

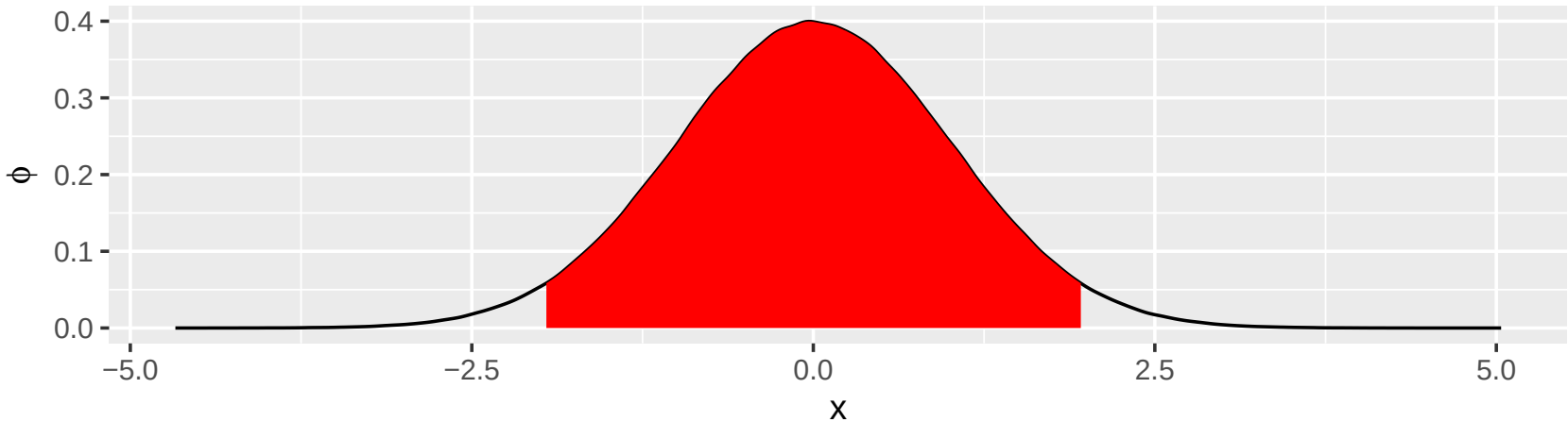
$$= \int_{-\infty}^{3} f_Q(q)dq \tag{2}$$

$$= \int_{0}^{3} f_Q(q)dq \quad \text{(Since } f_Q(q) = 0 \text{ for } q < 0\text{).} \tag{3}$$



Properties of the *standard* normal pdf and cdf

- ϕ (lowercase phi) denotes the pdf of a standard normal.
- Φ (uppercase phi) denotes the cdf of a standard normal.
- We should memorize that $\int_{-1.96}^{1.96} \phi(x)dx \approx 0.95$.



- $\Phi(1.96) \approx 0.975$.

Exercise 43. Let Q be a continuous rv. Is it true that $\mathrm{P}(Q \leq x) = \mathrm{P}(Q < x)$ for all $x \in \mathbb{R}$?

$$\begin{aligned} \mathrm{P}(Q \leq x) &= \mathrm{P}(Q < x) + \mathrm{P}(Q = x) \quad \text{(true for all rvs)} && (1) \\ &= \mathrm{P}(Q < x) + 0 \quad \text{(because } Q \text{ is continuous)} && (2) \\ &= \mathrm{P}(Q < x), \text{ so yes it is true.} && (3) \end{aligned}$$

Now recall Y, the #heads from 2 coin flips. $\mathrm{supp}(Y) = \{0, 1, 2\}$.

(a) $\mathrm{P}(Y \leq 1) = \mathrm{P}(Y = 0 \text{ or } Y = 1) = \mathrm{P}(Y = 0) + \mathrm{P}(Y = 1)$

(b) $\mathrm{P}(Y < 1) = \mathrm{P}(Y = 0)$ because the event $Y < 1$ is the same as the event $Y = 0$.

- Putting (a) and (b) together, we have: $\mathrm{P}(Y \leq 1) = \mathrm{P}(Y = 0) + \mathrm{P}(Y = 1) \neq \mathrm{P}(Y = 0) = \mathrm{P}(Y < 1)$.

In summary, $F_Q(1) = \mathrm{P}(Q < 1)$ but $F_Y(1) \neq \mathrm{P}(Y < 1)$.

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Learning objectives and outline

1. Example of a cdf for a discrete rv.
2. Equality of probability distribution.
    a. Helpful for understanding standardization (future lecture).
    b. Helpful for understanding iid (future lecture).
3. Learn *intuition* of expected value.

cdf exercise

Exercise 44. Now suppose we know the function $F_Q(x)$. Let's answer the question again of what the probability is that the coin stops spinning between 3 and 5 seconds. How can we write $\mathrm{P}(3 < Q < 5)$ in terms of F_Q?

$$\begin{aligned} \mathrm{P}(3 < Q < 5) &= \mathrm{P}(3 < Q \leq 5) \quad \text{(because } Q \text{ is continuous)} && (1)\\ &= \mathrm{P}(Q \leq 5) - \mathrm{P}(Q \leq 3) && (2)\\ &= F_Q(5) - F_Q(3). && (3) \end{aligned}$$

Remark 6. The equalities in (2) and (3) are true for both discrete and continuous rvs.

- However, regarding the equality in (1), if Q were discrete, it's not necessarily true that $\mathrm{P}(3 < Q < 5) = \mathrm{P}(3 < Q \leq 5)$.

Another cdf exercise

Exercise 45. What's the probability that the coin stops spinning after 5 seconds, in terms of F_Q?

First, how do we translate these words to math? What is the *event* we want the probability of?

- We want to find $\mathrm{P}(Q > 5)$.

Next hint:

$$\mathrm{P}(Q \leq 5) + \mathrm{P}(Q > 5) = 1.$$

Rearranging from the hint gives:

$$\begin{aligned} \mathrm{P}(Q > 5) &= 1 - \mathrm{P}(Q \leq 5) && (1) \\ &= 1 - F_Q(5). && (2) \end{aligned}$$

cdf example with discrete rv

Recall the rv Y, the # of heads from 2 fair coin flips.

$$\mathrm{P}(Y=0)=\frac{1}{4},\ \ \mathrm{P}(Y=1)=\frac{1}{2},\ \ \mathrm{P}(Y=2)=\frac{1}{4}.$$

How do we "define" the cdf of Y, which is the function F_Y? We first evaluate F_Y at a few strategic points, e.g., $-1, 0, 0.5, 1, 2, 73$:

$$F_Y(-1)=\mathrm{P}(Y\le -1)=0.$$

$$F_Y(0)=\mathrm{P}(Y\le 0)=\mathrm{P}(Y<0)+\mathrm{P}(Y=0)=0+\frac{1}{4}=\frac{1}{4}.$$

$$F_Y(0.5)=\mathrm{P}(Y\le 0.5)=\mathrm{P}(Y=0)=\frac{1}{4}.$$

$$F_Y(1)=\mathrm{P}(Y\le 1)=\mathrm{P}(Y=0 \text{ or } Y=1)=\frac{1}{4}+\frac{1}{2}=\frac{3}{4}.$$

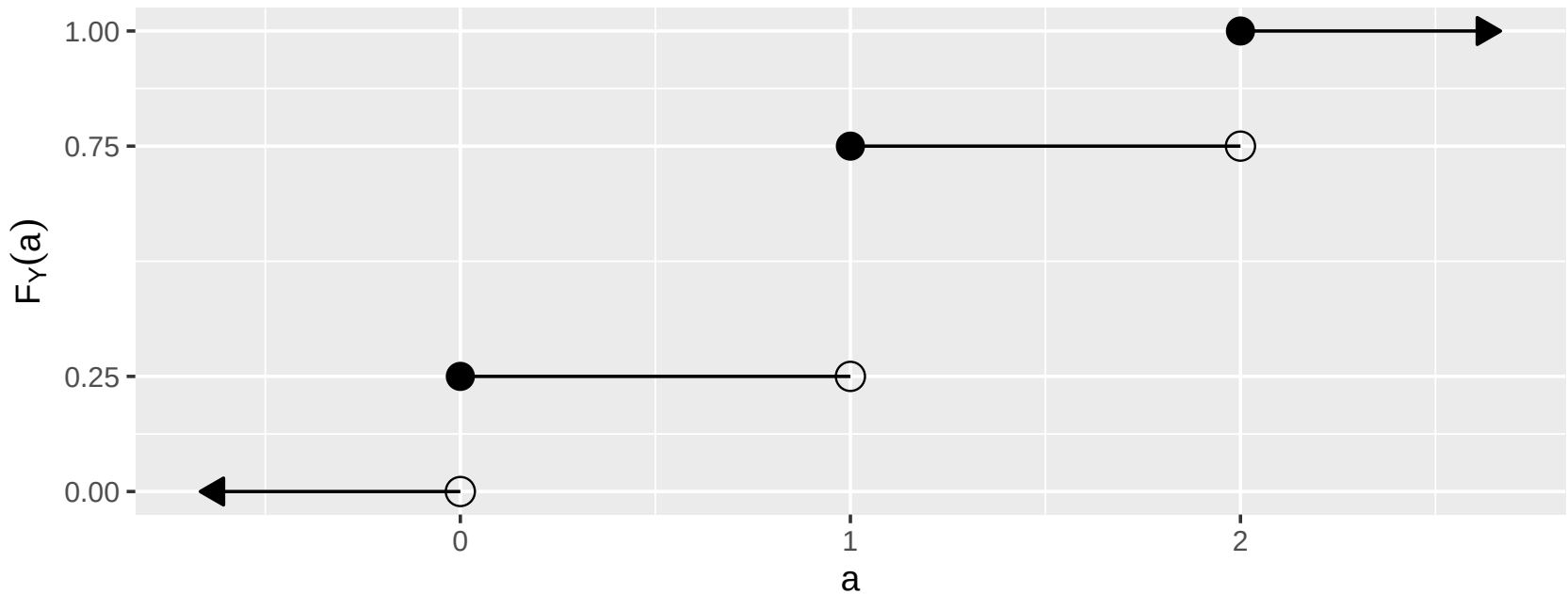
$$F_Y(2)=\mathrm{P}(Y\le 2)=\mathrm{P}(Y=0 \text{ or } Y=1 \text{ or } Y=2)=1.$$

$$F_Y(73)=\mathrm{P}(Y\le 73)=\mathrm{P}(Y\le 2)=1.$$

Exercise 46. Define the entire function $F_Y : \mathbb{R} \mapsto [0, 1]$. i.e., it should specify the output for *any* number.

$$F_Y(a) = \begin{cases} 0 & \text{for } a < 0 \\ \frac{1}{4} & \text{for } 0 \leq a < 1 \\ \frac{3}{4} & \text{for } 1 \leq a < 2 \\ 1 & \text{for } a \geq 2 \end{cases}$$

Example 17. Graph the cdf we just wrote down analytically.



Review: pmfs, pdfs, and cdfs

The table below summarizes some properties of pmfs, pdfs, and cdfs, where A is an event of a rv, and $a, c_1, c_2 \in \mathbb{R}$ with $c_1 < c_2$.

	Applicable to...	Gives $\mathrm{P}(X \in A)$ for...	Domain/range
pmf	discrete rvs	$A = \{a\}$	$\mathbb{R} \mapsto [0, 1]$
pdf	continuous rvs	$A = (c_1, c_2)$	$\mathbb{R} \mapsto [0, \infty)$
cdf	any rvs	$A = (-\infty, a]$	$\mathbb{R} \mapsto [0, 1]$

- The outputs of pmfs and cdfs are probabilities (that's why their range is $[0, 1]$).
- The output of a pdf is not directly probability; it is "density". We only get a probability if we integrate over the pdf.
- We can evaluate different types of events by combining (e.g., summing) several evaluations of the functions, as we show on the next slide.

Review: pmfs, pdfs, and cdfs

The table below summarizes some properties of pmfs, pdfs, and cdfs, where A is an event of a rv, and $a, c_1, c_2 \in \mathbb{R}$ with $c_1 < c_2$.

	Applicable to...	Gives $\mathrm{P}(X \in A)$ for...	Domain/range
pmf	discrete rvs	$A = \{a\}$	$\mathbb{R} \mapsto [0, 1]$
pdf	continuous rvs	$A = (c_1, c_2)$	$\mathbb{R} \mapsto [0, \infty)$
cdf	any rvs	$A = (-\infty, a]$	$\mathbb{R} \mapsto [0, 1]$

Exercise 47. Let Y have pmf f_Y, cdf F_Y, and $\mathrm{supp}(Y) = \{0, 1, 2\}$. Let $B := \{0, 1\}$.

1. In terms of f_Y, $\mathrm{P}(Y \in B) = f_Y(0) + f_Y(1)$.
2. Alternatively (also in terms of f_Y), $\mathrm{P}(Y \in B) = 1 - f_Y(2)$.
3. In terms of F_Y, $\mathrm{P}(Y \in B) = F_Y(1)$.

Review: pmfs, pdfs, and cdfs

The table below summarizes some properties of pmfs, pdfs, and cdfs, where A is an event of a rv, and $a, c_1, c_2 \in \mathbb{R}$ with $c_1 < c_2$.

	Applicable to...	Gives $\mathrm{P}(X \in A)$ for...	Domain/range
pmf	discrete rvs	$A = \{a\}$	$\mathbb{R} \mapsto [0, 1]$
pdf	continuous rvs	$A = (c_1, c_2)$	$\mathbb{R} \mapsto [0, \infty)$
cdf	any rvs	$A = (-\infty, a]$	$\mathbb{R} \mapsto [0, 1]$

Exercise 48. Let Q have pdf f_Q, cdf F_Q, and $\mathrm{supp}(Q) = [0, \infty)$. Let B be "$(0, 3)$ or $(5, \infty)$", i.e., $\{x \in \mathbb{R} | x \in (0, 3) \text{ or } x \in (5, \infty)\}$.

1. In terms of f_Q, $\mathrm{P}(Q \in B) = \int_0^3 f_Q(q)dq + \int_5^\infty f_Q(q)dq$.
2. Alternatively (also in terms of f_Q), $\mathrm{P}(Q \in B) = 1 - \int_3^5 f_Q(q)dq$.
3. In terms of F_Q,
   $\mathrm{P}(Q \in B) = 1 - \mathrm{P}(3 \leq Q \leq 5) = 1 - [F_Q(5) - F_Q(3)]$.

Probability distribution

- The probability distribution of a rv Z is an abstract term to refer to any function that can be used to evaluate the probability of any event of Z.

For example, how would we evaluate $\mathrm{P}(2 < X < 7)$?

- If X is discrete, we could use either the pmf or the cdf.
- If X is continuous, we could use either the pdf or the cdf.

Example of using term "probability distribution"

- Recall the example of the ice cream salesperson who has to choose Q_S, i.e., how much ice cream to buy as a function of the weather at his house.
- The optimal choice of Q_S depends on the (probability) distribution of weather at his house and at the beach.
- Weather was discrete, so we could say Q_S depends on the pmf of weather.
- But it is more common to say distribution, because the statement is valid if we change weather to be continuous (e.g., the temperature in Fahrenheit).

Defining "*same* (probability) distribution"

Definition 17. Let X and Y be rvs. X and Y have the same (probability) distribution, or are identically distributed if for any event A of the rvs, $\mathrm{P}(X \in A) = \mathrm{P}(Y \in A)$.

1. Equivalent: X and Y have the same cdfs.
2. Equivalent if X and Y are both discrete: the pmf of X is equal to the pmf of Y.
3. Equivalent if X and Y are both continuous: the pdf of X is equal to the pdf of Y.

Example: same distribution?

Let Y be the number of H from two fair coin flips, and let Z be the number of T from the same two coin flips.

Exercise 49. Consider the event $A := \{2\}$. Is it true that $\mathrm{P}(Y \in A) = \mathrm{P}(Z \in A)$?

- Hint: $\mathrm{P}(Y \in \{2\}) = \mathrm{P}(Y = 2)$, and $\mathrm{P}(Z \in \{2\}) = \mathrm{P}(Z = 2)$.
- Yes: $\mathrm{P}(Y = 2) = \mathrm{P}(\{(H, H)\}) = \mathrm{P}(\{(T, T)\}) = \mathrm{P}(Z = 2)$ because the flips are fair.
- And we could show that their pmfs are equal at all points.

Exercise 50. Do Y and Z have the same distribution? Yes.

- We can also say: "Y and Z are *identically distributed*".

Exercise 51. Do Y and Z equal each other? No. The functions are different, e.g., $Y((H, H)) = 2 \neq 0 = Z((H, H))$.

Motivation: ♡s and fives

Recall H, the rv of winnings from a game of ♡s and 5s.

- We draw 2 cards (without replacement) and you get \$1 for each card that is a heart and \$2 for each card that is a 5.
- Recall that $\text{supp}(H) = \{0, 1, 2, 3, 4, 5\}$.
- Poll: using intuition, how much money would you pay to play? Options: $<$ \$1, \$1 – \$2, \$2 – \$3, \$3 – \$4, \$4 – \$5, $>$ \$5 ?
- In practice, how would we figure out the correct answer?

Expected value

Intuition: the expected value of a rv X is the long-run average of observed outcomes of X, and is denoted $\mathrm{E}(X)$.

- Suppose the experiment corresponding to X is run 100 times, and we get a sequence of observed outcomes: $\omega_1, \ldots, \omega_{100}$.
- Each of these outcomes of the experiment gives an outcome of X. Let's denote this sequence of outcomes of X as: $x_1, \ldots, x_{100}$. e.g., $x_6 = X(\omega_6)$.
- Then, $\mathrm{E}(X) \approx \frac{1}{100} \sum_{i=1}^{100} x_i$.
- $\mathrm{E}(X) = \lim_{n \to \infty} \frac{1}{n} \sum_{i=1}^{n} x_i$, which we can think of as the average over an infinite amount of observed outcomes of X.

♡s and fives

Exercise 52. Recall H, the rv of winnings from a game of ♡s and 5s. How much money you should pay? (You can give an expression, i.e., without numbers.) $\mathrm{E}(H)$.

- This is a bit of a simplification, because we ignore risk aversion in this course.
- We will discuss how to evaluate $\mathrm{E}(H)$, but first we will do an easier example.

Easier game: rolling a die

Recall the rv D that is the number from rolling a fair 6-sided die.

- Suppose we play a game where you win \$$D$ from the roll. How much should you pay—for a fair bet—to play the game?
- We could simulate the game in R to calculate the long-run average.
- Simulation in R is intuitive, and would give a very close approximation, but sometimes it is nice to find the expected value without simulation.
- Indeed, there is an easier way.

Evaluating $\mathrm{E}(D)$ without simulation

- Suppose $d_1, \ldots, d_{100}$ are the 100 observed outcomes of D.
- We can approximate $\mathrm{E}(D)$ as: $\mathrm{E}(D) \approx \frac{1}{100} \sum_{i=1}^{100} d_i$.
- Recall that $\mathrm{supp}(D) = \{1, 2, 3, 4, 5, 6\}$.
- Let $n_j := \sum_{i=1}^{100} \mathrm{I}\{d_i = j\}$ for $j = 1, \ldots, 6$.

Exercise 53. How would you explain what n_4 is, in words? It is the number of 100 observed rolls that were 4.

Exercise 54. Give a guess for approximately what n_4 is. You may give an answer as a fraction.

- Hint: first solve an easier problem. e.g., approximately how many of 100 coin flips would you guess are heads?
- $n_4 \approx \frac{1}{6} \cdot 100 \approx 16.67$, because the probability of getting a 4 is $\frac{1}{6}$ so we would guess that about $\frac{1}{6}$ of the 100 rolls end up as 1.

Exercise 55. $n_3 \approx \frac{1}{6} \cdot 100$.

Approximating the sum of d_i

Recall: n_j is the amount of 100 observed rolls that were equal to j.

Exercise 56. What is $\sum_{j=1}^{6} n_j$? 100.

Exercise 57. $\sum_{i=1}^{100} d_i \approx$?. Leave answer as a numeric expression.

$$\sum_{i=1}^{100} d_i = 1 \cdot n_1 + 2 \cdot n_2 + 3 \cdot n_3 + 4 \cdot n_4 + 5 \cdot n_5 + 6 \cdot n_6 \tag{1}$$

$$\approx 1 \cdot \left(100 \cdot \frac{1}{6}\right) + 2 \cdot \left(100 \cdot \frac{1}{6}\right) + \cdots + 6 \cdot \left(100 \cdot \frac{1}{6}\right) \tag{2}$$

$$= 100 \cdot \frac{1}{6} \cdot (1 + 2 + \cdots + 6). \tag{3}$$

Average of the observed values

We just showed that $\sum_{i=1}^{100} d_i \approx 100 \cdot \frac{1}{6} \cdot (1 + 2 + \cdots + 6)$. Then,

$$\frac{1}{100}\sum_{i=1}^{100} d_i \approx \frac{1}{\cancel{100}} \cdot \cancel{100} \cdot \frac{1}{6} \cdot (1 + 2 + \cdots + 6) \tag{1}$$

$$= \frac{1}{6}(1 + 2 + \cdots + 6). \tag{2}$$

$$= 3.5. \tag{3}$$

- Further, if we replace the 100 above with 100,000, the 100,000s would cancel just like the 100s did.
  - i.e., the average of 100,000 rolls would also be ≈ 3.5.
- Thus, $\mathrm{E}(D) = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} d_i = 3.5$.
- Notice we did the above without actually observing the d_i!
  - What information did we use to evaluate $\mathrm{E}(D)$? The pmf of D.

Evaluating $\mathrm{E}(H)$

- To evaluate $\mathrm{E}(H)$ *analytically*, we would need to know the pmf of H.
- Solving for the pmf of H is a bit beyond the scope of our class.
- But recall that because of the intuition we have of $\mathrm{E}(H)$ being the long-run average, we do not need to solve for the pmf!

Monty Hall game

Rules of the game:

1. A prize is placed randomly behind one of 3 doors and it will not move for the duration of the game.
2. You initially pick one of the 3 doors.
3. The host then reveals one of the remaining two doors that does not contain the prize.
4. You then have the choice to switch or stay.

How many people think you should switch?

How many think you should stay?

How many think it doesn't matter?

People get very passionate about this problem. The only way to resolve arguments is to define things formally.

Monty Hall solution (1/2)

For simplicity, assume you choose door 2 as the initial choice.

Before defining rvs, what is the experiment?

- The experiment is the random parts: (1) placing the prize behind one of the three doors and (2) the door that is revealed.
- $\Omega = \{(1, H), (1, T), (2, H), (2, T), (3, H), (3, T)\}$, where the first element of the pair $\omega \in \Omega$ is the door where the prize is placed, and H means to reveal the lowest door number possible, and T the highest door possible.

Now let's define some rvs.

- Define Z to be the door number that the prize is behind. e.g., $Z(2, T) = 2$.
- Let D be the door number that you can switch to. e.g., $D(2, H) = 3$, $D(2, T) = 1$.

Monty Hall solution (2/2)

Recall: Z is the prize door, D is the door you can switch to.

Supppose you play the game. The host draws some $\omega^* \in \Omega$. Do you observe ω^*? No, you observe $D(\omega^*)$. Suppose $D(\omega^*) = 3$.

Exercise 58. What is $\mathrm{P}(Z = 3)$? (hint: use def. of P) $\frac{1}{3}$. This is the *long-run* proportion of times the prize is behind door 3.

Exercise 59. What we really want to know is $\mathrm{P}(Z = D)$.

ω	$Z(\omega)$	$Z(\omega) = 2$	$D(\omega)$	$Z(\omega) = D(\omega)$
$(1, H)$	1	$\times$	1	✓
$(1, T)$	1	$\times$	1	✓
$(2, H)$	2	✓	3	$\times$
$(2, T)$	2	✓	1	$\times$
$(3, H)$	3	$\times$	3	✓
$(3, T)$	3	$\times$	3	✓

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7. Expected value and variance

8. Standardization and joint distribution

Learning objectives and outline

1. Learn the analytical definition of expected value.
2. Learn the properties of expected value.
3. Learn the definition and intuition of variance.
4. Learn the properties of variance.

Definition of expected value

Definition 18 (Expected value of a discrete rv). Let X be a discrete rv with pmf f and $\operatorname{supp}(X) = \{w_1, w_2, \ldots, w_k\}$. Then, the expected value of X is:

$$\mathrm{E}(X) := \sum_{j=1}^{k} w_j f(w_j).$$

- We often use the notation $\mu_X := \mathrm{E}(X)$.
- We often refer to the expected value of X as the expectation of X or mean of X.

Definition of expected value

Definition 18 (Expected value of a discrete rv). Let X be a discrete rv with pmf f and $\operatorname{supp}(X) = \{w_1, w_2, \ldots, w_k\}$. Then, the expected value of X is:

$$\mathrm{E}(X) := \sum_{j=1}^{k} w_j f(w_j).$$

Exercise 60. Review: What does $\sum_{j=1}^{k} f(w_j)$ equal? 1.

Definition of expected value

Definition 18 (Expected value of a discrete rv). Let X be a discrete rv with pmf f and $\text{supp}(X) = \{w_1, w_2, \ldots, w_k\}$. Then, the expected value of X is:

$$\mathrm{E}(X) := \sum_{j=1}^{k} w_j f(w_j).$$

- We can substitute for $f(w_j)$ in the def.:

$$\mathrm{E}(X) = \sum_{j=1}^{k} w_j \mathrm{P}(X = w_j).$$

- So the expected value is a *weighted* average of the possible outcomes that X can take.
  - The weights are the probabilities that X takes those values.

Definition of expected value

Definition 18 (Expected value of a discrete rv). Let X be a discrete rv with pmf f and $\operatorname{supp}(X) = \{w_1, w_2, \ldots, w_k\}$. Then, the expected value of X is:

$$\mathrm{E}(X) := \sum_{j=1}^{k} w_j f(w_j).$$

Exercise 61. Is $\mathrm{E}(X)$ a rv or a constant?

A: A constant. We're just summing up numbers. Outcomes are numbers, probabilities are numbers. So $\mathrm{E}(X)$ is just a number.

Expected value of the die rv

Recall that $\mathrm{E}(X) := \sum_{j=1}^{k} w_j f(w_j)$.

Exercise 62. Let D be the rv that gives as output the number from rolling a fair 6-sided die. Evaluate $\mathrm{E}(D)$.

$$\mathrm{E}(D) = \underbrace{1 \times \frac{1}{6}}_{j=1} + \underbrace{2 \times \frac{1}{6}}_{j=2} + \cdots + \underbrace{6 \times \frac{1}{6}}_{j=6=k} \tag{1}$$

$$= \frac{1}{6}(1 + 2 + \cdots + 6) \tag{2}$$

$$= \frac{1}{6}(21) \tag{3}$$

$$= \frac{7}{2} \tag{4}$$

$$= 3.5. \tag{5}$$

Notes about expected value

1. Recall what we said that for any rv X, $\mathrm{E}(X)$ is just a number. For the distribution of D, $\mathrm{E}(D) = 3.5$.
2. $\mathrm{E}(X)$ is not necessarily a potential outcome of X. $\mathrm{E}(D) = 3.5$ is not a possible outcome of D when rolling a die.
    a. i.e., it's not necessarily true that $\mathrm{E}(X) \in \mathrm{supp}(X)$.
3. The expected value of a rv is not actually the value we "expect" to get.

Expected value of a *continuous* rv

Definition 19 (Expected value of a continuous rv). Let X be a continuous rv with pdf f. Then, the expected value of X is:

$$\mathrm{E}(X) := \int_{-\infty}^{\infty} x f(x) dx.$$

Exercise 63. $\int_{-\infty}^{\infty} f(x) dx = 1$.

Why is E called the "mean" of a rv?

We already know that $\mathrm{E}(X)$ is the long-run mean of a rv X. But there is another reason we call $\mathrm{E}(X)$ the "mean" of X.

Exercise 64. Suppose there is a population of 3 people with incomes (in \$1,000's) 20, 25, 45. Let X be the income of someone drawn at random. Evaluate $\mathrm{E}(X)$.

What values can X take? 20, 25, 45. i.e., $\mathrm{supp}(X) = \{20, 25, 45\}$.

$$\begin{aligned}
\mathrm{E}(X) &= 20 \cdot \mathrm{P}(X = 20) + 25 \cdot \mathrm{P}(X = 25) + 45 \cdot \mathrm{P}(X = 45) && (1)\\
&= 20 \cdot \frac{1}{3} + 25 \cdot \frac{1}{3} + 45 \cdot \frac{1}{3} && (2)\\
&= \frac{1}{3}(20 + 25 + 45) && (3)\\
&= 30. && (4)
\end{aligned}$$

So $\mathrm{E}(X)$ is the mean of the population.

Properties of expected value

For any constants a, b, c and any rvs X, Y:

1. $\mathrm{E}(c) = c$.
2. $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$.
3. $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$. This is called the "distributive property of expected value".

Recall that expected value (for a discrete rv) is defined in terms of summations so many of the properties of expectation come from properties of summation.

Properties of expected value

For any constants a, b, c and any rvs X, Y:

1. $\mathrm{E}(c) = c$.
2. $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$.
3. $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$. This is called the "distributive property of expected value".

Can we simplify a quotient of rvs using these properties?

Exercise 65. $\mathrm{E}(\frac{X}{Y}) \stackrel{?}{=} \frac{\mathrm{E}(X)}{\mathrm{E}(Y)}$

A: No. The intuition is the same as we discussed about summations.

Properties of expected value

For any constants a, b, c and any rvs X, Y:

1. $\mathrm{E}(c) = c$.
2. $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$.
3. $\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$. This is called the "distributive property of expected value".

What if the denominator is a constant?
Exercise 66. $\mathrm{E}(\frac{X}{c}) \stackrel{?}{=} \frac{\mathrm{E}(X)}{c}$ for $c \neq 0$?

A: Yes, by Property 2 that $\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$. What are a and b in this case? $a = \frac{1}{c}, b = 0$.

Example: expected revenue

Expected revenue is long-run average revenue.

Suppose a coffee shop sells only small and large coffees.

- Their revenue, R, is the sum of revenue from small coffees, R_S, and the revenue from large coffees, R_L. That is, $R := R_S + R_L$.
- Suppose that $\mathrm{E}(R_S) = 120$ and $\mathrm{E}(R_L) = 160$.

Exercise 67. What is $\mathrm{E}(R)$?

$$\begin{aligned} \mathrm{E}(R) &= \mathrm{E}(R_S + R_L) && (1)\\ &= \mathrm{E}(R_S) + \mathrm{E}(R_L) \quad \text{by prop. 3} && (2)\\ &= 120 + 160 && (3)\\ &= 280. && (4) \end{aligned}$$

Note that we did not need the pmf of R to evaluate $\mathrm{E}(R)$.

Useful theorem for calculating expected values

Theorem 2. Let X be a discrete rv with $\operatorname{supp}(X) = \{w_1, \ldots, w_k\}$ and pmf f_X. Let $Y := g(X)$, where $g : \mathbb{R} \to \mathbb{R}$. Then,

$$\mathrm{E}[Y] = \mathrm{E}[g(X)] = \sum_{i=1}^{k} g(w_i) f_X(w_i).$$

Exercise 68. Let D be the number from rolling a fair 6-sided die. Evaluate $\mathrm{E}(D^2)$.

Hint: $g(x) = x^2$. Now, plug in g into the formula of Theorem 2.

Recall for D, we have: $w_1 = 1, w_2 = 2, \ldots, w_6 = 6$. i.e., $w_i = i$.

Solution: we apply Theorem 2 using $g(x) = x^2$ to get:

$$\mathrm{E}(D^2) = \mathrm{E}[g(D)] = \sum_{i=1}^{k} g(w_i) f(w_i) \tag{1}$$

$$= \sum_{i=1}^{k} w_i^2 f(w_i) \tag{2}$$

$$= \sum_{i=1}^{6} (i^2) \frac{1}{6} \qquad \text{(because } w_i = i\text{)} \tag{3}$$

$$= \frac{1}{6}(1^2 + 2^2 \cdots + 6^2) \tag{4}$$

$$= \frac{91}{6}. \tag{5}$$

Definition of variance

Definition 20. Let X be a rv and $\mu_X := \mathrm{E}(X)$. Then, the variance of X is:

$$\mathrm{V}(X) := \mathrm{E}\left[(X - \mu_X)^2\right].$$

Common notation is $\sigma_X^2 := \mathrm{V}(X)$.

- $\mathrm{V}(X)$ measures how far away X is from its mean, on average.
- Note in the definition that variance *is* an expected value: Let $Q := (X - \mu_x)^2$. Then, $\mathrm{V}(X) = \mathrm{E}(Q)$.
- You can think of Q as a "distance rv": each time you do the experiment (e.g., roll a die), Q is the "distance from the mean of X".
- Is $\mathrm{V}(X)$ a constant or a rv? A constant, since it's an expected value.

Equivalent expression for variance

Recall that we proved an equivalent expression for the sum of squared deviation from the mean of a sequence, x_i:

$$\sum_{i=1}^{n}(x_i - \overline{x})^2 = \sum_{i=1}^{n} x_i^2 - n(\overline{x})^2.$$

There's a similarly useful equivalence for variance of a rv:

Theorem 3. For any rv X,

$$\mathrm{V}(X) = \mathrm{E}\left[(X - \mu_X)^2\right] = \mathrm{E}(X^2) - \mu_X^2.$$

Proof that $\mathrm{E}\left[(X-\mu_X)^2\right] = \mathrm{E}(X^2) - \mu_X^2$

$$
\begin{aligned}
\mathrm{E}\left[(X-\mu_X)^2\right] &= \mathrm{E}\left[X^2 - 2X\mu_X + \mu_X^2\right] && (1)\\
&= \mathrm{E}(X^2) - \mathrm{E}(2X\mu_X) + \mathrm{E}(\mu_X^2) && (2)\\
&= \mathrm{E}(X^2) - 2\mu_X\mathrm{E}(X) + \mu_X^2 && (3)\\
&= \mathrm{E}(X^2) - 2\mu_X^2 + \mu_X^2 && (4)\\
&= \mathrm{E}(X^2) - \mu_X^2. && (5)
\end{aligned}
$$

- In (1) we just break up the square inside the expectation.
- (1) -> (2) uses the distributive property of expected value: $\mathrm{E}(X+Y) = \mathrm{E}(X) + \mathrm{E}(Y)$.
- (2) -> (3) uses $\mathrm{E}(aX) = a\mathrm{E}(X)$. Recall: μ_X is a constant!
- (3) -> (4) uses $\mu_X := \mathrm{E}(X)$.
- (4) -> (5) combines terms.

Evaluating $\mathrm{V}(D)$

Recall the example where D is the roll from a fair 6-sided die.

- We evaluated $\mathrm{E}(D) = \frac{7}{2}$ in Exercise 62.
- We evaluated $\mathrm{E}(D^2) = \frac{91}{6}$ in Exercise 68.

Exercise 69. Evaluate $\mathrm{V}(D)$.

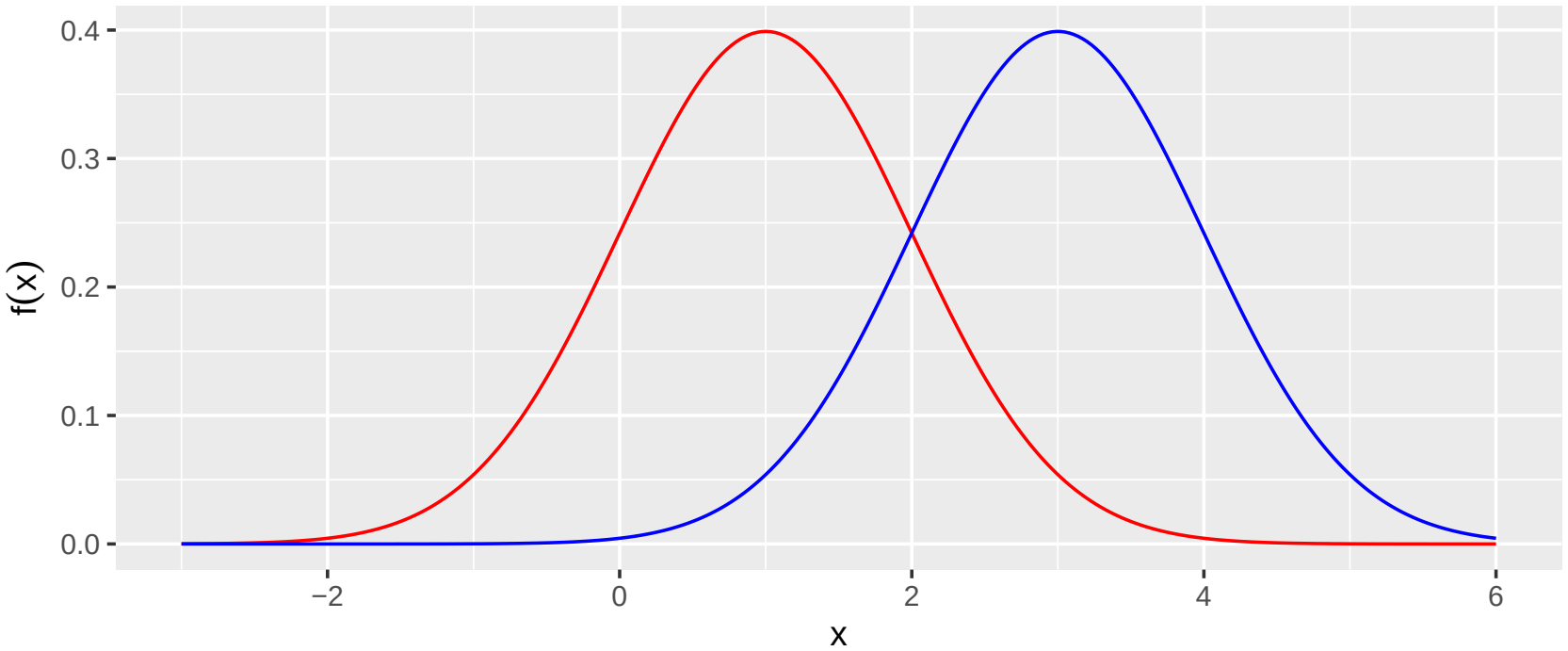
The simplest solution is to apply Theorem 3. Recall that $\mu_D := \mathrm{E}(D)$. Then, $\mathrm{V}(D) = \mathrm{E}(D^2) - {\mu_D}^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$.

Properties of variance

For any constant c and any rv X:

1. $\mathrm{V}[c] = 0$.
   - $\mathrm{E}\left[(c - \mathrm{E}(c))^2\right] = \mathrm{E}\left[(c - c)^2\right] = \mathrm{E}\left[0^2\right] = 0$.
2. $\mathrm{V}[X + c] = \mathrm{V}[X]$.
   - e.g., let $Z := D + 100$. Then, $\mathrm{supp}(Z) = \{101, 102, 103, 104, 105, 106\}$.
   - Is the distribution of Z is more spread out than D? No.
3. $\mathrm{V}[cX] = c^2\mathrm{V}[X]$.
   - Let $W := 2D$. Then, $\mathrm{supp}(W) = \{2, 4, 6, 8, 10, 12\}$.
   - Is the distribution of W is more spread out than D? Yes. $\mathrm{V}(W) = \mathrm{V}(2D) = 2^2\mathrm{V}(D) = 4\mathrm{V}(D)$.

Visualizing $\mathrm{V}(X + c)$

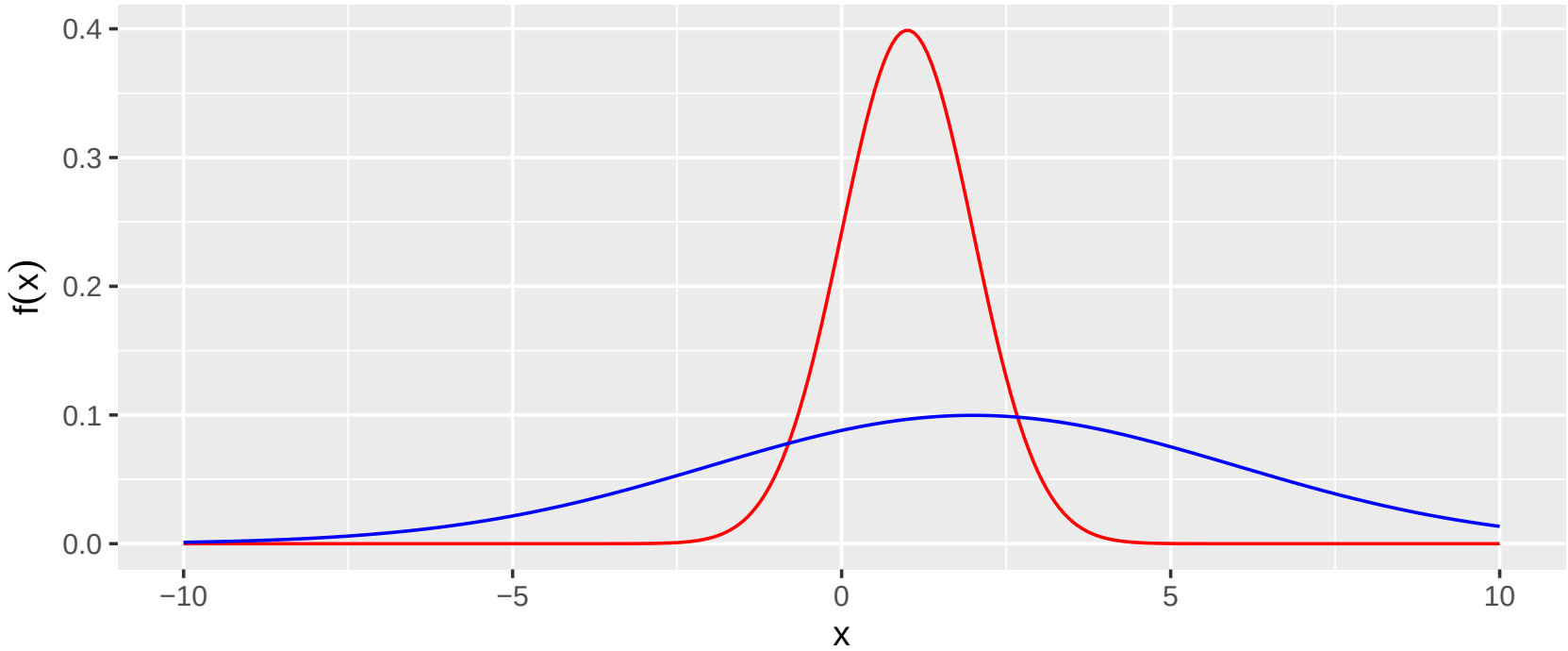


- The red curve is the pdf of a rv X with $\mathrm{E}(X) = 1, \mathrm{V}(X) = 1$.
- The blue curve is the pdf of the rv $W := X + 2$.

Exercise 70. $\mathrm{E}(W) = \mathrm{E}(X + 2) = \mathrm{E}(X) + \mathrm{E}(2) = 1 + 2 = 3$.

Exercise 71. $\mathrm{V}(W) = \mathrm{V}(X + 2) = \mathrm{V}(X) = 1$.

Visualizing $\mathrm{V}(cX) = c^2\mathrm{V}(X)$



- The red curve is the pdf of a rv X with $\mathrm{E}(X) = 1, \mathrm{V}(X) = 1$.
- The blue curve is the pdf of the rv $W := 2X$.

Exercise 72. $\mathrm{E}(W) = \mathrm{E}(2X) = 2 \cdot \mathrm{E}(X) = 2 \cdot 1 = 2$.

Exercise 73. $\mathrm{V}(W) = \mathrm{V}(2X) = 2^2 \cdot \mathrm{V}(X) = 4 \cdot 1 = 4$.

Income inequality

- Suppose that variance of income in a country is our measure of income inequality.
- Suppose that Iran and Turkmenistan both discover huge amounts of new oil reserves.
- Iran decides to double everyone's income.
- Turkmenistan decides to give \$10,000 to every individual.

Exercise 74. What happens to the measure of income inequality in Iran and, separately, in Turkmenistan?

- Let X be income of an individual chose at random from Iran.
- $\mathrm{V}(2X) = 2^2\mathrm{V}(X) = 4\mathrm{V}(X)$, so inequality quadruples in Iran.
- Let Y be the income of an individual chosen at random from Turkemnistan.
- $\mathrm{V}(Y + 10{,}000) = \mathrm{V}(Y)$, so inequality stays the same in Turkmenistan.

Standard deviation

The standard deviation of X is

$$\mathrm{SD}(X) := \sqrt{\mathrm{V}(X)}$$

Common notation is $\sigma_X := \mathrm{SD}(X)$.

Reviewing new notation

- $\mu_X := \mathrm{E}(X)$ and is pronounced "mu x".
- $\sigma_X^2 := \mathrm{V}(X)$ and is pronounced "sigma squared x" or "sigma x squared".
- $\sigma_X := \mathrm{SD}(X)$ and is pronounced "sigma x".

Notation

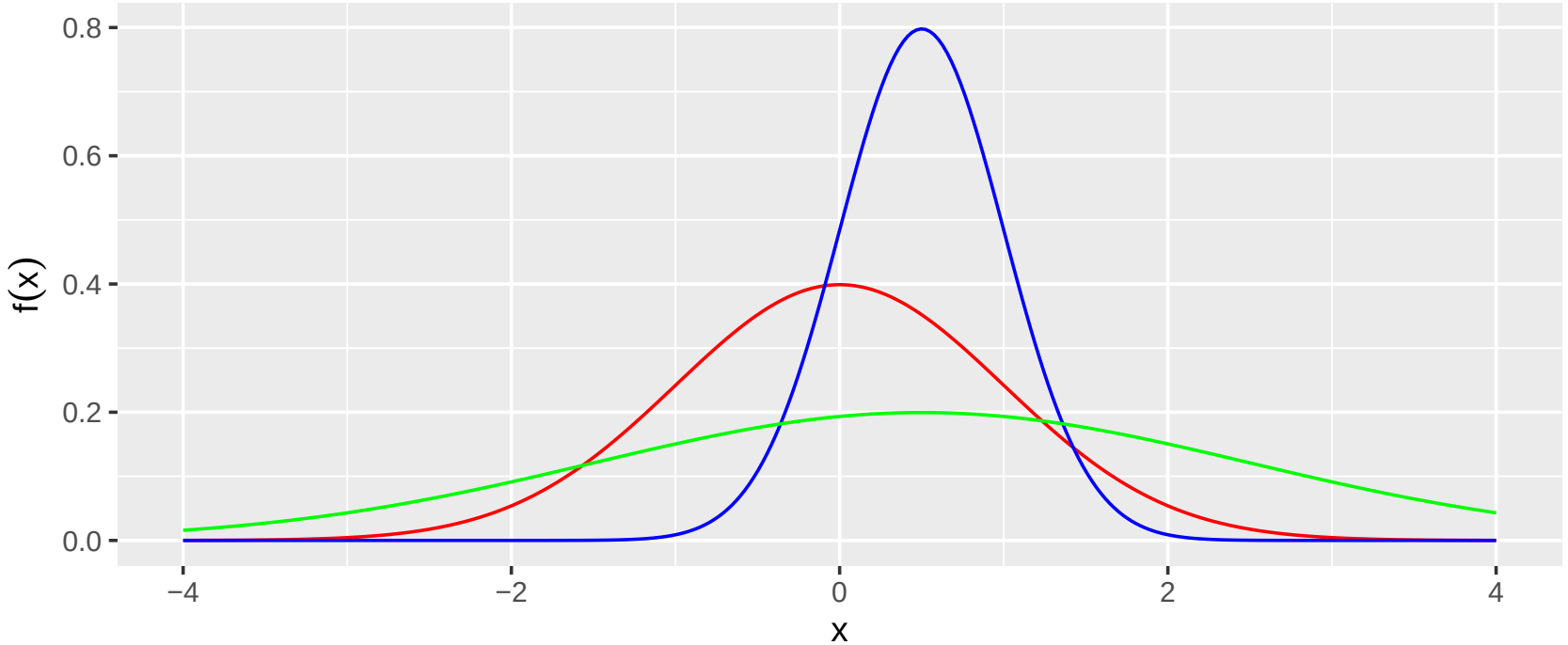
- Recall: the normal distribution has two parameters, μ and σ:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- It is not a coincidence that we refer to the parameter μ as the "mean" and the parameter σ as the "standard deviation".
- That is, if X is normally distributed, then the mean and variance of X completely determine the distribution.
- We thus write $N(a, b^2)$ to refer to a normal distribution with mean a and variance b^2 (i.e., standard deviation b).
- $N(0, 1)$ is called the *standard* normal distribution.

Notation 1. We use $\sim$ to mean "has the distribution". e.g., $X \sim N(0, 1)$ means "X has the standard normal distribution."

Name that normal



Exercise 75. Densities of the following three distributions are shown above: $N(0,1)$, $N(\frac{1}{2},\frac{1}{4})$, $N(\frac{1}{2},4)$. Which is which?

A: $N(0,1)$ is the red, $N(\frac{1}{2},\frac{1}{4})$ is the blue, and $N(\frac{1}{2},4)$ is the green.

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8. Standardization and joint distribution

Learning objectives

1. Learn what "standardization" is and why it's useful.
   - Standardization serves two purposes:
     a. To find the probability of events of normal rvs.
     b. (later in this course) To construct confidence intervals.
   - We will also use standardization as an opportunity to get practice with properties of expected value and variance.
2. Introduction to joint distributions.
   - We will define joint pmfs.

Standardization

- Suppose we have a rv X with $\mathrm{E}(X) = 5$, $\mathrm{V}(X) = 4$.
- The goal of standardization is to create a ***new rv*** that stretches/squeezes the distribution of X and moves it around so the distribution of the new rv has mean 0 and variance 1.

Exercise 76. How can we create a rv Q that perfectly depends on X but is shifted to have mean 0?

Subtract 5: $Q := X - 5$.

$$\mathrm{E}(Q) = \mathrm{E}(X - 5) = \mathrm{E}(X) - \mathrm{E}(5) = 5 - 5 = 0.$$

Exercise 77. What is $\mathrm{V}(Q)$? $\mathrm{V}(Q) = \mathrm{V}(X - 5) = \mathrm{V}(X) = 4$.

Standardization (recall $\mathrm{E}(X) = 5, \mathrm{V}(X) = 4$)

Exercise 78. What about variance? We want a rv to take the same "shape" as X but to be squeezed/stretched. Define $Y := cX$ for a constant c. What value of c makes $\mathrm{V}(Y)$ equal to 1?

The general solution is $c := \frac{1}{\mathrm{SD}(X)}$. Let's verify this solution:

$$
\begin{aligned}
& V\left(\frac{1}{\mathrm{SD}(X)} \cdot X\right) && && (1)\\
&= \left[\frac{1}{\mathrm{SD}(X)}\right]^2 \mathrm{V}(X) && \text{by prop: } \mathrm{V}(cX) = c^2\mathrm{V}(X) && (2)\\
&= \frac{1^2}{[\mathrm{SD}(X)]^2}\mathrm{V}(X) && \text{by } \left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2} && (3)\\
&= \frac{1}{\mathrm{V}(X)}\mathrm{V}(X) && && (4)\\
&= 1. && && (5)
\end{aligned}
$$

Standardization: summary

- We can create a new rv, W that has the same "shape" as X but has mean 0 and variance 1:

$$W := \frac{X - \mathrm{E}(X)}{\sqrt{\mathrm{V}(X)}}$$

- In summary, standardization gives you a new rv with:
  - Expected value (mean) = 0.
  - Variance = 1.
- Does W have a normal distribution?
  - Only if X has a normal distribution.
  - Specifically, standardization does not turn a non-normal rv into a normal rv.

Why we standardize normals

Definition 21. The Z table is the (tabulated) cdf of a $N(0,1)$.

- Suppose $Z \sim N(0,1)$. Unfortunately, there is no "nice" formula for F_Z, so computers must numerically integrate to calculate, e.g., $\mathrm{P}(Z \leq 1.5) = F_Z(1.5)$.
- The cdf is thus tabulated (as the result of numerical integration) in the back of every probability textbook.
- But what if $X \sim N(4,9)$? We can't have the cdf of all possible normal distributions.
- Standardization allows us to calculate F_X by using F_Z!
- Recall: we use Φ (capital Phi) to denote the cdf of a $N(0,1)$.

Exercise on the probability of an event of interest

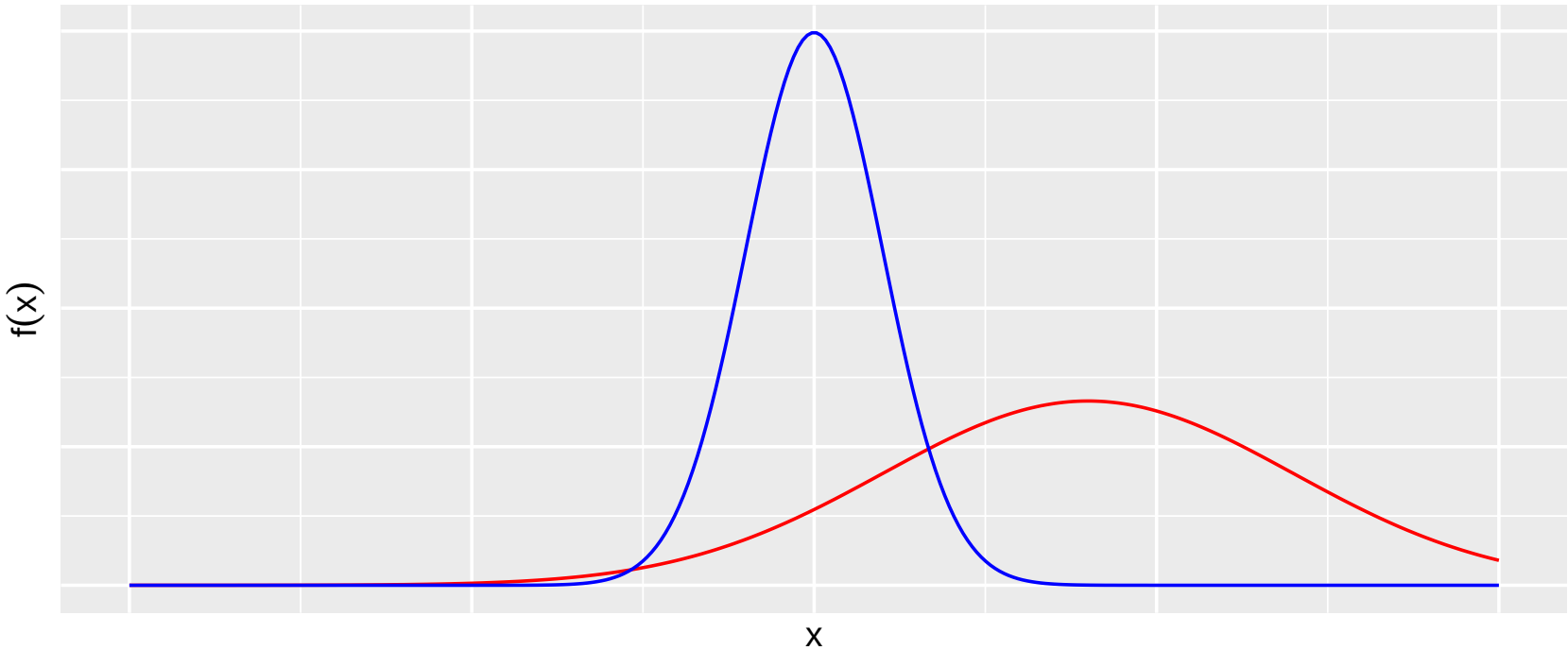
- Suppose that the distribution of income (in 10's of thousands of euros) in France is $N(4, 9)$. i.e., the average income in France is 40,000€. Let X be an income drawn at random from France.
- We often want to know the probability of an event of X.

Exercise 79. What does $\mathrm{P}(X < 2)$ tell us in this case? Intuitively, what would it mean about the population if $\mathrm{P}(X < 2)$ were 0.25?

It gives the proportion of incomes in France less than 20,000€.

We need cdfs to calculate probabilities of events

- If $X \sim N(4, 9)$ and $Z \sim N(0, 1)$, then $\mathrm{P}(X < 2) \neq \mathrm{P}(Z < 2)$, so we would need to look up a different cdf for X and Z.



- We want the area to the left of 2 under the *red* curve, not the blue curve.
- But we can't put the cdfs of all normal distributions in the back of the book because there are an infinite amount.

Employing standardization (recall $X \sim N(4, 9)$)

- The solution is we only need the cdf of a $N(0, 1)$, because:
  - $\mathrm{P}(X < 2) = \mathrm{P}(X - 4 < 2 - 4)$.
  - Also, $\mathrm{P}(X < 2) = \mathrm{P}(\frac{X-4}{\sqrt{9}} < \frac{2-4}{\sqrt{9}})$.
  - The distribution of $W := \frac{X-4}{\sqrt{9}}$ is $N(0, 1)$ by standardization.
  - So $F_X(2) = P(X < 2) = P(W < \frac{2-4}{\sqrt{9}}) = F_Z(\frac{2-4}{\sqrt{9}})$, where F_Z is the cdf of a $N(0, 1)$, and is in the back of the book.
  - In essence, we "translate" an event of a *non*-standard normal to an event of a standard normal.
  - The event that X is less than 2 has the same probability of occurring as a standard normal being less than $\frac{2-4}{\sqrt{9}}$.
- To finish the ex.: $F_Z(\frac{2-4}{\sqrt{9}}) = F_Z(\frac{-2}{3}) \approx F_Z(-0.67)$. We then use the Z table to approx. F_Z at a point: $F_Z(-0.67) \approx 0.25$.
- With R, we can use the pnorm() function:

```
> pnorm(-2/3)
[1] 0.2524925
```

Joint distributions

- Up until now we've only been talking about one rv at a time.
- Now we're going to talk about how certain rvs are related.
- First we're going to talk about a couple of silly examples with coins and then with dice.
  - Then we'll study the joint distribution of weather in the ice cream salesperson example.
  - On the problem set you'll look at the relationship between rain and commuting times.

Coin example

Suppose that for **one** experiment we flip a (not necessarily fair) coin 10 times. What does **one** outcome of the experiment look like?

- One potential outcome is $\omega^* := (H, T, T, T, H, T, T, H, H, T)$.
- How many elements would the sample space have? i.e., how many possible distinct outcomes are there? $2^{10} = 1024$. 
- Let Y be the number of heads from 10 coin flips, and let Z be the number of tails from those same coin flips. For example,

$$Y(\omega^*) = 4, \quad Z(\omega^*) = 6.$$

Exercise 80. Is there a relationship between the functions Y and Z? i.e., can we put $Z(\omega)$ on the LHS of an '=' and $Y(\omega)$ on the RHS?

Yes: for all $\omega \in \Omega$, $Y(\omega) + Z(\omega) = 10$. Thus, $Z(\omega) = 10 - Y(\omega)$.

We more commonly write this as: $Z = 10 - Y$.

Why important?

The relationship between Y and Z is important when evaluating joint probabilities.

- Consider the following probability of a joint event of Y and Z:

$$\mathrm{P}(Y = 3\ , Z = 3) = \mathrm{P}(Y = 3 \textit{ and } Z = 3) = 0.$$

- Translated to English: "What is the probability that we flip a coin 10 times and get (exactly) 3 heads and (exactly) 3 tails?"
- If we look at the individual events of Y and Z separately, they have positive probabilities:

$$\mathrm{P}(Y = 3) > 0, \quad \mathrm{P}(Z = 3) > 0.$$

- But looking at the *joint* event, the probability is 0.

What does the multiplication give?

Exercise 81. When solving the previous problem, it's tempting to think that $\mathrm{P}(Y = 3, Z = 3)$ equals

$$\mathrm{P}(Y = 3) \cdot \mathrm{P}(Z = 3).$$

- It is not true.
- But what does the above multiplication of probabilities solve?
- i.e., the number $\mathrm{P}(Y = 3) \cdot \mathrm{P}(Z = 3)$ is (in English) "the probability that ..."

  the probability that we do 10 coin flips and get 3 heads, *and* we do 10 ***new*** coin flips and get 3 tails.

Joint probability mass function

We simply extend the concept of a probability mass function:

Definition 22. For discrete rvs X and Y, the joint probability mass function (joint pmf) is the function $f_{XY} : \mathbb{R}^2 \mapsto [0, 1]$ that maps a joint outcome to the probability that it occurs:
$f_{XY}(x, y) = \mathrm{P}(X = x, Y = y)$.

Putting the previous exercise into joint pmf notation, we have $f_{YZ}(3, 3) = 0$.

For your curiosity, Y (and Z) has a *binomial distribution*.

- You can use R to evaluate the pmf. e.g., $f_{YZ}(4, 6) = f_Y(4) \approx 0.205$, as calculated in the following command: dbinom(4, size = 10, prob = 0.5)

Die example

- Consider an experiment of rolling a fair 6-sided die and define $T = 1$ if an even number comes up, 0 otherwise; and D is the actual number that comes up.

Joint events of T and D

- Each line below is *one* joint event of T and D:
  1. $T = 1, D = 4$ (die is even and 4)
  2. $T = 0, D \leq 4$ (die is odd and less than or equal to 4)
  3. $T \leq 1, D \leq 4$ (die is even or odd, and less than or equal to 4)

Joint pmf of T and D

Example 18. What is the joint pmf of T and D?

- First think of all the possible joint outcomes that T and D can theoretically take by combining the possible individual outcomes each can take. There are 12, in principle:

$$(0,1),(0,2),(0,3),(0,4),(0,5),(0,6)$$
$$(1,1),(1,2),(1,3),(1,4),(1,5),(1,6)$$

Now define a function $f_{TD} : \mathbb{R}^2 \mapsto [0, 1]$ as follows:

$$f_{TD}(0, 1) = \mathrm{P}(T = 0, D = 1) = \mathrm{P}(D = 1) = \frac{1}{6}.$$
$$f_{TD}(0, 2) = \mathrm{P}(T = 0, D = 2) = 0.$$
$$f_{TD}(0, 3) = \mathrm{P}(T = 0, D = 3) = \frac{1}{6}.$$
$$f_{TD}(0, 4) = \mathrm{P}(T = 0, D = 4) = 0.$$
$$f_{TD}(0, 5) = \mathrm{P}(T = 0, D = 5) = \frac{1}{6}.$$
$$f_{TD}(0, 6) = \mathrm{P}(T = 0, D = 6) = 0.$$
$$f_{TD}(1, 1) = \mathrm{P}(T = 1, D = 1) = 0.$$
$$f_{T,D}(1, 2) = \mathrm{P}(T = 1, D = 2) = \frac{1}{6}.$$
$$f_{TD}(1, 3) = \mathrm{P}(T = 1, D = 3) = 0.$$
$$f_{TD}(1, 4) = \mathrm{P}(T = 1, D = 4) = \frac{1}{6}.$$
$$f_{TD}(1, 5) = \mathrm{P}(T = 1, D = 5) = 0.$$
$$f_{TD}(1, 6) = \mathrm{P}(T = 1, D = 6) = \frac{1}{6}.$$
$$f_{TD}(t, d) = 0 \text{ if } t \notin \{0, 1\} \text{ or } d \notin \{1, \ldots, 6\}.$$

The above ***function*** f_{TD} is the joint pmf of T and D.

Marginal pmf

Definition 23 (marginal pmf). Let X and Y be discrete rvs. Then, the marginal pmf of X is the function f_X defined by $f_X(a) = \mathrm{P}(X = a)$.

- The "marginal pmf of X" is essentially how we used the term "pmf of X" *before* we introduced the concept of *joint* pmfs.
- e.g., we might want to talk about the marginal pmfs of T and D, which would be f_T and f_D, as opposed to the *joint* pmf of T and D, which would be f_{TD}.

Example: ice cream and weather

- Recall our ice cream salesperson, Luca.
- $\omega = (w_L, w_B)$, where $w_L, w_B \in \{\textit{cloudy}, \textit{sunny}\}$.
- Let $S_L := \mathrm{I}\{w_L = \textit{sunny}\}$, $S_B := \mathrm{I}\{w_B = \textit{sunny}\}$.

Joint pmf of S_L, S_B			
	$S_B(\omega) = 0$	$S_B(\omega) = 1$	
$S_L(\omega) = 0$	0.05	0.25	
$S_L(\omega) = 1$	0.30	0.40	0.70

- e.g., the probability it is cloudy at Luca's house and sunny at the beach is $f_{S_L S_B}(0, 1) = \mathrm{P}(S_L = 0, S_B = 1) = 0.25$.

Exercise 82. Find the (marginal) probability it's sunny at L's house.

$\mathrm{P}(S_L = 1) = \sum_{i=0}^{1} \mathrm{P}(S_L = 1, S_B = i)$
$= \mathrm{P}(S_L = 1, S_B = 0) + \mathrm{P}(S_L = 1, S_B = 1) = 0.30 + 0.40 = 0.70.$

Probability distribution

The joint probability distribution of rvs *completely determines* important properties:

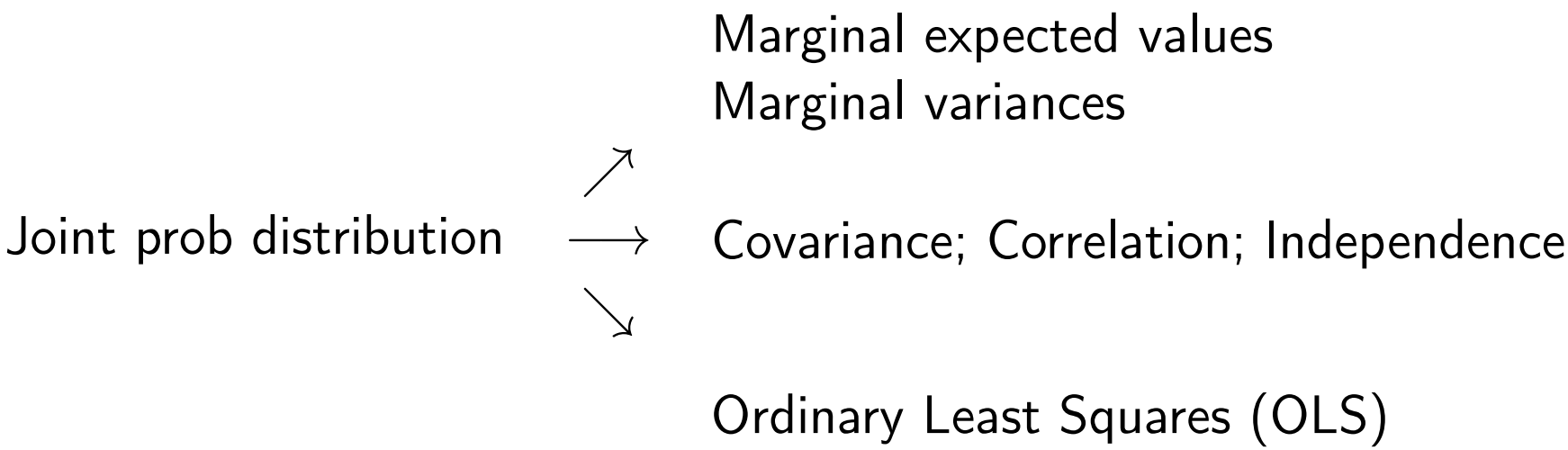


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Learning objectives

1. Learn intuition and definition of independence of rvs.
2. Introduce definitions and motivate covariance/correlation.
    a. Do a numerical exercise on covariance of education and income.
    b. Study the properties of covariance/correlation.
3. Make an important connection between the distribution of a rv and the "population".
4. A quick first exercise on "story telling".

Dependence as information

- Intuition of independence: knowing the outcome of one gives you zero ***information*** about the probability of the outcome of another rv.
- Suppose one experiment consists of rolling a die and flipping a coin. Does knowing whether the coin flip was heads give you any idea about the probability that the die was 2?
  - If yes, then the corresponding rvs are not independent.

Definition of independence of rvs

Definition 24 (Independence of discrete rvs). Let X and Y be two discrete rvs with marginal pmfs f_X and f_Y (respectively), and joint pmf f_{XY}. X and Y are independent if for all $x, y \in \mathbb{R}$,

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y).$$

- We use the following notation to mean that X and Y are independent: $X \perp\!\!\!\perp Y$.
- If X and Y are not independent, they are dependent.
- The condition for independence is equivalent to:

$$\mathrm{P}(X = x, Y = y) = \mathrm{P}(X = x) \cdot \mathrm{P}(Y = y).$$

Definition of independence of rvs

Definition 24 (Independence of discrete rvs). Let X and Y be two discrete rvs with marginal pmfs f_X and f_Y (respectively), and joint pmf f_{XY}. X and Y are independent if for all $x, y \in \mathbb{R}$,

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y).$$

- It is clear from the definition that whether two rvs are independent is determined by their joint pmf.
  - Recall the marginal pmfs can be obtained from the joint pmf.

Exercise

Recall the rvs T and D. T is 1 if the roll of a fair 6-sided die is even and 0 otherwise; D is the number on the die.

Exercise 83. Is $T \perp\!\!\!\perp D$? If not, prove it.

T and D are not independent:

1. $\mathrm{P}(T = 0, D = 2) = 0$.
2. $\mathrm{P}(T = 0) \cdot \mathrm{P}(D = 2) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$.
3. And $0 \neq \frac{1}{12}$, so $\mathrm{P}(T = 0, D = 2) \neq \mathrm{P}(T = 0) \cdot \mathrm{P}(D = 2)$.

Equivalently, we can write this chain of logic all at once:

$$\mathrm{P}(T = 0, D = 2) = 0 \neq \frac{1}{12} = \mathrm{P}(T = 0) \cdot \mathrm{P}(D = 2).$$

Intuition of covariance

- We are often interested in how two variables move together.
- Suppose we take one person at random in CAT and we know that person's education is above the average education in CAT.
  - Does that tell us that the person's income is also *likely* above the average income in CAT?
  - Yes. So we would say that income and education have "positive covariance."
- Here we will define covariance, which is an easy way to summarize the relationship (i.e., joint distribution) of X and Y with one number.

Definition of covariance and intuition

Definition 25. For any rvs X, Y, let $\mu_X := \mathrm{E}(X)$, $\mu_Y := \mathrm{E}(Y)$. The covariance of X and Y is:

$$\mathrm{cov}(X, Y) := \mathrm{E}[\underbrace{(X - \mu_X)}_{\text{1st term}}\underbrace{(Y - \mu_Y)}_{\text{2nd term}}].$$

- Notation: We often use $\sigma_{XY} := \mathrm{cov}(X, Y)$.

Definition of covariance and intuition

Definition 25. For any rvs X, Y, let $\mu_X := \mathrm{E}(X)$, $\mu_Y := \mathrm{E}(Y)$. The covariance of X and Y is:

$$\mathrm{cov}(X, Y) := \mathrm{E}[\underbrace{(X - \mu_X)}_{\text{1st term}}\underbrace{(Y - \mu_Y)}_{\text{2nd term}}].$$

Exercise 84. Suppose X is income of a person from CAT, and Y is that person's years of education. For a specific person (i.e., a specific ω^*), what does it mean if $X(\omega^*) - \mu_X > 0$?

It helps to move μ_X to the other side of the inequality:

$$X(\omega^*) - \mu_X > 0 \implies X(\omega^*) > \mu_X.$$

In other words, it means that person ω^*'s income is higher than the average income of the population.

Definition of covariance and intuition

Definition 25. For any rvs X, Y, let $\mu_X := \mathrm{E}(X)$, $\mu_Y := \mathrm{E}(Y)$. The covariance of X and Y is:

$$\mathrm{cov}(X, Y) := \mathrm{E}[\underbrace{(X - \mu_X)}_{\text{1st term}}\underbrace{(Y - \mu_Y)}_{\text{2nd term}}].$$

- If $\sigma_{XY} > 0$, then on average (that's the expected value):
  - when X is above its mean, Y is also above its mean.
  - when X is below its mean, Y is also below its mean (a negative times a negative is positive).
  - In summary, the 1st term and 2nd term have the same sign.
- If $\sigma_{XY} < 0$, the opposite is true. *On average*:
  - The 1st term and 2nd term have opposite signs.

Covariance example, and underlying experiment

Example 19. Consider the population of employed people in CAT. Let X be monthly income of a person chosen at random from the population, measured in 1,000€s; and let Y be the number of years of education of the person. How would we evaluate $\mathrm{cov}(X, Y)$?

- For simplicity, suppose the *entire pop* has only 5 people. What's the underlying experiment of the rvs X and Y?
- It's selecting a person at random from the population.
- What is Ω? $\Omega = \{Ela,\ Amir,\ Hank,\ Nia,\ Anne\}$.

Recall: $\sigma_{XY} := \mathrm{E}\left[(X - \mu_X)(Y - \mu_y)\right]$. X is monthly income, measured in 1,000€s, Y is years of education.

ω	$X(\omega)$	$Y(\omega)$	$X(\omega) - \mu_X$	$Y(\omega) - \mu_Y$	$[X(\omega) - \mu_X][Y(\omega) - \mu_Y]$
Ela	3.7	10	-0.6	-4.6	2.76
Amir	6.5	16	2.2	1.4	3.08
Hank	4.6	14	0.3	-0.6	-0.18
Nia	3.8	21	-0.5	6.4	-3.20
Anne	2.9	12	-1.4	-2.6	3.64
avg.	$\mu_X = 4.3$	$\mu_Y = 14.6$			$\sigma_{XY} = 1.22$

1. First, we calculate the means, μ_X, μ_Y.
2. Second, we calculate the deviations from the means, $X(\omega) - \mu_X$ and $Y(\omega) - \mu_Y$.
3. Third, we multiply the deviations.
4. Finally, we average the multiplied deviations.

Alternative expression for covariance

- Just like variance, there is a convenient alternative expression:

$$\operatorname{cov}(X, Y) := \mathrm{E}\left[(X - \mu_X)(Y - \mu_Y)\right] = \mathrm{E}\left[XY\right] - \mu_X \mu_Y.$$

- The definition is better to look at when thinking about how to *interpret* what covariance captures.
- The alternative expression is often easier to use when *calculating* the covariance.

Recall: $\sigma_{XY} := \mathrm{E}\left[(X - \mu_X)(Y - \mu_y)\right] = \mathrm{E}(XY) - \mu_X\mu_Y$. X is monthly income, measured in 1,000€s, Y is years of education.

ω	$X(\omega)$	$Y(\omega)$	$X(\omega)Y(\omega)$
Ela	3.7	10	37.0
Amir	6.5	16	104.0
Hank	4.6	14	64.4
Nia	3.8	21	79.8
Anne	2.9	12	34.8
avg.	$\mu_X = 4.3$	$\mu_Y = 14.6$	$\mathrm{E}(XY) = 64$

1. First, we calculate $X(\omega)Y(\omega)$.
2. Second, we calculate the means, μ_X, μ_Y, and $\mathrm{E}(XY)$.
3. Finally, we plug in: $\mathrm{E}(XY) - \mu_X\mu_Y = 64 - 4.3 \times 14.6 = 1.22$.

Duality of population and distribution

Recall that $\mathrm{E}(X) =$ "the mean of the *population* (of incomes) the person was drawn from". The following all equal the same number:

1. The mean of X, i.e., $\mathrm{E}(X)$.
2. The mean of the distribution of X.
3. The mean of income in the population.

(3) does not even involve a rv. The key insight is that the population can be viewed as the distribution of the rv X!

Recall that the word "distribution" just means we can use it to evaluate the probability of any event of a rv.

Consider the event "$X < 2$". Recall: $\mathrm{P}(X < 2) =$ "the proportion of the population with monthly income less than 2,000€".

- Note the RHS interpretation does not involve a rv.

Remark 7. The connection pop $\longleftrightarrow$ dist only holds if, in the underlying experiment of the rv, the person is drawn ***at random***.

The population is the distribution

This concept extends beyond just $\mathrm{E}(X)$. e.g., consider the phrase "the covariance of income and education in CAT is positive".

- There is no explicit rv in this phrase, and yet covariance is defined in terms of rvs!
- The joint distribution of X, Y ***is*** the population (of income and education).
- Implicitly, the above phrase means the same as the more verbose rv interpretation involving the experiment of drawing a person at random from the population.

Exercise on definitions

Recall $\sigma_{XY} \coloneqq \mathrm{E}\left[(X - \mu_X)(Y - \mu_Y)\right]$.

Exercise 85. Simplify $\mathrm{cov}(X, X)$.

$$\begin{aligned} \mathrm{cov}(X, X) &= \mathrm{E}\left[(X - \mu_X)\,(X - \mu_X)\right] && (1) \\ &= \mathrm{E}\left[(X - \mu_X)^2\right] && (2) \\ &= \mathrm{V}(X). && (3) \end{aligned}$$

Properties of covariance

For any rvs X, Y, and Z the following properties hold:

1. $\mathrm{cov}(X, Y) = \mathrm{cov}(Y, X)$ (by properties of multiplication)
2. If $X \perp\!\!\!\perp Y$, then $\mathrm{cov}(X, Y) = 0$.
3. $\mathrm{cov}(X, X) = \mathrm{V}(X)$.
4. $\mathrm{cov}(X + Y, Z) = \mathrm{cov}(X, Z) + \mathrm{cov}(Y, Z)$.
5. For any constants a_1, a_2, b_1, b_2:

$$\mathrm{cov}(a_1 X + b_1, a_2 Y + b_2) = a_1 a_2 \mathrm{cov}(X, Y).$$

Note that b_1 and b_2 do not affect the covariance, but multiplying by a_1 and a_2 does.

Magnitude vs. sign

Definition 26. The magnitude of a number x is how far it is away from 0, i.e., its absolute value $|x|$.

- e.g., the magnitude of -100 is larger than the magnitude of 8 because $|-100| = 100 > 8 = |8|$.
- It's often helpful to distinguish the magnitude of a number from its *sign* (whether its positive or negative).
- When a number represents an effect in later lectures:
  - The sign tells us whether the effect is positive or negative (e.g., beneficial or harmful).
  - The magnitude tells us whether the effect is "large" or "small".

Covariance is difficult to interpret (1/3)

Example 20 (income and education). Suppose that X is income of a person and Y is the person's number of years of education. Are X and Y really closely related if $\mathrm{cov}(X, Y) = 1.22$? Is that magnitude large?

It's really hard to interpret that magnitude. We need to know:

- Is X *monthly* income or *yearly* income?
- What is income measured in? Euros? US dollars? Japanese Yen?
- If income is measured in US dollars, is it dollars from 1980, when the data was collected? Or dollars in 2025, after we've had inflation?

Just like $\mathrm{V}(X)$ is affected by the units of X, so is $\mathrm{cov}(X, Y)$.

Covariance is difficult to interpret (2/3)

Exercise 86. Suppose that Z is monthly income measured in €'s, and X is monthly income measured in 1,000€s. e.g., if $X(\omega^*) = 3$, then $Z(\omega^*) = 3000$. If $\text{cov}(X, Y) = 1.22$, what is $\text{cov}(Z, Y)$?

The first step is specifying the relationship between Z and X as an equation with Z on the LHS and X on the RHS: $Z = 1000X$.

$$\begin{aligned} \text{cov}(Z, Y) &= \text{cov}(1000X, Y) && (1)\\ &= 1000 \cdot \text{cov}(X, Y) && (2)\\ &= 1000 \cdot 1.22 && (3)\\ &= 1220. && (4) \end{aligned}$$

Covariance is difficult to interpret (3/3)

So we have to be very careful when interpreting covariance.

- Property (5) that we showed before, where a_1 and a_2 change the covariance, is not a desirable property.
- It would be great if the RHS had no a_1 and no a_2.
- What is the solution that we're looking for? Correlation!

Correlation

Definition 27. For rvs X and Y, Pearson's correlation coefficient of X and Y is:

$$\operatorname{corr}(X, Y) := \frac{\operatorname{cov}(X, Y)}{\operatorname{SD}(X) \cdot \operatorname{SD}(Y)} = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}.$$

- We will use "correlation" for short in this course.
- If $\operatorname{corr}(X, Y) = 0$, we say that X and Y are not correlated or *un*correlated.

Correlation

Definition 27. For rvs X and Y, Pearson's correlation coefficient of X and Y is:

$$\operatorname{corr}(X, Y) := \frac{\operatorname{cov}(X, Y)}{\operatorname{SD}(X) \cdot \operatorname{SD}(Y)} = \frac{\sigma_{XY}}{\sigma_X \cdot \sigma_Y}.$$

Exercise 87. Which of the terms σ_{XY}, σ_X, σ_Y can be negative?

A: only σ_{XY}.

- So the sign of $\operatorname{corr}(X, Y)$ is the same as the sign of $\operatorname{cov}(X, Y)$. If one of them is positive, the other is also.
- Thus, we often use some phrases interchangeably, such as: "X and Y have negative covariance" and "X and Y are negatively correlated".

Properties of correlation

1. $-1 \leq \text{corr}(X, Y) \leq 1$.
2. For constants a_1, a_2, b_1, b_2 with $a_1 a_2 > 0$,

$$\text{corr}(a_1 X + b_1, a_2 Y + b_2) = \text{corr}(X, Y).$$

The magnitude of correlation is easier to interpret than covariance.

How does the magic work for the a's to go away?

$$\begin{aligned}
\text{corr}(a_1 X + b_1, a_2 Y + b_2) &:= \frac{\text{cov}(a_1 X + b_1, a_2 Y + b_2)}{\text{SD}(a_1 X + b_1) \cdot \text{SD}(a_2 Y + b_2)} \\
&= \frac{a_1 a_2 \text{cov}(X, Y)}{a_1 \text{SD}(X) \cdot a_2 \text{SD}(Y)} \\
&= \frac{\text{cov}(X, Y)}{\text{SD}(X) \cdot \text{SD}(Y)} \\
&= \text{corr}(X, Y).
\end{aligned}$$

Review of notation

Let X and Y be rvs.

- $\sigma_X^2 \coloneqq \mathrm{V}(X)$.
- $\sigma_X \coloneqq \mathrm{SD}(X)$.
- $\sigma_{XY} \coloneqq \mathrm{cov}(X, Y)$.

In our class, we won't use a symbol for $\mathrm{corr}(X, Y)$. Some books use $\rho_{XY} \coloneqq \mathrm{corr}(X, Y)$.

- ρ is the Greek letter rho.

Story telling

Let G be the high school GPA of a person chosen at random from the population of UPF students. Let X be the person's PAU score.

Exercise 88. Would you guess $\mathrm{cov}(G, X)$ is pos. or neg.? What's the mechanism?

One story: $\mathrm{cov}(G, X) > 0$, because hard-working people, on average, get a high GPA and study a lot for a high PAU score. People who are not as hard-working get low GPA and low PAU scores, on average, because both require a lot of effort to study for.

- Alternative story: *intelligence* is the mechanism, not hard work.

Exercise 89. Give a story (as believable as possible) for why $\mathrm{cov}(G, X)$ could *in theory* be negative.

One story: the more time students dedicate to achieving a high GPA, the less time they have to study for the PAU.

- Here, the driving mechanism is that time is a finite resource.

Table of Contents

1. Summation operator and the mean

2. Distributions

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7. Expected value and variance

8. Standardization and joint distribution

Learning objectives

1. Variance of sums of rvs.
    a. Prove general result for $\mathrm{V}(X + Y)$.
2. Independent and identically distributed (iid).
3. Example: investment portfolio.

rvs as mechanisms

- Let W be a function that outputs how many km I walk on Friday nights, which depends (only) on the weather.
- Suppose that the weather is either rainy or clear. i.e., $\Omega = \{\text{rainy}, \text{clear}\}$.
- If rainy, I stay inside and watch a movie: $W(\text{rainy}) = 2$.
- If clear, I walk through the streets of Barcelona: $W(\text{clear}) = 9$.

Suppose that last Friday, it rained: $\omega_{\text{last friday}} = \text{rainy}$.

- $W(\omega_{\text{last friday}}) = 2$.

Exercise 90. Is W a rv or a number? rv.

- W represents the *mechanism* by which I choose what to do on Fridays.

E and V of sums of rvs

- The first thing to know is that $\mathrm{V}(X+Y)$ does not *necessarily* equal $\mathrm{V}(X)+\mathrm{V}(Y)$.
- This is different from expected value, because recall that for *any* X and Y:

$$\mathrm{E}(X+Y)=\mathrm{E}(X)+\mathrm{E}(Y).$$

- So what *does* $\mathrm{V}(X+Y)$ equal in general?
- Recall the following properties of covariance for any rvs Q_1, Q_2, Q_3:
  1. $\mathrm{cov}(Q_1, Q_1)=\mathrm{V}(Q_1)$.
  2. $\mathrm{cov}(Q_1+Q_2, Q_3)=\mathrm{cov}(Q_1, Q_3)+\mathrm{cov}(Q_2, Q_3)$.
  3. $\mathrm{cov}(Q_1, Q_2)=\mathrm{cov}(Q_2, Q_1)$.

$\mathrm{V}(X+Y) = ?$

1. $\mathrm{cov}(Q_1, Q_1) = \mathrm{V}(Q_1)$.
2. $\mathrm{cov}(Q_1 + Q_2, Q_3) = \mathrm{cov}(Q_1, Q_3) + \mathrm{cov}(Q_2, Q_3)$.
3. $\mathrm{cov}(Q_1, Q_2) = \mathrm{cov}(Q_2, Q_1)$.

- What property should we use first?
- Using property (1), what does $\mathrm{V}(X+Y)$ equal?

First, we define $W := X + Y$ to apply prop. (1) directly.

$$
\begin{aligned}
\mathrm{V}(X+Y) &= \mathrm{V}(W) && (1)\\
&= \mathrm{cov}(W, W) \quad \text{by prop. (1)} && (2)\\
&= \mathrm{cov}(X+Y, W) \quad \text{by substitution} && (3)\\
&= \mathrm{cov}(X, W) + \mathrm{cov}(Y, W) \quad \text{by prop. (2)} && (4)\\
&= \mathrm{cov}(X, X+Y) + \mathrm{cov}(Y, X+Y) \quad \text{by subst.} && (5)\\
&= \mathrm{cov}(X+Y, X) + \mathrm{cov}(X+Y, Y) \quad \text{by prop. (3)} && (6)
\end{aligned}
$$

$\mathrm{V}(X+Y) = ?$

1. $\mathrm{cov}(Q_1, Q_1) = \mathrm{V}(Q_1)$.
2. $\mathrm{cov}(Q_1 + Q_2, Q_3) = \mathrm{cov}(Q_1, Q_3) + \mathrm{cov}(Q_2, Q_3)$.
3. $\mathrm{cov}(Q_1, Q_2) = \mathrm{cov}(Q_2, Q_1)$.

$$
\begin{aligned}
&= \mathrm{cov}(X+Y, X) + \mathrm{cov}(X+Y, Y) \text{ copied from prev. slide} && (1)\\
&= \mathrm{cov}(X, X) + \mathrm{cov}(Y, X) + \mathrm{cov}(X, Y) + \mathrm{cov}(Y, Y) && (2)\\
&= \mathrm{V}(X) + \mathrm{V}(Y) + \mathrm{cov}(Y, X) + \mathrm{cov}(X, Y) \quad \text{by prop. (1)} && (3)\\
&= \mathrm{V}(X) + \mathrm{V}(Y) + \mathrm{cov}(X, Y) + \mathrm{cov}(X, Y) \quad \text{by prop. (3)} && (4)\\
&= \mathrm{V}(X) + \mathrm{V}(Y) + 2\mathrm{cov}(X, Y). && (5)
\end{aligned}
$$

Variance of sums

We've just proved the following theorem:

Theorem 4. For any two rvs, X and Y,

$$\mathrm{V}(X+Y) = \mathrm{V}(X) + \mathrm{V}(Y) + 2\mathrm{cov}(X,Y).$$

- For the special case where $X \perp\!\!\!\perp Y$, then $\mathrm{cov}(X,Y) = 0$.
- So $\mathrm{V}(X+Y)$ does equal $\mathrm{V}(X) + \mathrm{V}(Y)$ when $X \perp\!\!\!\perp Y$.

Corollary 1. For rvs X_1, X_2, and X_3 that are independent,

$$\mathrm{V}(X_1 + X_2 + X_3) = \mathrm{V}(X_1) + \mathrm{V}(X_2) + \mathrm{V}(X_3).$$

That is, the variance of the sum equals the sum of the variances.

Example where $\mathrm{V}(X+Y) \neq \mathrm{V}(X)+\mathrm{V}(Y)$

Exercise 91. Let Y be the number of heads from two coin flips, and Z the number of tails from the (same) flips. What is $\mathrm{V}(Y+Z)$?

Hint: define $Q := Y+Z$. What is $\mathrm{supp}(Q)$?

ω	$Y(\omega)$	$Z(\omega)$	$Q(\omega)$
(H,H)	2	0	2
(H,T)	1	1	2
(T,H)	1	1	2
(T,T)	0	2	2

- Alternatively, observe that $Z = 2-Y$.
  - Then, $\mathrm{V}(Y+Z) = \mathrm{V}(Y+(2-Y)) = \mathrm{V}(2) = 0$.
- Are $\mathrm{V}(Y)$ and $\mathrm{V}(Z)$ equal to 0?
  - No, so $\mathrm{V}(Y+Z) = 0 \neq \mathrm{V}(Y)+\mathrm{V}(Z)$ in this case.

Example where $\mathrm{V}(X+Y) \neq \mathrm{V}(X)+\mathrm{V}(Y)$

Exercise 91. Let Y be the number of heads from two coin flips, and Z the number of tails from the (same) flips. What is $\mathrm{V}(Y+Z)$?

Hint: define $Q := Y+Z$. What is $\mathrm{supp}(Q)$?

ω	$Y(\omega)$	$Z(\omega)$	$Q(\omega)$
(H,H)	2	0	2
(H,T)	1	1	2
(T,H)	1	1	2
(T,T)	0	2	2

Exercise 92. Is it true that $\mathrm{E}(Y+Z) = \mathrm{E}(Y)+\mathrm{E}(Z)$?

Yes. First, $\mathrm{E}(Y) = \frac{1}{4}(2+1+1+0) = 1$. Also, $\mathrm{E}(Z) = 1$.

Then, $\mathrm{E}(Y+Z) = \mathrm{E}(Q) = \mathrm{E}(2) = 2 = 1+1 = \mathrm{E}(Y)+\mathrm{E}(Z)$.

Independent and identically distributed (iid) rvs

Definition 28 (iid). The rvs $X_1, \ldots, X_n$ are independent and identically distributed (iid) if the following two criteria are satisfied:

1. The $X_1, \ldots, X_n$ are independent.
2. The X_i all have the same (marginal) distribution.

Reminder about P

- Let V_1, V_2, V_3 be an iid collection of coin flips from a fair coin, where $V_i := \mathrm{I}\{\text{i'th flip is heads}\}$.
- What is $\mathrm{P}(V_2 = 1)$? $\frac{1}{2}$.
- We run the experiment and happen to get $\omega^* := (T, H, H)$.

$$V_1(\omega^*) = 0,\ V_2(\omega^*) = 1,\ \ V_3(\omega^*) = 1.$$

Exercise 93. What is $\mathrm{P}(V_2 = 1)$? $\frac{1}{2}$.

- Recall how we *defined* P. What does $\mathrm{P}(V_2 = 1)$ mean? If we run the experiment an infinite amount of times, it's the proportion of times the event $V_2 = 1$ happens.
- Recall the experiment underlying V_2... It's not taking a sample from a sample. It's flipping the coin three (new) times.
- $V_2(\omega^*) = 1$, $\mathrm{P}(V_2(\omega^*) = 1) = \mathrm{P}(1 = 1) = 1$, $\mathrm{P}(V_2 = 1) = \frac{1}{2}$.

Variance of the sum of iid rvs

Exercise 94. Suppose rvs $X_1, \ldots, X_n$ are iid. Simplify $\mathrm{V}\left(\sum_{i=1}^{n} X_i\right)$.

Hint: if it looks complicated, do it for $n = 2$, then for $n = 3$.

$$\mathrm{V}\left(\sum_{i=1}^{n} X_i\right) = \mathrm{V}(X_1 + X_2 + \cdots + X_n) \tag{1}$$

$$= \mathrm{V}(X_1) + \mathrm{V}(X_2) + \cdots + \mathrm{V}(X_n) \quad \text{(by indep.)} \tag{2}$$

$$= \sum_{i=1}^{n} \mathrm{V}(X_i). \tag{3}$$

Thus, if iid, the variance of the sum equals the sum of the (individual) variances.

Let $\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i$. i.e., $\bar{X}(\omega) := \frac{1}{n}\sum_{i=1}^{n} X_i(\omega)$.

Suppose we flip 2 fair coins. $\Omega = \{(H,H),(H,T),(T,H),(T,T)\}$. Define X_1 and X_2 as follows:

ω	$X_1(\omega)$	$X_2(\omega)$	$\bar{X}(\omega)$
(H,H)	$X_1((H,H)) = 1$	$X_2((H,H)) = 1$	1
(H,T)	$X_1((H,T)) = 1$	$X_2((H,T)) = 0$	$\frac{1}{2}$
(T,H)	$X_1((T,H)) = 0$	$X_2((T,H)) = 1$	$\frac{1}{2}$
(T,T)	$X_1((T,T)) = 0$	$X_2((T,T)) = 0$	0

Exercise 95. How can we then explicitly define the rv $\bar{X}$ as a function from Ω? Note for $n = 2$, $\bar{X}(\omega) = \frac{1}{2}(X_1(\omega) + X_2(\omega))$.

$$\bar{X}((H,H)) = \frac{1}{2}(X_1((H,H)) + X_2((H,H))) = \frac{1}{2}(1+1) = 1.$$
$$\bar{X}((H,T)) = \frac{1}{2}(X_1((H,T)) + X_2((H,T))) = \frac{1}{2}(1+0) = \frac{1}{2}.$$

Let $\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i$. i.e., $\bar{X}(\omega) := \frac{1}{n}\sum_{i=1}^{n} X_i(\omega)$.

Suppose we flip 2 fair coins. $\Omega = \{(H,H),(H,T),(T,H),(T,T)\}$. Define X_1 and X_2 as follows:

ω	$X_1(\omega)$	$X_2(\omega)$	$\bar{X}(\omega)$
(H,H)	$X_1((H,H)) = 1$	$X_2((H,H)) = 1$	1
(H,T)	$X_1((H,T)) = 1$	$X_2((H,T)) = 0$	$\frac{1}{2}$
(T,H)	$X_1((T,H)) = 0$	$X_2((T,H)) = 1$	$\frac{1}{2}$
(T,T)	$X_1((T,T)) = 0$	$X_2((T,T)) = 0$	0

Exercise 95. How can we then explicitly define the rv $\bar{X}$ as a function from Ω? Note for $n = 2$, $\bar{X}(\omega) = \frac{1}{2}(X_1(\omega) + X_2(\omega))$.

$$\bar{X}((T,H)) = \frac{1}{2}(X_1((T,H)) + X_2((T,H))) = \frac{1}{2}(0+1) = \frac{1}{2}.$$
$$\bar{X}((T,T)) = \frac{1}{2}(X_1((T,T)) + X_2((T,T))) = \frac{1}{2}(0+0) = 0.$$

Thinking of $\bar{X}$ as one rv

- We can think of $\bar{X}$ as its own random variable!
- We could have defined $\bar{X}$ without first defining X_1 and X_2:

$$\bar{X} := \begin{cases} 0 & \text{if both coin flips are T} \\ 1 & \text{if both coin flips are H} \\ \frac{1}{2} & \text{if one of the coin flips was H and the other T} \end{cases}$$

Recall that to get one outcome of $\bar{X}$, we have to flip two coins. So two coin flips is *one* experiment.

Exercise 96. Suppose $\omega^* = (HT)$. Is $\bar{X}$ a rv or a number?

- $\bar{X}$ is a rv. $\bar{X}(\omega^*) = \frac{1}{2}$ is a number.
- Analogy: Let f be defined by $f(x) = x^2$. Then, f is a function, but $f(5)$ is a number.
- Sometimes we use a *lowercase* $\bar{x}$ to refer to the outcome for a specific sample: $\bar{x} = \bar{X}(\omega^*) = \frac{1}{2}$.

Expected value of $\bar{X}$

ω	$X_1(\omega)$	$X_2(\omega)$	$\bar{X}(\omega)$
(H,H)	$X_1((H,H))=1$	$X_2((H,H))=1$	$\bar{X}((H,H))=1$
(T,T)	$X_1((T,T))=0$	$X_2((T,T))=0$	$\bar{X}((T,T))=0$
(H,T)	$X_1((H,T))=1$	$X_2((H,T))=0$	$\bar{X}((H,T))=\frac{1}{2}$
(T,H)	$X_1((T,H))=0$	$X_2((T,H))=1$	$\bar{X}((T,H))=\frac{1}{2}$

Exercise 97. Evaluate $\mathrm{E}(X_1)$, $\mathrm{E}(X_2)$, $\mathrm{E}(\bar{X})$.

$\mathrm{E}(X_1)=\frac{1}{4}(1+0+1+0)=\frac{1}{2}$. $\mathrm{E}(X_2)=\frac{1}{4}(1+0+0+1)=\frac{1}{2}$.

$\mathrm{E}(\bar{X})=\frac{1}{4}(1+0+\frac{1}{2}+\frac{1}{2})=\frac{1}{2}$.

Alternatively (and more easily):
$\mathrm{E}(\bar{X})=\mathrm{E}\left[\frac{1}{2}(X_1+X_2)\right]=\frac{1}{2}\left[\mathrm{E}(X_1)+\mathrm{E}(X_2)\right]=\frac{1}{2}\left[\frac{1}{2}+\frac{1}{2}\right]=\frac{1}{2}$.

- We'll show it's not a coincidence that $\mathrm{E}(\bar{X})=\mathrm{E}(X_i)$.

Evaluating $\mathrm{E}(\bar{X})$ for any n (1/2)

Consider the previous exercise of evaluating $\mathrm{E}(\bar{X})$.

- The first way involved first finding the pmf of $\bar{X}$ and then using the definition of expected value.
- However, imagine how the table would look if n were, e.g., 10. One ω would be $(H, H, T, T, T, H, H, T, T, T)$.
- That is, the table would have $2^{10} = 1024$ rows!

When n is large, the "alternative" way is easier. We apply properties of expected value and we just need to know $\mathrm{E}(X_i)$ rather than working with the pmfs.

Evaluating $\mathrm{E}(\bar{X})$ for any n (2/2)

Suppose $X_1, \ldots, X_n$ are iid. Then,

$$\mathrm{E}(\bar{X}) = \mathrm{E}\Big(\frac{1}{n}\sum_{i=1}^{n} X_i\Big) \tag{1}$$

$$= \frac{1}{n}\mathrm{E}\Big(\sum_{i=1}^{n} X_i\Big) \tag{2}$$

$$= \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}(X_i) \tag{3}$$

$$= \frac{1}{n} n \cdot \mathrm{E}(X_i) \tag{4}$$

$$= \mathrm{E}(X_i). \tag{5}$$

What can we say about $\mathrm{E}(X_1)$ and $\mathrm{E}(X_7)$? $\mathrm{E}(X_1) = \mathrm{E}(X_7)$ so we can write $\mathrm{E}(X_i)$ for all since they are all the same.

Exercise 98. Suppose that we have n rvs, $X_1, \ldots, X_n$, and that they are iid. Suppose $n = 25$ and $\mathrm{V}(X_3) = 100$. What is $\mathrm{V}(\bar{X})$?

1. The first thing you might wonder is: how can we solve this problem if we only know the variance of X_3?
   - The key is that the X_i are iid so they have the *same distribution* and variance is a property of the distribution so they all have the same variances:

$$\mathrm{V}(X_1) = \mathrm{V}(X_2) = \mathrm{V}(X_3) = \cdots = \mathrm{V}(X_{25}) = 100.$$

     - In this case, we often use the notation "$\mathrm{V}(X_i) = 100$".

   a. The second thing you might wonder is: how can we solve this problem if we don't know the distribution of $\bar{X}$?
      - Even if we knew the distribution of the X_i, it might be a lot of work to figure out the distribution of $\bar{X}$.
      - Thankfully, we can use properties of variance to solve the problem without knowing the distribution of $\bar{X}$.

Solution (recall $n = 25, \mathrm{V}(X_3) = 100$)

$$\mathrm{V}(\bar{X}) = \mathrm{V}\Big(\frac{1}{n}\sum_{i=1}^{n} X_i\Big) \tag{1}$$

$$= \frac{1}{n^2}\mathrm{V}\Big(\sum_{i=1}^{n} X_i\Big) \qquad \text{by } \mathrm{V}(aY) = a^2\mathrm{V}(Y) \tag{2}$$

$$= \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{V}(X_i) \qquad \text{by independence} \tag{3}$$

$$= \frac{1}{25^2}\sum_{i=1}^{25} \mathrm{V}(X_i) \tag{4}$$

$$= \frac{1}{25^2}\sum_{i=1}^{25} \mathrm{V}(X_3) \qquad \text{by identically distributed} \tag{5}$$

$$= \frac{1}{25^2}\sum_{i=1}^{25} 100 \tag{6}$$

Solution: final steps

Make sure you can solve the previous expression out to a final answer. We'll see the final answer below:

$$\frac{1}{25^2}\sum_{i=1}^{25} 100 = \frac{1}{25^2} \cdot 25 \cdot 100 \tag{1}$$

$$= \frac{100}{25} \tag{2}$$

$$= 4. \tag{3}$$

If confused about (1) recall $\sum_{i=1}^{n} a = na$. In this case, n is 25 and a is 100.

General results for distribution of $\bar{X}$

Theorem 5. Let $X_1, \ldots, X_n$ be iid. Then,

$$\mathrm{V}(\bar{X}) = \frac{\mathrm{V}(X_i)}{n}$$
$$\mathrm{E}(\bar{X}) = \mathrm{E}(X_i).$$

The Law of Large Numbers combines the above two equalities to provide a powerful result.

Theorem 6 (Law of Large Numbers (LLN)). Let $X_1, \ldots, X_n$ be iid. Then, for any $d > 0$,

$$\mathrm{P}\left[\bar{X} \in (\mu_X - d, \mu_X + d)\right] \to 1.$$

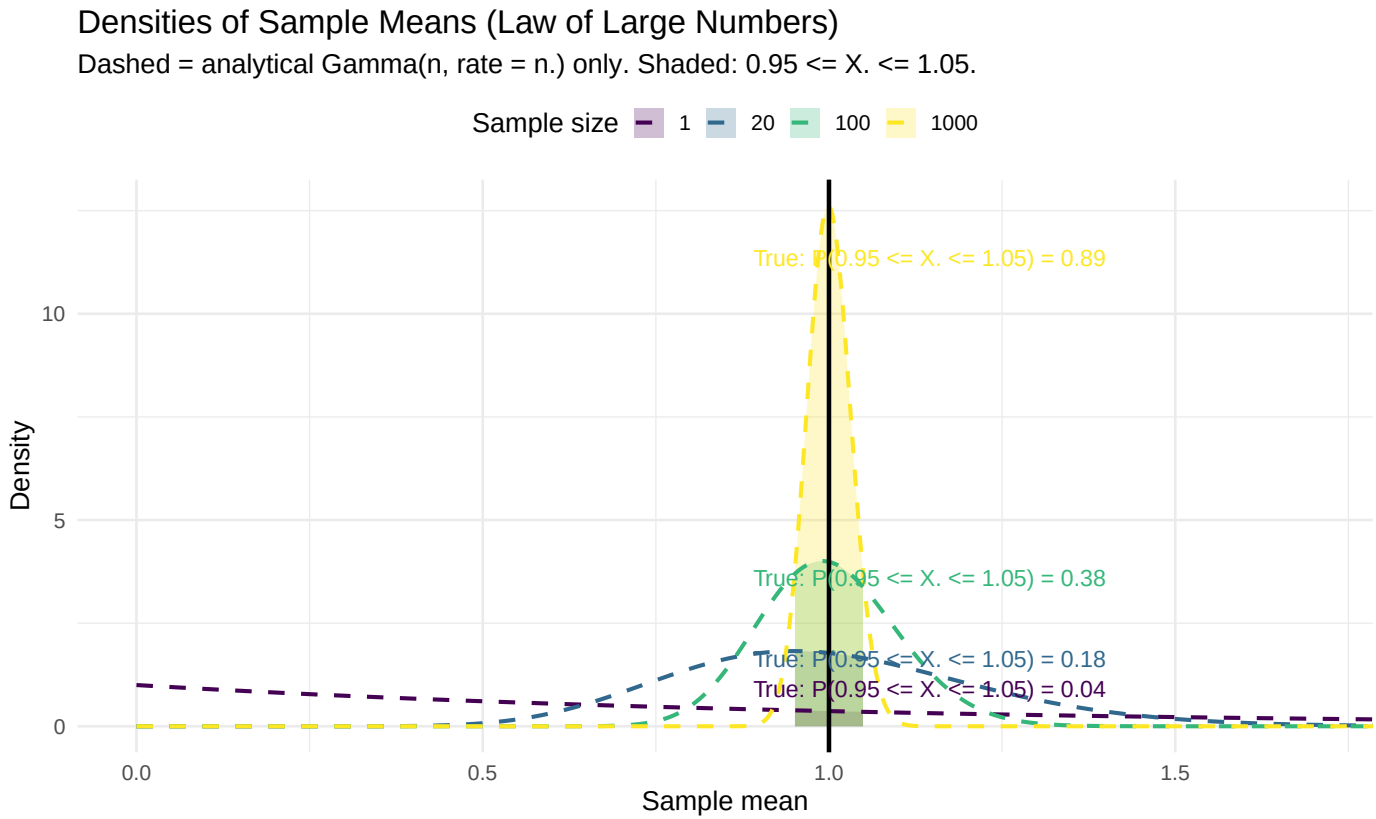
Equivalently: $\lim_{n\to\infty} \mathrm{P}(\bar{X} - d < \mathrm{E}(X_i) < \bar{X} + d) = 1.$

Equivalently: $\mathrm{P}\left[\left|\bar{X} - \mathrm{E}(X_i)\right| < d\right] \to 1.$

Key intuition: For large n, it is very likely that $\bar{X}$ is close to $\mathrm{E}(X_i)$.

LLN

Let $X_1, \ldots, X_n$ be iid from a population that has distribution $exp(1)$. e.g., $\mathrm{E}(X_i) = 1$.



Investment portfolio

Let W, X, Y, Z be rvs that represent profit from a unit of "stock w", "stock x", "stock y", and "stock z". Suppose that W, X, Y, Z are independent, each with expected value 1 and variance 5.

Portfolio 1: one unit each of stocks x, y, z.
Portfolio 2: three units of stock w.

Exercise 99. What is the expected profit for portfolio 1, and separately for portfolio 2?

Portfolio 1: $\mathrm{E}(X + Y + Z) = \mathrm{E}(X) + \mathrm{E}(Y) + \mathrm{E}(Z) = 3$.

Portfolio 2: $\mathrm{E}(W + W + W) = \mathrm{E}(3W) = 3\mathrm{E}(W) = 3(1) = 3$.

Investment portfolio

Let W, X, Y, Z be rvs that represent profit from a unit of "stock w", "stock x", "stock y", and "stock z". Suppose that W, X, Y, Z are independent, each with expected value 1 and variance 5.

Portfolio 1: one unit each of stocks x, y, z.
Portfolio 2: three units of stock w.

Exercise 100. What is the variance of profit for portfolio 1, and separately for portfolio 2?

Portfolio 1: $\mathrm{V}(X+Y+Z) = \mathrm{V}(X)+\mathrm{V}(Y)+\mathrm{V}(Z)$ (by indep.)
$= 5+5+5 = 15.$

Portfolio 2: $\mathrm{V}(W+W+W) = \mathrm{V}(3W) = 3^2\mathrm{V}(W) = 9(5) = 45.$
$\mathrm{V}(W+W+W) \neq \mathrm{V}(W)+\mathrm{V}(W)+\mathrm{V}(W).$

Investment portfolio

These exercises show the importance of "portfolio diversification".

Let's take this a step further. If we could have any dependence structure among our assets, is independence what we ideally want?

- Let's look at our general formula again:

$$\mathrm{V}(X+Y) = \mathrm{V}(X) + \mathrm{V}(Y) + 2\mathrm{cov}(X,Y).$$

- Can we do any better than independence?
- Yes, if covariance of our assets is *negative*, then we can get even lower variance in our overall profits.
- You can think of this concept as "hedging your bets".

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Learning objectives

1. Introduce conditional probability.
2. Introduce conditional distributions.
3. Alternate perspective on independence and joint distributions.
4. Examples.
    a. College graduation and employment.
    b. Shooting free throws.
    c. Luca the ice cream salesperson.
5. Introduce conditional expected value, which will be one of the focal points going forward in the course.

Chance of rain

Suppose the weather app on your phone says "There's a 70% chance of rain in the next hour".

Exercise 101. What does that mean?

- Of all the many times with the exact same measurements of certain conditions (e.g., same current temperature, humidity, wind speed, atmospheric pressure) as the current conditions, the event "it rains in the next hour" has happened 70% of those times.
- There's no *inherent* 70% chance of rain.
  - In this class, we are not thinking about randomness the same way as quantum mechanics does (i.e., randomness coming from the atomic and sub-atomic levels).
  - In 200 years, we'll have better technology, and even if the unconditional distribution were the same (e.g., no global warming), forecasts will be closer to 0% or 100%.

Using information under uncertainty

- We often have *information* that is helpful for making a decision under uncertainty.
- In order to make a "decision", recall that we use the probability distribution of rvs.
  - e.g., we used the distribution of H to decide how much to pay to play ♡s and 5s.
- How do we incorporate this information to make a more informed decision? Answer: conditional probability.
  - Incorporating information means to use the outcome of one or more rvs to obtain a more relevant probability distribution for the specific situation.

Decision: should I take my umbrella?

- Suppose the decision I need to make is whether to take my umbrella, based on whether I think it will rain.
  - Maybe there is a 20% chance on any given day that it will rain in Barcelona.
  - However, if I wake up and see dark clouds, I know the probability must be higher.
  - What I really want to know is: out of all of the days in Gainesville where there are dark clouds in the morning, how many of *those* times does it rain?

Example: UF football

- Suppose we are betting on UF winning a football game.
- Perhaps the "unconditional" probability of UF winning a game is 0.60.
- What are examples of relevant information that we might want to condition on?

1. Whether the game is home or away.
2. Whether it's raining.
3. Perhaps we're making the bet at half-time. Then we might condition on whether UF is winning at half time, or its margin of victory.

Example with T and D

Recall the rv T that is 1 if the roll of a fair 6-sided die is even, and 0 if odd; and D that is the number on the rolled die.

Exercise 102. Suppose we know that $T(\omega^*) = 1$ for a particular outcome of the experiment ω^*. Knowing this *information*, what is the probability that D is 4?

A: $\frac{1}{3}$, because D must be 2, 4, or 6, and each is equally likely.

- We will now introduce the formal framework of *conditional probability* and use it to solve this same problem.
- Then we will do more complicated examples that are not as easy to solve intuitively (i.e., without the conditional probability framework) as this one is.

Conditional probability for discrete rvs

Definition 29 (conditional probability of discrete rvs). Let X, Y be two discrete rvs. Let x and y be any numbers such that $\mathrm{P}(Y = y) > 0$. We define the conditional probability that X equals x given that Y equals y as:

$$\mathrm{P}(X = x | Y = y) := \frac{\mathrm{P}(X = x, Y = y)}{\mathrm{P}(Y = y)}.$$

- We are introducing new notation for this definition. We read the '|' as "given that" or "conditional on".
- Joint probability is symmetric:

$$\mathrm{P}(X = x, Y = y) = \mathrm{P}(Y = y, X = x).$$

- But conditional probability is typically not symmetric:

$$\mathrm{P}(X = x | Y = y) \neq \mathrm{P}(Y = y | X = x).$$

Conditional pmf

Definition 30 (conditional pmf). Let X, Y be two discrete rvs with joint pmf f_{XY}. Let f_Y be the pmf of Y and let x and y be any numbers such that $f_Y(y) > 0$. We define the conditional probability mass function of X given that Y equals y as:

$$f_{X|Y}(x|y) := \frac{f_{XY}(x,y)}{f_Y(y)}.$$

- Note that "$f_{X|Y}$" is just the name of the function.
- $f_{X|Y}(\cdot|y) : \mathbb{R} \mapsto [0,1]$. For each y, there is a different conditional pmf.

Revisiting the motivating example

Exercise 103. Suppose we know that $T(\omega^*) = 1$ for a particular outcome of the experiment ω^*. Knowing this *information*, what is the probability that D is 4? Solve this using conditional probability.

$$\mathrm{P}(D = 4 | T = 1) = \frac{\mathrm{P}(D = 4, T = 1)}{\mathrm{P}(T = 1)} \tag{1}$$

$$= \frac{1/6}{1/2} \tag{2}$$

$$= \frac{1}{6} \cdot \frac{2}{1} = \frac{1}{3}. \tag{3}$$

- Using notation of conditional pmf: $f_{D|T}(4|1) = \frac{1}{3}$.
- $\mathrm{P}(T = 1 | D = 4) = 1 \neq \frac{1}{3} = \mathrm{P}(D = 4 | T = 1)$.

Alternative definition of independence

We can now provide an alternative, and more intuitive, definition of independence.

Definition 31 (Alternative definition of independence). Let rvs X and Y have joint pmf f_{XY}. Let Y have pmf f_Y. X and Y are independent if

$$f_{X|Y}(x|y) = f_X(x),$$

for all values of y such that $f_Y(y) > 0$, and all values of x.

- Intuition: knowing more information (on the outcome of the other rv) does not help.

Showing not independent

Exercise 104. Using the new definition of independence, show that T is not independent of D.

In the previous exercises, we showed that $\mathrm{P}(D = 4|T = 1) = \frac{1}{3}$. The unconditional probabilility, $\mathrm{P}(D = 4)$, is $\frac{1}{6}$.

Thus, $\mathrm{P}(D = 4|T = 1) = \frac{1}{3} \neq \frac{1}{6} = \mathrm{P}(D = 4)$.

Or using conditional pmf notation: $f_{D|T}(4|1) = \frac{1}{3} \neq \frac{1}{6} = f_D(4)$.

Joint distribution of employment status and college degree in the US population aged 25 and greater, 2008		
	unemp. $(Y(\omega) = 0)$	emp. $(Y(\omega) = 1)$
No degree $(X(\omega) = 0)$	0.037	0.622
Degree $(X(\omega) = 1)$	0.009	0.332

Exercise 105. Suppose we choose someone at random from the above-defined population and they have a college degree. What is the probability that the person is employed?

$$\mathrm{P}(Y = 1|X = 1) = \frac{\mathrm{P}(Y = 1, X = 1)}{\mathrm{P}(X = 1)} \tag{1}$$

$$= \frac{\mathrm{P}(Y = 1, X = 1)}{\mathrm{P}(X = 1, Y = 0) + \mathrm{P}(X = 1, Y = 1)} \tag{2}$$

$$= \frac{0.332}{0.009 + 0.332} = \frac{0.332}{0.341} \approx 0.974. \tag{3}$$

Rearranging to solve for a joint probability

The definition of a conditional pmf can be rearranged.

- Recall the definition $f_{X|Y}(x|y) := \frac{f_{XY}(x,y)}{f_Y(y)}$.
- We solve for $f_{XY}(x,y)$ by multiply both sides by $f_Y(y)$ and switch the LHS and RHS to get:

$$f_{XY}(x,y) = f_Y(y) f_{X|Y}(x|y)$$

- Note also that we could have used $f_{Y|X}(y|x)$ instead. That is,

$$f_{XY}(x,y) = f_Y(y) f_{X|Y}(x|y) = f_X(x) f_{Y|X}(y|x).$$

- In case it helps, let's write the above using P notation:

$$\begin{aligned} \mathrm{P}(X = x, Y = y) &= \mathrm{P}(Y = y) \cdot \mathrm{P}(X = x | Y = y) \\ &= \mathrm{P}(X = x) \cdot \mathrm{P}(Y = y | X = x). \end{aligned}$$

Revisit a joint probability

Recall: $\mathrm{P}(X = x, Y = y) = \mathrm{P}(Y = y) \cdot \mathrm{P}(X = x | Y = y)$.

Recall D is the number from a fair 6-sided die, and T is 1 if the number is even, 0 otherwise.

Exercise 106. Evaluate $\mathrm{P}(D = 4, T = 1)$ using conditional probability.

$$\begin{aligned}\mathrm{P}(D = 4, T = 1) &= \mathrm{P}(D = 4) \cdot \mathrm{P}(T = 1 | D = 4) \\ &= \frac{1}{6} \cdot 1 \\ &= \frac{1}{6}.\end{aligned}$$

Free throw example

Let's define an experiment to pick someone at random and have them shoot 2 free throws. Letting "N" mean "no" (didn't make) and "Y" mean "yes" (did make), we have: $\Omega = \{YY, NN, YN, NY\}$.

- Let $F(\omega) = 1$ if the person makes the first, 0 otherwise; and let $S(\omega) = 1$ if the (same) person makes the second, 0 otherwise.
- Suppose that $\mathrm{P}(F = 1) = \mathrm{P}(S = 1) = 0.5$.

Exercise 107. What's the prob a person makes both free throws?



A: We don't know. This question is about $\mathrm{P}(F = 1, S = 1)$, and we don't have information on the joint (or conditional) probabilities.

$$\mathrm{P}(F = 1, S = 1) = \mathrm{P}(F = 1) \cdot \mathrm{P}(S = 1 | F = 1).$$

$\mathrm{P}(S = 1|F = 1)$ only equals $\mathrm{P}(S = 1)$ if $F \perp\!\!\!\perp S$.

Two (of many) possibilities

Recall: $\mathrm{P}(F = 1) = \mathrm{P}(S = 1) = 0.5$.

Extreme 1: $\mathrm{P}(S = 1|F = 1) = 0$.
$\mathrm{P}(F = 1, S = 1) = \mathrm{P}(F = 1) \cdot \mathrm{P}(S = 1|F = 1) = 0.5 \cdot 0 = 0$.

Extreme 2: $\mathrm{P}(S = 1|F = 1) = 1$.
$\mathrm{P}(F = 1, S = 1) = \mathrm{P}(F = 1) \cdot \mathrm{P}(S = 1|F = 1) = 0.5 \cdot 1 = 0.5$.

Therefore, what is the set of possible values $\mathrm{P}(F = 1, S = 1)$ could take? $\mathrm{P}(F = 1, S = 1) \in [0, 0.5]$.

Does the other conditional make sense?

Exercise 108. Does it even make sense to talk about $\mathrm{P}(F = 1|S = 1)$?

- Yes. Remember, one experiment is two free throw shots. So an ω doesn't even exist until both free throws are completed.
- How can we conceptualize this? Well, suppose I tell you that a player shot two free throws and made the second. What's the probability that they made the first?
- So we could have approached the previous exercise using $\mathrm{P}(F = 1, S = 1) = \mathrm{P}(S = 1) \cdot \mathrm{P}(F = 1|S = 1)$, but it would have been more difficult to think through it.

Ice cream profits

- Recall our ice cream salesperson, Luca.
- $\omega = (w_L, w_B)$, where $w_L, w_B \in \{\mathit{cloudy}, \mathit{sunny}\}$.
- Let $S_L := \mathrm{I}\{w_L = \mathit{sunny}\}$, $S_B := \mathrm{I}\{w_B = \mathit{sunny}\}$.
- Knowing $S_L(\omega)$ is equivalent to knowing ω_L, and knowing $S_B(\omega)$ is equivalent to knowing ω_B.
- Recall π is profit. We could write π as a function of w_L and w_B or alternatively as a function of S_L and S_B.

Exercise 109. How many elements does the set $\mathrm{supp}(\pi)$ have (at most)?

π depends completely on S_L and S_B, and each of them can take 2 values, so π can take at most 4 values.

Joint pmf of S_L, S_B		
	$S_B(\omega) = 0$	$S_B(\omega) = 1$
$S_L(\omega) = 0$	0.05	0.25
$S_L(\omega) = 1$	0.30	0.40

We did not fully work out Luca's optimal choice of how much ice cream to buy, but suppose the following is true:

$$\pi(S_L = 1, S_B = 1) = \pi(w_L = \textit{sunny}, w_B = \textit{sunny}) = 30 \quad (1)$$
$$\pi(S_L = 1, S_B = 0) = \pi(w_L = \textit{sunny}, w_B = \textit{cloudy}) = 10 \quad (2)$$
$$\pi(S_L = 0, S_B = 1) = \pi(w_L = \textit{cloudy}, w_B = \textit{sunny}) = 20 \quad (3)$$
$$\pi(S_L = 0, S_B = 0) = \pi(w_L = \textit{cloudy}, w_B = \textit{cloudy}) = 15 \quad (4)$$

- These numbers are made up.
- $\operatorname{supp}(\pi) = \{10, 15, 20, 30\}$.

Exercise 110. $\mathrm{P}(\pi = 30) = \mathrm{P}(S_L = 1, S_B = 1) = 0.40$.

Intuition of conditional randomness

- Conditional on weather at Luca's house, profits are still random.
- What does it mean to be random? The value can be different for different ω.
- Specifically, even if we know w_L we don't know w_B and π depends on both w_L and w_B.

Exercise 111. Order (without evaluating) the following 3 numbers: $\mathrm{P}(\pi = 30)$, $\mathrm{P}(\pi = 30|S_L = 0)$, and $\mathrm{P}(\pi = 30|S_L = 1)$.
$\mathrm{P}(\pi = 30|S = 1) \geq \mathrm{P}(\pi = 30) \geq \mathrm{P}(\pi = 30|S = 0) = 0.$

- Analogy: The prob of UF winning a football game, conditional on winning at half-time is greater than the unconditional prob of winning the game... which in turn is greater than the prob of winning conditional on trailing at half-time.

Joint pmf of S_L, S_B		
	$S_B(\omega) = 0$	$S_B(\omega) = 1$
$S_L(\omega) = 0$	0.05	0.25
$S_L(\omega) = 1$	0.30	0.40

Exercise 112. Recall $\pi(w_L = \textit{sunny}, w_B = \textit{sunny}) = 30$. Evaluate $\mathrm{P}(\pi = 30 | S_L = 1)$.

$$\begin{aligned}
\mathrm{P}(\pi = 30 | S_L = 1) &= \mathrm{P}(S_L = 1, S_B = 1 | S_L = 1) && (1)\\
&= \mathrm{P}(S_B = 1 | S_L = 1) && (2)\\
&= \frac{\mathrm{P}(S_B = 1, S_L = 1)}{\mathrm{P}(S_L = 1)} && (3)\\
&= \frac{0.40}{0.30 + 0.40} = \frac{0.40}{0.70} = \frac{4}{7}. && (4)
\end{aligned}$$

So when Luca wakes up and sees it's sunny, he knows there is a $\frac{4}{7}$ prob of a \$30 profit.

Indeed, $\mathrm{P}(\pi = 30 | S_L = 1) = \frac{4}{7} > \frac{4}{10} = \mathrm{P}(\pi = 30)$.

$\mathrm{P}(\pi = 30|S_L = 1) = \frac{4}{7}$ (copied from prev. slide).

We can find $\mathrm{P}(\pi = 10|S_L = 1)$ in 2 ways:

1. We could do similar calculations to how we evaluated $\mathrm{P}(\pi = 30|S_L = 1)$.
2. Alternatively, recall that conditional on $S_L = 1$ happening, π has only 2 potential outcomes (S_B can be 0 or 1), and so the probabilities must sum to 1. Thus, $\mathrm{P}(\pi = 10|S_L = 1) = 1 - \mathrm{P}(\pi = 30|S_L = 1) = 1 - \frac{4}{7} = \frac{3}{7}$.

We then define the conditional pmf, $f_{\pi|S_L}$, by:
$f_{\pi|S_L}(30|1) = \frac{4}{7}, f_{\pi|S_L}(10|1) = \frac{3}{7}$, $f_{\pi|S_L}(s|1) = 0$ for $s \notin \{10, 30\}$.

- As mentioned above, it's not a coincidence that $f_{\pi|S_L}(30|1) + f_{\pi|S_L}(10|1) = 1$.

Exercise 113. $\mathrm{P}(S_B = 0, S_L = 1) + \mathrm{P}(S_B = 1, S_L = 1) = \mathrm{P}(S_L = 1)$.

Exercise 114. Evaluate $f_{S_B|S_L}(0|1) + f_{S_B|S_L}(1|1)$.

$$
\begin{aligned}
&= \mathrm{P}(S_B = 0 | S_L = 1) + \mathrm{P}(S_B = 1 | S_L = 1) \\
&= \frac{\mathrm{P}(S_B = 0, S_L = 1)}{\mathrm{P}(S_L = 1)} + \frac{\mathrm{P}(S_B = 1, S_L = 1)}{\mathrm{P}(S_L = 1)} \quad \text{(by def. of cond. prob.)} \\
&= \frac{\mathrm{P}(S_B = 0, S_L = 1) + \mathrm{P}(S_B = 1, S_L = 1)}{\mathrm{P}(S_L = 1)} \\
&= \frac{\mathrm{P}(S_L = 1)}{\mathrm{P}(S_L = 1)} \quad \text{(by Exercise 113)} \\
&= 1.
\end{aligned}
$$

It's not surprising the sum is 1. In fact, the conditional pmf $f_{S_B|S_L}(s|1)$ *is* a pmf.

The conditional pmf given 0, i.e., $f_{S_B|S_L}(s|0)$, is a *different* pmf.

Properties of a conditional pmf

A conditional probability *is* a probability. And a conditional pmf $f_{X|Y}(x|y^*)$ *is* a pmf. What do we mean by this?

Let X and Y be discrete rvs and let $\text{supp}(X) = \{w_1, \ldots, w_k\}$. For each y^* with $f_Y(y^*) > 0$, we have that:

1. $0 \leq f_{X|Y}(x|y^*) \leq 1$ for all w_i.
2. $\sum_{i=1}^{k} f_{X|Y}(w_i|y^*) = 1$.
3. There is an associated expected value, called the conditional expected value of X given $Y = y^*$, written as: $\text{E}(X|Y = y^*)$.

Conditional distribution $\to$ conditional expectation

We now use the conditional distribution, given $S_L = 1$, to evaluate the conditional expected value, given $S_L = 1$.

$$\begin{aligned}
\mathrm{E}(\pi|S_L = 1) &= 30 \cdot \mathrm{P}(\pi = 30|S_L = 1) + 10 \cdot \mathrm{P}(\pi = 10|S_L = 1) \\
&= 30 \cdot \frac{4}{7} + 10 \cdot \frac{3}{7}. \\
&= \frac{120 + 30}{7} \\
&= \frac{150}{7} \approx 21.4.
\end{aligned}$$

Exercise 115. Recall our interpretation of E as a long-run average. How can we interpret $\mathrm{E}(\pi|S_L = 1)$ in this way? It is the long-run profit of the subset of days ω where $S_L(\omega) = 1$.

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Learning objectives

1. Understand the difference between the worlds of "probability" and "statistics".
   a. We have different information.
   b. Poverty rate example.
2. Fully understand a "random sample" by connecting to the probability framework we've built up.
   a. What is the sample space, Ω?
   b. How do we mathematize that we want a sample to be "representative"?
3. Consequences of sampling with vs. without replacement.
4. Formally define a "statistic".
5. Brief introduction to parameters and estimators.

Running example: parental income

Suppose we are interested in studying the parental income of students in a classroom of size 30.

- Let Ω be the set of all people in the classroom (so Ω has 30 elements).
- Suppose we choose someone at random from the classroom, ω.
- Let $X(\omega)$ be that person's parental income.

Probability

- Suppose we are interested in the value $\mathrm{P}(X \leq 25{,}000)$.
  - Recall 25,000 is the poverty level for a family of 4.

Exercise 116. Is $\mathrm{P}(X \leq 25{,}000)$ a number or a rv? It's a number.

Exercise 117. How do we interpret what $\mathrm{P}(X \leq 25{,}000)$ means?

There are two (equivalent) interpretations:

1. It is the probability that if we choose someone at random from the room, that person's parental income is $\leq$ 25,000€.
2. A more intuitive interpretation is that it is the *proportion* of students in this class whose parental income is $\leq$ 25,000€.

An interesting question in *probability*

Suppose we know the distribution of X. An interesting question in *probability* would be:

> *"What is* $\mathrm{P}(X \leq 25{,}000)$*? Calculate it."*

1. We just use the probability distribution of X. That's what it's for! If we had the cdf F_X of X, we just plug in 25,000 to F_X, or look it up in the cdf table.
2. Equivalently, for a finite pop, we could just count how many people in the pop have parental incomes $\leq$ 25,000€.

- The result of the calculation is a number and there is no uncertainty attached to the number.

An interesting question in *statistics*

An interesting question in *statistics* would be:

> *"What is your best* ***guess*** *for* $\mathrm{P}(X \leq 25{,}000)$*?"*

- Why do we have to guess now instead of calculate? Because in statistics, ***we don't have data on the population***.
- We don't know the poverty rate in the population.

$$\Longleftrightarrow$$

We can't calculate the probability that a random draw from the population is below the poverty rate.

Example 21. What if we assume that X is normal? Can we then calculate $\mathrm{P}(X \leq 25{,}000) = \mathrm{P}\left(\frac{X-\mathrm{E}(X)}{\sqrt{V(X)}} \leq \frac{25{,}000-\mathrm{E}(X)}{\sqrt{\mathrm{V}(X)}}\right)$?

No. $\mathrm{E}(X)$ is the mean income in *the pop*, which we can't calculate. Similarly, we don't know the variance in the pop.

In statistics, what do we have to work with?

- In statistics, we do not have data on the population (i.e., the entire class of 30 students).
- Instead, suppose we have data on parental incomes from **only 5** of the 30 students in the class. Data on a *subset* of the population is called a sample.
- Statistics in a nutshell is about the following: How can we make a good *guess* at what the poverty rate in the ***population*** is, using only the data in our ***sample***.

Estimation (poverty rate example)

Suppose we have the following observed values from a random sample (these values were in the vector of length 30 we looked at):

$$x_1 = 23{,}972$$
$$x_2 = 50{,}909$$
$$x_3 = 40{,}464$$
$$x_4 = 20{,}782$$
$$x_5 = 45{,}454$$

And remember, since we are doing statistics, we do not know the other incomes in the population (the classroom). The above sample of size 5 is all we have to work with.

Reminder of the big picture

Let's go back to the question that we considered in probability. The question was: what is $\mathrm{P}(X \leq 25{,}000)$?

- In probability, we assumed we knew the cdf of X.
- In statistics, our object of interest is still $\mathrm{P}(X \leq 25{,}000)$.
  - In this new situation, where we only observe a sample from the population, what is the distribution of X?
  - Well, it's the same as if we observe the population. But we just don't know what it is.
  - We don't know the distribution, and we don't know, for example, $\mathrm{P}(X \leq 25{,}000)$.
  - In statistics, we use the data we do have (on the sample) to make a *guess* about the population.

So what is our guess?

Recall our sample:

$$x_1 = 23{,}972$$
$$x_2 = 50{,}909$$
$$x_3 = 40{,}464$$
$$x_4 = 20{,}782$$
$$x_5 = 45{,}454$$

Exercise 118. Using the sample, what is your guess for what $\mathrm{P}(X \leq 25{,}000)$ is?

The most intuitive answer is $\frac{2}{5} = 0.40$.

- We use the proportion in the sample to make a guess about the proportion in the population.
- That's statistics! If you want to be fancy, change the word "guess" to "estimate", but this idea is the heart of statistics.

Next steps

- We guessed that the percentage of students in the classroom with parental incomes $\leq$ 25,000€ was 40% when in reality it was only 10%.
- In a later lecture, we will discuss how we can incorporate into our guess that we only had a sample of size 5 to work with.
  - When we guessed 40% we should not have been *confident* that our guess was close to the population value.
- Nonetheless, we might claim that although our guess was far from the population value, *with the information we had* (the 5 numbers), we made the best guess we could.
  - In other words, although our actual guess was bad in some sense, we can argue that our *method* of guessing was good.

Simplifications just for lecture exposition

- To get a full understanding of sampling, we will consider a sample of size 3 from a pop of size 4 in the following slides.
- Typically the pop and sample will both be much larger, but we use this simple setting to get a complete understanding of Ω.

Suppose we have a pop of size 4: Laia, Oriol, Sara, and Pablo.

The sampling experiment is choosing 3 people at random and with replacement from the pop. Each potential sample, ω, is 3 people. Here are examples of potential samples:

$\omega_1 := (Laia, Sara, Pablo)$.

$\omega_2 := (Sara, Laia, Pablo)$. Note $\omega_2 \neq \omega_1$ (order is important).

$\omega_3 := (Sara, Sara, Sara)$. Recall we sample *with* replacement.

Exercise 119. How many elements are in Ω? A: $4 \times 4 \times 4 = 4^3 = 64$.

- Suppose we obtain a sample, ω, of 3 people.
- How do we explore the properties of the sample? Also, how do we connect the sample back to the population?
- The first step is that we need to work with numbers. How can we turn people into numbers? Random variables.

Defining X_1, X_2, X_3 on Ω

Now that Ω is defined, we define X_1, X_2, X_3 on Ω:
$X_i :=$ income of the i'th person in ω.

For example, recall $\omega_1 := (\textit{Laia}, \textit{Sara}, \textit{Pablo})$. Then,

$$X_1(\omega_1) = \text{Laia's income}$$
$$X_2(\omega_1) = \text{Sara's income}$$
$$X_3(\omega_1) = \text{Pablo's income.}$$

Exercise 120. Is $X_1(\omega_1)$ a rv?

No. It's a number. There is nothing random about Laia's income.

X_1 is a rv, but once there is a fixed ω^*, $X_1(\omega^*)$ is also fixed.

e.g., define f st $f(x) = x^2$. Then, f is a function, but $f(3) = 9$ is a number.

Defining X_1, X_2, X_3 on Ω

Now that Ω is defined, we define X_1, X_2, X_3 on Ω:
$X_i :=$ income of the i'th person in ω.

We now generalize the above example and provide a new definition.

Definition 32. A random sample is a sequence of rvs $Y_1, \ldots, Y_n$, defined on a sample chosen at random and with replacement.

- It's important to understand $Y_1, \ldots, Y_n$ is ***one*** random sample.
- e.g., the X_1, X_2, X_3 above is one random sample of size 3.
- X_2, for example, is the second element of the random sample.
- A random sample is iid and we'll use "random sample" and "iid sample" interchangeably.

Representative sample

We want the sample to be representative of the population. But what does "representative" mean?

- In a random sample of 10 people from CAT, there's a <25% chance we get a perfectly representative observed sample in terms of sex (5 women, 5 men). ( ▸ How to calculate the <25%?)
- Also, we typically don't have data on the population, so we can't check if the observed sample is representative.

The solution is that instead of focusing on the observed sample, we focus on the *method* being representative.

- Any given observed sample is 75% likely to over-represent one sex and under-represent the other... but on average the random sample will produce five men and five women.
- We'll formalize this on the next slide. To make the main point more clear, we'll use a sample of size 3 instead of 10.

- Let F_1, F_2, F_3 be a random sample from CAT, where F_i is 1 if the i'th person in the sample is female, and 0 otherwise.
- Similarly, $M_i := \mathrm{I}\{i\text{'th person in } \omega \text{ is male}\}$.
- Let $T_F := \sum_{i=1}^{3} F_i$ and $T_M := \sum_{i=1}^{3} M_i$. $\mathrm{supp}(T_F) = \{0, 1, 2, 3\}$.
- Assume for simplicity $\mathrm{P}(F_i = 1) = \mathrm{P}(M_i = 1) = \frac{1}{2}$.

Exercise 121. What is $\mathrm{P}(T_F = T_M)$? Observe that $T_M = 3 - T_F$.

$$\begin{aligned} \mathrm{P}(T_F = T_M) &= \mathrm{P}(T_F = 3 - T_F) && (1)\\ &= \mathrm{P}(2 \cdot T_F = 3) && (2)\\ &= \mathrm{P}\left(T_F = \frac{3}{2}\right) = 0. && (3) \end{aligned}$$

- There is a 0% chance the sample is balanced on sex.
- Although each observed sample has more men or women, there's no systematic oversampling of men or women across samples: $\mathrm{E}(T_F) = \mathrm{E}(T_M) = \frac{3}{2}$.
- This is what the "on average" means in the previous slide.

Consequences of samping without replacement

- Suppose we have the same pop: Laia, Oriol, Sara, and Pablo.
- Laia's income is 20,000, Oriol's is 30,000, Sara's is 40,000, and Pablo's is 50,000.
- Let Z_1, Z_2, Z_3 be a sample of incomes ***without*** replacement from this population.

Exercise 122. Is Ω the same as it was with replacement?

No. e.g., $(\textit{Sara}, \textit{Sara}, \textit{Sara}) \notin \Omega$.

Example 22. Are Z_1, Z_2, Z_3 iid? $\mathrm{P}(Z_2 = 40{,}000) = \frac{1}{4}$.

$$\mathrm{P}(Z_2 = 40{,}000 | Z_1 = 40{,}000) = 0 \neq \frac{1}{4} = \mathrm{P}(Z_2 = 40{,}000).$$

So $Z_1 \not\perp\!\!\!\perp Z_2$. Thus, Z_1, Z_2, Z_3 are not iid.

Why is iid so important?

If the X_i are iid, then:

1. $\mathrm{V}(\bar{X}) = \frac{\mathrm{V}(X_i)}{n}$. Otherwise, there are a lot of hard-to-work-with covariance terms. Generally (whether X_i are iid or not),

$$\mathrm{V}(\bar{X}) = \frac{1}{n^2}\left(\sum_{i=1}^{n}\mathrm{V}(X_i) + 2\sum_{1\leq i<j\leq n}\mathrm{cov}(X_i, X_j)\right).$$

2. The Law of Large Numbers applies.
3. The Central Limit Theorem applies.

Big picture: Sampling without replacement $\Longrightarrow$ the X_i are not iid $\Longrightarrow$ the above convenient properties do not trivially hold.

Sampling with replacement is silly, but we assume it anyway

- It would be silly and inefficient to collect potentially duplicate data by sampling with replacement.
  - Thus, *in practice*, samples are are collected without replacement.
- Since the math is cleaner, we typically *assume* the sample was collected with replacement.
  - Even though this assumption is usually wrong, it makes almost no difference unless the sample is very small relative to the population.

Definition of a statistic

In statistics, ω is a sample.

- A statistic is something that (potentially) changes when the sample changes. i.e., when ω changes.

Definition 33 (statistic). A statistic is a function of (only) the sample, ω.

Exercise 123. Is a statistic a rv? Yes. It is a function of ω.

Function of a random sample

- Let $X_1, \ldots, X_n$ be an iid sample. Let T be a statistic.
- Then by definition, T is a function of $X_1, \ldots, X_n$.
- What does "is a function of" mean?
- Informally: for a fixed ω^*, the number $T(\omega^*)$ can be calculated using only the numbers $X_1(\omega), \ldots, X_n(\omega)$.
- Example: $\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{n}(X_1 + \cdots + X_n)$ is a statistic.
- Example: we defined $T_F := \sum_{i=1}^{3} F_i$ which represents the total number of women in the sample.
  - If we take a new sample, ω, $F_i(\omega)$ may change and thus $T_F(\omega)$ may change. So T_F is a rv.

Parameter

Definition 34. Let $X_1, \ldots, X_n$ be a random sample. A parameter (of a population) is a function of the *distribution* of X_i.

- A parameter is a function of the population data.
- What does "is a function of" mean above? It means if we have data on the population then we can calculate the parameter.
- A parameter is something we *would* like to know.
- But we don't know it, because we don't have data on the entire population.
- We have data on a sample.
- Thus, a parameter is one number, and it exists, but we don't know what it is.
  - Just like we don't know the exact day of Socrates's death.
  - Or which (two) days Mount Vesuvius destroyed Pompeii.
  - The number of tritons del Montseny at this point in time.

Examples of parameters

Suppose the pop of interest is Catalunya.

Example 23. Give 4 examples of parameters for this population.

1. The mean income of CAT.
2. The median income of CAT.
3. The maximum income in CAT.
4. The variance of incomes in CAT. Why is this parameter interesting?
    a. Recall: variance is a measure of income inequality.

Example 24. Give an ex. of a parameter that involves *two* variables.

1. The covariance of income and years of education in CAT.
2. The *correlation* of income and years of education in CAT.

Average income in the classroom

- Let $X_1, \ldots, X_5$ be a random sample of incomes from this classroom of size 30.
- Is $\mathrm{E}(X_3)$ a statistic or a parameter? A parameter.
  - Note that the X_i are iid so $\mathrm{E}(X_3) = \mathrm{E}(X_1)$, etc.
  - The X_i all have the same distribution. That is, the population is the distribution.
  - Thus, it should make sense that $\mathrm{E}(X_3)$ is the mean of the distribution, and thus the mean of the population.

$\mathrm{E}(X_3)$ can be calculated from the pop

- If $\mathrm{E}(X_3)$ is a parameter, by definition (of a parameter) we must be able to calculate it from the population.
- We will now evaluate $\mathrm{E}(X_3)$ using the distribution of X_3 and show explicitly that this equals the mean of the population.
- Suppose that the classroom incomes are $k_1, \ldots, k_{30}$, and for simplicity that all incomes are distinct.

Exercise 124. What's the mean income in the classroom in terms of the k_i? $\bar{k} = \frac{1}{30} \sum_{i=1}^{30} k_i$.

- So then the question is: does $\mathrm{E}(X_3) = \frac{1}{30} \sum_{i=1}^{30} k_i$?

Showing that $\mathrm{E}(X_3) = \frac{1}{30}\sum_{i=1}^{30} k_i$

Recall $k_1, \ldots, k_{30}$ are the distinct incomes in the classroom (= pop).

- $\mathrm{supp}(X_3) = \{k_1, \ldots, k_{30}\}$. By def. of expectation:

$$\begin{aligned}
\mathrm{E}(X_3) &= k_1 \cdot \mathrm{P}(X_3 = k_1) + \cdots + k_{30} \cdot \mathrm{P}(X_3 = k_{30}) && (1)\\
&= k_1 \cdot \frac{1}{30} + \cdots + k_{30} \cdot \frac{1}{30} \qquad \text{(by iid)} && (2)\\
&= \sum_{i=1}^{30} k_i \frac{1}{30} && (3)\\
&= \frac{1}{30}\sum_{i=1}^{30} k_i. && (4)
\end{aligned}$$

So $\mathrm{E}(X_3) = \frac{1}{30}\sum_{i=1}^{30} k_i$, the mean of the classroom (= *population*). Similarly, $\mathrm{V}(X_3)$ is $\frac{1}{30}\sum_{i=1}^{30}\left[\left(k_i - \bar{k}\right)^2\right]$, the var in the classroom (= *population*). ▸ See proof

Estimators

Definition 35 (estimator). An estimator is a statistic whose purpose is to make a guess about a specific parameter.

Exercise 125. Is an estimator a function of only the sample? Yes, because an estimator is a statistic.

Exercise 126. Is an estimator a rv? Yes, because the random sample is a collection of rvs (so any function of the rvs will also be a rv).

Example 25. Recall in a sample of size n, we defined $T_F := \sum_{i=1}^{n} F_i$ to be the total number of women in the sample. T_F is a statistic.

- It's not an estimator since it does not directly make a guess about a param. (e.g., the number of women in the pop)

Example 26. $\bar{T} := \frac{T_F}{n}$ is an estimator because the proportion of women in the sample can be used to make a guess about the proportion of women in the population.

The poverty rate

Let $X_1, \ldots, X_5$ be an iid sample from the pop of CAT incomes.

Let α be the proportion of incomes in CAT that are $\leq$ 25,000.

Exercise 127. Is α a parameter or a statistic? It's a parameter, since we need data on the entire population in order to calculate it.

Exercise 128. Is $\mathrm{P}[X_3 \leq 25000]$ a parameter or a statistic?

- Intuitively, what does $\mathrm{P}[X_3 \leq 25000]$ represent?
- It's the prob that the 3rd element of the sample is $\leq$ 25000.
- Suppose $\alpha = 0.10$. If 10% of the pop is poor, and we choose someone at random from the pop, what's the prob that person is poor? It's 0.10.
- Thus, $\mathrm{P}[X_3 \leq 25000] = \alpha$, which we agreed is a parameter.

Why did we choose X_3 in previous exercise?

- We just showed that $\mathrm{P}[X_3 \leq 25000] = \alpha$.
- Why did I choose X_3? Specifically, what is $\mathrm{P}[X_2 \leq 25000]$ equal to?
- $\mathrm{P}[X_2 \leq 25000] = \alpha$ also. Why?
- Because $X_1, \ldots, X_5$ are iid, so all X_i have the same marginal distribution. In fact, the distribution is the population.
- If they have the same distribution, then they have the same expected value, variance, etc.
- In this case, we often write $\mathrm{P}[X_i \leq 25000]$.

Let $X_1, \ldots X_5$ be an iid sample. Imagine a table where each row is an observed sample. Suppose we have N observed samples of size 5.

ω	$X_1(\omega)$	$X_2(\omega)$	$X_3(\omega)$	$X_4(\omega)$	$X_5(\omega)$	$X_6(\omega)$	$X_7(\omega)$
ω_1							
ω_2							
ω_2							
ω_2							
$\vdots$							
ω_N							

A related (more simple) problem (1/2)

- In order to understand that $\mathrm{P}[X_3 \leq 25000]$ is a parameter, let's come back to a more simple context.
- Let V_1, V_2, V_3 be an iid sample of coin flips from a fair coin, where $V_i := \mathrm{I}\{\text{i'th flip is heads}\}$.
- What is $\mathrm{P}(V_2 = 1)$? $\frac{1}{2}$.
- We collect a sample and happen to get $\omega^* := (T, H, H)$.

$$V_1(\omega^*) = 0,\ V_2(\omega^*) = 1,\ \ V_3(\omega^*) = 1.$$

Exercise 129. What is $\mathrm{P}(V_2 = 1)$? $\frac{1}{2}$.

- Recall how we *defined* P. What does $\mathrm{P}(V_2 = 1)$ mean? If we run the experiment an infinite amount of times, it's the proportion of times the event $V_2 = 1$ happens.
- Recall the experiment underlying V_2... It's not taking a sample from a sample. It's flipping the coin three (new) times.
- $V_2(\omega^*) = 1$, $\mathrm{P}(V_2(\omega^*) = 1) = \mathrm{P}(1 = 1) = 1$, $\mathrm{P}(V_2 = 1) = \frac{1}{2}$.

A related (more simple) problem (2/2)

- To know that $\mathrm{P}(V_2 = 1)$ is $\frac{1}{2}$ we need to know the "population", which in this case means knowing that it's a fair coin (i.e., the distribution of V_2).
- We can't say that $\mathrm{P}(V_2 = 1)$ is $\frac{1}{2}$ based only on the sample, so $\mathrm{P}(V_2 = 1)$ can't be a statistic.
- It seems counterintuitive, since we see an *outcome* of X_3. But the *probability* that $X_3 \leq 25000$ can't be calculated from the sample.
- We only observe $X_3(\omega^*)$, which is just a number. We don't know the *distribution* of X_3 from just the sample, because the distribution of X_3 is the population.

Where does the 25% come from?

Back to lecture

We first reformulate the problem: Let X be the number of heads from 10 fair coin flips. What is $\mathrm{P}(X = 5)$?

X has a binomial distribution with parameters $n = 10$ and $p = 0.5$.

We use the pmf of a binomial distribution to then get:

$$\mathrm{P}(X = 5) = f_X(5; n = 10, p = 0.5) = \binom{10}{5} 0.5^5 (1 - 0.5)^{10-5}$$

$$= \binom{10}{5} 0.5^{10} = \frac{(10)!}{(10-5)! \cdot 5!} \left(\frac{1}{2}\right)^{10} = \frac{63}{2^8} \approx 0.246.$$

This can also be calculated with R:

```
> dbinom(x = 5, size = 10, prob = 0.5)

[1] 0.2460938
```

$\mathrm{V}(X_3)$ is var in the *pop* Back to lecture

$$
\begin{aligned}
\mathrm{V}(X_3) &= \mathrm{E}\left[(X_3 - \mathrm{E}(X_3))^2\right] && (1)\\
&= \mathrm{E}\left[(X_3 - \bar{k})^2\right] && (2)\\
&= \sum_{i=1}^{30}\left[(k_i - \bar{k})^2\right]\mathrm{P}(X_3 = k_i) && (3)\\
&= \sum_{i=1}^{30}\left[(k_i - \bar{k})^2\right]\frac{1}{30} && (4)\\
&= \frac{1}{30}\sum_{i=1}^{30}\left[(k_i - \bar{k})^2\right], && (5)
\end{aligned}
$$

which is the formula for var in the pop.

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7. Expected value and variance

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Learning objectives

1. More exercises to understand the difference between a parameter and an estimator.
2. Review: connect expected value to population mean.
3. Learn the basics about what makes an estimator "good".
4. Learn the difference between an "estimator" and an "estimate".
5. Walk through 3 examples about the bias of estimators.
6. Use simulation to get an understanding of the properties of estimators.
    a. We will simulate estimating the unemployment rate.

Suppose we have an iid sample of incomes $X_1, \ldots, X_5$.

- Can someone remind us, in words, how can we estimate the poverty rate parameter, $\alpha := \mathrm{P}[X_3 \leq 25000]$?
- We would use "the proportion of the ***sample*** that is ≤ 25000".

Example 27. How do we define our estimator using math?
$Z_i := \mathrm{I}\{X_i \leq 25000\}$.

Then, our estimator of α is $\bar{Z}$. Recall $\bar{Z} := \frac{1}{n}\sum_{i=1}^{n} Z_i$.

We could even skip defining Z_i and directly define our estimator:

$$\frac{1}{n}\sum_{i=1}^{n} \mathrm{I}\{X_i \leq 25000\}.$$

Applying our estimator, $\bar{Z}$, to a sample

Recall $Z_i := \mathrm{I}\{X_i \leq 25000\}$, $\bar{Z} := \frac{1}{n}\sum_{i=1}^{n} Z_i$.

Suppose we obtain a sample, ω^*, of size 5, and the values we observe are the following:

$$
\begin{aligned}
x_1 &:= X_1(\omega^*) = 23,972 & \qquad z_1 &:= Z_1(\omega^*) = 1 \\
x_2 &:= X_2(\omega^*) = 50,909 & \qquad z_2 &:= Z_2(\omega^*) = 0 \\
x_3 &:= X_3(\omega^*) = 40,464 & \qquad z_3 &:= Z_3(\omega^*) = 0 \\
x_4 &:= X_4(\omega^*) = 20,782 & \qquad z_4 &:= Z_4(\omega^*) = 1 \\
x_5 &:= X_5(\omega^*) = 45,454 & \qquad z_5 &:= Z_5(\omega^*) = 0
\end{aligned}
$$

$$\bar{z} := \bar{Z}(\omega^*) = \frac{2}{5}$$

It's standard to use lowercase letters to record observed values.

Exercise 130. Is $\bar{Z}$ a rv? Yes. Consider the function $f(x) = x^2$. f is a function (not a number), but $f(7)$ is a number. Similarly, $\bar{Z}$ is a rv and $\bar{Z}(\omega^*)$ is a number.

Estimator vs. estimate

Definition 36 (estimate). An estimate is the outcome of an estimator for one *specific* sample.

- The estimator is our *method* of guessing.
- The estimate is our (numeric) *guess* for a specific sample.
- In the poverty example we just did, $\bar{Z}$ is our estimator, and $\bar{z} = \bar{Z}(\omega^*) = \frac{2}{5}$ is our estimate using the sample we observed.

Exercise 131. Is an estimator a rv?

- Yes. An estimator is a rv because it's a *function* of the (random) sample. If you draw a new sample, you get a new potential outcome, ω, and thus $\bar{Z}(\omega)$ will be different.
- An estimate is a number, because it is for a specific potential outcome ω^* (i.e., a specific sample).

Exercise 132. Let's continue our running example. $X_1, \ldots, X_5$ are a random sample of incomes, and $Z_i := \mathrm{I}\{X_i \leq 25{,}000\}$. What about $\mathrm{E}(Z_3)$? Is it a parameter or a statistic?

$\mathrm{supp}(Z_3) = \{0, 1\}$. Then,

$$\begin{aligned} \mathrm{E}(Z_3) &= 0 \cdot \mathrm{P}(Z_3 = 0) + 1 \cdot \mathrm{P}(Z_3 = 1) && (1) \\ &= \mathrm{P}(Z_3 = 1) && (2) \\ &= \mathrm{P}(X_3 \leq 25000) && (3) \\ &= \alpha. && (4) \end{aligned}$$

So $\mathrm{E}(Z_3)$ is a parameter.

- Similarly, recall that expected value gives a *number*.
- A statistic would be a rv.

Exercise 133. What's our estimator of $\mathrm{E}(Z_3)$? It's $\bar{Z}$.

Estimator for pop var. (1/2)

- Let $X_1, \ldots, X_n$ be a random sample.

Exercise 134. $\bar{X}$ is a natural estimator for $\mu_X := \mathrm{E}(X_i)$. Why?
The LLN tells us that $\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mathrm{E}(X_i)$.

Example. What is a natural estimator for $\mathrm{V}(X_i)$?

- First let's define $Z_i := (X_i - \mu_X)^2$. Then, the Z_i are iid because the X_i are iid.
- Then, by the LLN we know that $\frac{1}{n}\sum_{i=1}^{n} Z_i \xrightarrow{p} \mathrm{E}(Z_i)$. And $\mathrm{E}(Z_i) = \mathrm{E}\left[(X_i - \mu_X)^2\right] = \mathrm{V}(X_i)$.

Exercise 135. Why was it important to argue that the Z_i are iid?
It's an assumption needed to apply the LLN.

Exercise 136. Is $\frac{1}{n}\sum_{i=1}^{n} (X_i - \mu_X)^2$ thus an estimator of $\mathrm{V}(X_i)$?

- No, an estimator must depend on only the sample, and μ_X depends on the population.

Estimator for pop var. (2/2)

- Recall that $\frac{1}{n}\sum_{i=1}^{n} Z_i \xrightarrow{p} \mathrm{E}\left[(X_i - \mu_X)^2\right] = \mathrm{V}(X)$.
- But μ_X is a *parameter*, so we can't have that in the estimator we are constructing.
- We apply the plug-in principle of statistics: when needed, replace parameters by their sample analogs:
  - Define $\hat{\sigma}^2 := \frac{1}{n}\sum_{i=1}^{n}\left[\left(X_i - \bar{X}\right)^2\right]$.
- Note that $Q_i := \left(X_i - \bar{X}\right)^2$ are not iid so the LLN does not directly apply.
- However, it can still be shown that $\hat{\sigma}^2 \xrightarrow{p} \mathrm{V}(X_i)$.
  - We won't prove this in class, but you can take this result as given.

"good" estimators

Let $X_1, \ldots, X_5$ be an iid sample of incomes from this classroom of size 30.

- What is our estimator of $\mathrm{E}(X_i)$? It is $\bar{X}$.
- But is $\bar{X}$ a "good" estimator?

Exercise 137. Is an estimator "good" only if it equals the param.? No.

- First of all, $\bar{X}$ is a rv and $\mathrm{E}(X_i)$ is a constant. So it is strange to talk about a rv equaling a constant.
- Second, it is not realistic to expect to ***exactly*** calculate the parameter (a feature of the population) from only a sample.

Video discussion

- Why can't we just talk about the estimates? e.g., the $\frac{2}{5}$?
- Why would we want to talk about the estimator, $\bar{Z}$?

We'll now watch a video that will drive in the point of why we need to talk about both the estimator and the estimate.

Is this a "good" shot?

[Show the video of Damian Lillard's 2019 "bad" shot].

Discussion: is it a good shot?

- Paul George claims it was a "bad shot". He is clearly talking about the "estimator" interpretation of a "good shot".
- Damian Lillard claims it was a "good shot". He could be talking about either the estimator or estimate.
- The good shot estimator is about whether Lillard was likely to make that shot at the moment he took it.
  - i.e., it is about the *distribution* of the shot over many shots in the same conditions.
    - if he took that shot 1,000 times in those ***same conditions***, does he make it 60% of the time?
  - In non-technical terms, was it good shot *selection*?
- The good shot estimate is simply about whether it went in or not.

Good estimators versus good estimates

- As opposed to basketball, in econometrics we never know whether an estimate is good.
- To know that, we would need to know the parameter, which we never observe.
- However, we *can* know whether an estimator is good, because the concept of "good" does not require observing the parameter.
- Rather, it shows that certain properties of the distribution of an estimator hold ***no matter what the actual parameter is***.
  - Consistency is one property of the distribution of estimators.
  - Now we will see another.
- Remember: estimators have distributions. Estimates are numbers.

Consistency

In the following definitions, let $\hat{\gamma}$ be an estimator of a parameter γ.

Definition 37. $\hat{\gamma}$ is a consistent estimator of γ if $\hat{\gamma} \xrightarrow{p} \gamma$.

Exercise 138. What examples have we seen so far?

1. $\hat{\gamma} = \bar{X}$, $\gamma = \mathrm{E}(X_i)$.
   - i.e., $\bar{X}$ is a consistent estimator of $\mathrm{E}(X_i)$.
2. $\hat{\gamma} = \hat{\sigma}^2$, $\gamma = \mathrm{V}(X_i)$.
   - $\hat{\sigma}^2 := \frac{1}{n}\sum_{i=1}^{n}\left[\left(X_i - \bar{X}\right)^2\right]$ is a consistent estimator of $\mathrm{V}(X_i)$.

Consistency is about the probability of an estimator being close to the parameter.

- That is, it is about the *distribution* of an estimator as the sample size increases.

Unbiasedness

Example 28. What other property should we require for an estimator to be "good"?

- That $\bar{X}$ does not *systematically* give estimates above $\mathrm{E}(X_i)$ or systematically below.
- In other words, we want $\bar{X}$ to be correct (equal the param.) *on average*.

Exercise 139. What does "on average" mean? An average over *what*?

An average over many samples.

Exercise 140. How do we state that in math? $\mathrm{E}(\bar{X}) = \mathrm{E}(X_i)$.

Bias of estimators

In the following definitions, let $\hat{\gamma}$ be an estimator of a parameter γ.
Let wrt := "with respect to".

Definition 38. The bias of $\hat{\gamma}$ (wrt γ) is: $\text{bias}(\hat{\gamma}) := \text{E}(\hat{\gamma}) - \gamma$.

Definition 39. $\hat{\gamma}$ is unbiased (wrt γ) if $\text{bias}(\hat{\gamma}) = 0$.
i.e., if $\text{E}(\hat{\gamma}) = \gamma$.

Definition 40. $\hat{\gamma}$ is biased downwards (wrt γ) if the bias of $\hat{\gamma}$ is negative, i.e.; $\text{E}(\hat{\gamma}) < \gamma$.

Definition 41. $\hat{\gamma}$ is biased upwards (wrt γ) if the bias of $\hat{\gamma}$ is positive, i.e.; $\text{E}(\hat{\gamma}) > \gamma$.

Again, we see that this property of unbiasedness is about the *distribution* of $\hat{\gamma}$.

but... we only view one sample

- It's important to understand we only observe one sample.
  - And we only have one observation of $\bar{X}$, e.g., $\bar{x}_1$.
  - And our $\bar{x}_1$ is likely not going to equal $\mathrm{E}(X_i)$.
    - Recall when we estimated the poverty rate, the parameter was 0.10 and our estimate was 0.40.
  - But we want to know that our *method* of estimation is not *inherently* biased.
- It's like shooting a basketball.
  - If 80% of your misses are to the left of the basket, there is something inherently biased about your shot technique.
  - You don't have to make the shot every time. But you should miss $\approx 50\%$ to the left, and $\approx 50\%$ to the right.

Example 1: $\bar{X}$ from an iid sample

- Recall $X_1, \ldots, X_5$ are an iid sample of incomes from a classroom of size 30. An estimator of $\mathrm{E}(X_i)$ is $\bar{X}$.

Example 29. Is $\bar{X}$ an *unbiased* estimator of $\mu_X := \mathrm{E}(X_i)$?

$$\mathrm{E}(\bar{X}) = \mathrm{E}\left(\frac{1}{5}\sum_{i=1}^{5} X_i\right) = \frac{1}{5}\sum_{i=1}^{5} \mathrm{E}(X_i) \tag{1}$$

$$= \frac{1}{5}\sum_{i=1}^{5} \mu_X \qquad \text{(by identically distributed)} \tag{2}$$

$$= \frac{1}{5} 5 \cdot \mu_X \qquad \left(\text{by } \sum_{i=1}^{n} a = na\right) \tag{3}$$

$$= \mu_X, \text{ so yes } \bar{X} \text{ is unbiased.} \tag{4}$$

- It's amazing we can prove that $\mathrm{E}(\bar{X}) = \mu_X$ even though we don't know what μ_X is!

Example 2: a sample of volunteers

- Suppose we are interested in the mean income in a classroom of size 30 and let μ be the mean income of the classroom.
- Let $W_1, \ldots, W_8$ be a random sample from the 20 people in the classroom who raise their hands to participate.

Exercise 141. Is $\bar{W}$ an unbiased estimator?

- The same steps in the previous exercise apply to show that $\mathrm{E}(\bar{W}) = \mathrm{E}(W_i)$. That is, $\bar{W}$ is an unbiased estimator of $\mathrm{E}(W_i)$.

Exercise 142. How can we interpret $\mathrm{E}(W_i)$? It is the mean income of the ***volunteers***.

- If those with higher income are more likely to volunteer, then $\mathrm{E}(W_i) > \mu$, so $\bar{W}$ is biased upwards as an estimator of the mean income in the classroom, μ.

Example 3: estimating variance (1/3)

- Let $X_1, \ldots, X_n$ be an iid sample of incomes.
- Recall the intuitive estimator (inspired by the plug-in principle) of $\sigma^2 := \mathrm{V}(X_i)$:

$$\hat{\sigma}^2 := \frac{1}{n} \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2.$$

- We won't prove the following because it involves a lot of steps, but $\mathrm{E}\left(\hat{\sigma}^2\right) = \frac{n-1}{n}\sigma^2 \neq \sigma^2$. So $\hat{\sigma}^2$ is a *biased* estimator of σ^2.
- To get unbiasedness, we need to make the $\frac{n-1}{n}$ disappear.
- We define $S^2 := \frac{n}{n-1}\hat{\sigma}^2$.

Example 3: estimating variance (2/3)

Recall that $\mathrm{E}\left(\hat{\sigma}^2\right) = \frac{n-1}{n}\sigma^2$ and $S^2 := \frac{n}{n-1}\hat{\sigma}^2$.

Exercise 143. Verify that S^2 is indeed unbiased.

$$\begin{aligned}
\mathrm{E}\left(S^2\right) &= \mathrm{E}\left(\frac{n}{n-1}\hat{\sigma}^2\right) && \text{(by def. of } S^2\text{)} && (1)\\
&= \frac{n}{n-1}\mathrm{E}\left(\hat{\sigma}^2\right) && && (2)\\
&= \frac{n}{n-1}\left(\frac{n-1}{n}\sigma^2\right) && && (3)\\
&= \sigma^2. && && (4)
\end{aligned}$$

Example 3: estimating variance (3/3)

- Recall:

$$\hat{\sigma}^2 := \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

Let's express S^2 directly in terms of the X_i:

$$S^2 := \frac{n}{n-1} \hat{\sigma}^2 \tag{1}$$

$$= \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})^2 \right) \tag{2}$$

$$= \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2 . \tag{3}$$

Sources of bias, and corrections

We've seen that estimators can be biased from two sources:

1. The sampling process (e.g., if volunteers as in Example 2).
   - Typically we cannot "correct" bias from sampling to get an unbiased estimator.
2. The estimator itself (e.g., $\hat{\sigma}^2$ as in Example 3).
   - For $\hat{\sigma}^2$ we could correct the bias since $\mathrm{E}\left(\hat{\sigma}^2\right) = \frac{n-1}{n}\sigma^2$, and we know what n is so we could just cancel the $\frac{n-1}{n}$.
   - But that's not always the case. Imagine an estimator $\tilde{\sigma}^2$ such that $\mathrm{E}\left(\tilde{\sigma}^2\right) = \frac{\mu-1}{\mu}\sigma^2$. Since we do not observe μ, it's not easy to remove the bias.

Consistency vs. unbiasedness

- Consistency and unbiasedenss are two properties of the distributions of estimators that capture different things.
- Consistency is about the behavior of the distribution of an estimator as n gets larger.
  - That is, it is an asymptotic property.
- Unbiasedness is a property of the distribution that holds for any n.

Efficiency

- Recall that unbiasedness is about one feature of the distribution of $\hat{\gamma}$, in particular $\mathrm{E}(\hat{\gamma})$.
- We now look at a property about a different feature of the distribution of $\hat{\gamma}$.

Definition 42. Let $\hat{\gamma}_1$ and $\hat{\gamma}_2$ be two estimators of some parameter γ. We say that $\hat{\gamma}_1$ is a more efficient estimator (wrt $\hat{\gamma}_2$) if $\mathrm{V}(\hat{\gamma}_1) < \mathrm{V}(\hat{\gamma}_2)$.

- In general, we are often interested in the Mean Squared Error (MSE) of estimators: $\mathrm{E}\left[(\hat{\gamma} - \gamma)^2\right]$ which is affected by both $\mathrm{E}(\hat{\gamma})$ and $\mathrm{V}(\hat{\gamma})$.

$MSE(\hat{\gamma}) = \mathrm{bias}(\hat{\gamma})^2 + \mathrm{V}(\hat{\gamma})$.

Simulation of unbiasedness

We will do a simulation in R.

A few essentials from recent lectures

- Difference between an estimator and an estimate.
- Make sure you understand that $\bar{Z}$ is a rv and $\bar{Z}(\omega^*)$ is a number.
- Why it's important to talk about an estimator, not just an estimate.
- Study this lecture until you know cold what is an estimator and what is a parameter.
- Make sure you know *why* $\mathrm{E}(X_3)$ and $\mathrm{P}(X_3 \leq 25000)$ are parameters, and that we need the population to be able to calculate them.

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Learning objectives

1. Understand the concept of a random interval.
    a. Just like an estimator is a random variable, a confidence interval is a random *interval*.
2. Motivate why confidence intervals solve a problem.
3. Understand how to interpret a confidence interval.
    - What exactly does it mean to be "95% sure"?

Review of probability and being 95% sure

Suppose you and your bestie roommate just left the house, jumped in the car, and left for a camping trip.

Your roommate then asks: "Did you remember to lock the door of the apartment?". You respond: "I'm 95% sure I did."

Exercise 144. Your anxious friend asks: "What does 95% sure" mean?

1. One possibility: Out of all the times in your life you have said you're 95% sure, you've been right 95% of those times.
2. Alternatively: Out of all the times in the past with the *same set of* circumstances (you have the key in your pocket, you remembered to take your hat, etc.), you indeed locked the door of the apartment 95% of those times.

Motivating example

Example 30 (Idescat boss). Suppose you work for the Institut d'Estadística de Catalunya (Idescat), and your boss told you to estimate the average gross income in CAT. Suppose you have a random sample of incomes, $X_1, \ldots, X_n$. Let's denote the average income (in 1,000€'s) in CAT as μ_X.

- To estimate μ_X, you use the estimator $\bar{X}$, and suppose that for your sample ω^*, $\bar{X}(\omega^*)$ is 29.591.
- Your boss then asks: how *confident* are you in your estimate?
  - How can you answer your boss's question?

Attempt to define "confidence"

Recall: $\bar{X}(\omega^*) = 29.591$.

It ***would be*** nice if we could tell her something like the following:

$$\mathrm{P}(\mu_X = 29.591) = 0.8.$$

Exercise 145. Can we define "confidence" in the above sense, i.e., to mean "I'm 80% sure that the true parameter is 29.591" ?

- No. μ_X is a constant, so $\mathrm{P}(\mu_X = 29.591) \in \{0, 1\}$.
- Recall, for example: $\mathrm{P}(20 = 23) = 0$, $\mathrm{P}(23 = 23) = 1$.

Next idea to try

What if we use our estimator? It would be nice to say the following:

$$\mathrm{P}(\bar{X} = \mu_X) = 0.8,$$

which would mean that if we take a sample, there is an 80% chance that our estimate equals the parameter.

Exercise 146. What can we say about $\mathrm{P}(\bar{X} = \mu_X)$?

- If $\bar{X}$ is normally distributed, then $\mathrm{P}(\bar{X} = \mu_X) = 0$.
- If $\bar{X}$ is discrete, then $\mathrm{P}(\bar{X} = \mu_X) \approx 0$.
  - Recall when we estimated the unemployment rate with a random sample of size 3: $\mathrm{supp}(\bar{X}) = \left\{0, \frac{1}{3}, \frac{2}{3}, 1\right\}$ so unless $\mathrm{E}(X_i)$ happens to be in the support, the prob is 0.
  - Analogy: if we flip a fair coin 100 million times, the prob of exactly 50 million heads is ≈ 0.

Let's take a step back

Our boss asked us how confident we are about our estimate. What do we think she really wants to know?

- She's probably not asking us what the chance is our estimate *equals* the population value *exactly*.
- Rather, she just wants to have an idea of whether we are confident our estimate is *close* to the population value.
- We want to say that $\bar{X}$ is "close" to μ_X with high probability.
  - The tool that gives this information is a ***confidence interval***.
- For example, suppose we could find a value h such that

$$\mathrm{P}(\bar{X} - h \leq \mu_X \leq \bar{X} + h) = 0.8.$$

i.e., $\bar{X}$ is close (i.e., within h) to μ_X 80% of the time.

- How do we interpret the P above? It is the long-run proportion of times both inequalities are true over many samples.

CI vs ci (just like estimator vs. estimate)

- Just like the sample mean $\bar{X}(\omega)$ is different for each sample, ω, the outcome of the CI is also different for each sample.
- For *each* sample, ω, there is an observed confidence interval. i.e., the CI is a function of ω and is thus a *random* interval.
- We use a lowercase *ci* for the *observed CI* of a sample, ω^*:

$$ci := CI(\omega^*) = \left(\bar{X}(\omega^*) - h,\ \bar{X}(\omega^*) + h\right) = (\bar{x} - h,\ \bar{x} + h).$$

The "coverage event"

- Consider the following event: $\mu_X \in (\bar{X} - h,\ \bar{X} + h)$.
- If the event happens, we say the ci covers the parameter.
- e.g., if $\mu_X = 40$, then...
  - The ci $(20, 50)$ covers μ_X because $40 \in (20, 50)$.
  - The ci $(60, 90)$ does not cover μ_X because $40 \notin (60, 90)$.
- The event $\mu_X \in (\bar{X} - h, \bar{X} + h)$ is equivalent to the event $\bar{X} - h \leq \mu_X \leq \bar{X} + h$.
  - e.g., above, $40 \in (20, 50)$ is equivalent to $20 < 40 < 50$.
- The following 2 statements are equivalent

$$\mathrm{P}\left[\mu_X \in (\bar{X} - h, \bar{X} + h)\right] = 0.8 \tag{1}$$

$$\mathrm{P}(\bar{X} - h \leq \mu_X \leq \bar{X} + h) = 0.8. \tag{2}$$

Pretend exercise

Example 31 (Pretend exercise). Suppose for simplicity that:

1. The sample size, n, is 4.
2. $h(0.8) = 10$. i.e., $\mathrm{P}(\bar{X} - 10 \leq \mu_X \leq \bar{X} + 10) = 0.8$.
   - Then, $CI = (\bar{X} - 10, \bar{X} + 10)$, or for a specific sample, ω^*, $ci = CI(\omega^*) = (\bar{x} - 10, \bar{x} + 10)$.
   - Note that to *construct* the CI, we do not need to know μ_X. Thus, it is a valid (interval) estimator.
3. We somehow know that $\mu_X = 40$.
4. We have 3 observed samples, ω_a, ω_b, ω_c, each of size 4.

Practice with *CI* and *ci*

First, let's just look at the first two samples and calculate the observed statistics:

	sample 1	sample 2
x_1	67	26
x_2	30	44
x_3	52	35
x_4	48	28
$\bar{x}$	$\bar{x}_1 = \bar{X}(\omega_1) = 49.25$	$\bar{x}_2 = \bar{X}(\omega_2) = 33.25$
80% *ci*	$ci_1 = CI(\omega_1) = (39.25, 59.25)$	$ci_2 = CI(\omega_2) = (23.25, 43.25)$

For example:

$$\begin{aligned} ci_2 = CI(\omega_2) &= (\bar{X}(\omega_2) - 10, \bar{X}(\omega_2) + 10) \\ &= (33.25 - 10, 33.25 + 10) \\ &= (23.25, 43.25). \end{aligned}$$

Exercise 147. Fill in the 80% *ci* row, and the $\mathrm{I}[\mu_X \in ci]$ row. Recall: units are 1,000€'s and $h(0.8) = 10$.

	sample 1	sample 2	sample 3
x_1	67	26	34
x_2	30	44	40
x_3	52	35	65
x_4	48	28	80
$\bar{x}$	49.25	33.25	54.75
80% *ci*	(39.25, 59.25)	(23.25, 43.25)	(44.75, 64.75)
μ_X	40	40	40
$\mathrm{I}[\mu_X \in ci]$	1	1	0

Exercise 147. Fill in the 80% *ci* row, and the $\mathrm{I}[\mu_X \in ci]$ row. Recall: units are 1,000€'s and $h(0.8) = 10$.

	sample 1	sample 2	sample 3
x_1	67	?	?
x_2	30	?	?
x_3	52	?	?
x_4	48	?	?
$\bar{x}$	49.25	?	?
80% *ci*	(39.25, 59.25)	?	?
μ_X	?	??	??
$\mathrm{I}[\mu_X \in ci]$	?	??	??

In practice, things are not as simple as the above because:

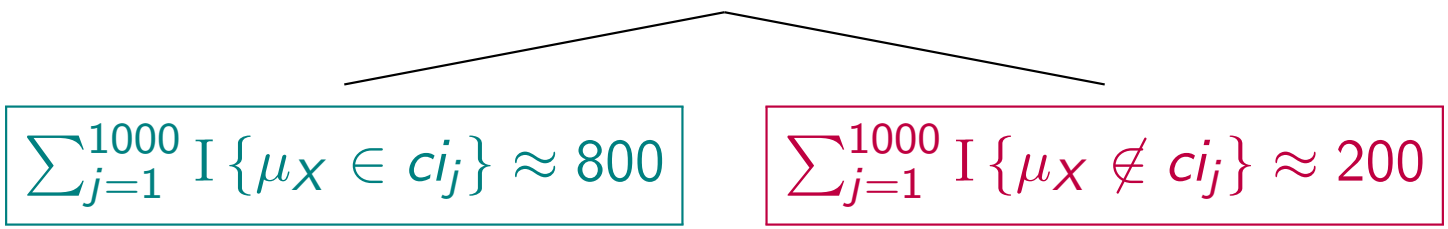
1. We only observe one sample.
2. We don't observe μ_X (and thus we don't know whether $\mu_X \in ci$).

Coverage exercise

- Now, suppose that instead of 3 samples, we do the pretend exercise with **1,000** samples of size 4.
- Because we have 1,000 samples, we also have 1,000 ci's. Let ci_j be the ci of the j'th sample. e.g., $ci_3 = (44.75, 64.75)$.
- Continuing with $p = 0.8$, let's do an exercise to summarize what an 80% CI means and how to interpret it.

Exercise 148. How can we break down the 1,000 80% ci's into two categories? i.e., what are the two categories? And about how many of the ci's are in each category?

$$\#\text{ of 80\% confidence intervals} = 1000$$



First steps to thinking about h

- We can think of h as a sort of margin for error.
- We start with the equality from the previous slide:

$$\mathrm{P}(\bar{X} - h \leq \mu_X \leq \bar{X} + h) = 0.8.$$

- The h we choose depends on the RHS number.

Exercise 149. If the RHS is larger, e.g., 0.9, will h be larger or smaller?

Larger. The intuition is that if our boss tells us we need to be ***more*** sure, then we need a ***larger*** margin of error.

- Analogy: suppose my doctor asks how many cups of water I drank today, I might say "I'm not sure, maybe 4 or 5".
- Now suppose the doctor says my response is critical, and if I'm wrong, the antibiotic they prescribe me for my infection might not work.
- Then I want a high confidence level, and accordingly I would give a wider range, like "Somewhere between 3 and 7".

Extreme values of h

In order to get a handle on exercises like these, it's often useful to look at what happens at the extremes.

Exercise 150. $\mathrm{P}(\bar{X} - \infty \leq \mu_X \leq \bar{X} + \infty) = \mathrm{P}(-\infty \leq \mu_X \leq \infty) = 1$.

Exercise 151. On the other extreme, what if h is very small? Let's say h is 0. Then, $\mathrm{P}(\bar{X} - 0 \leq \mu_X \leq \bar{X} + 0) = \mathrm{P}(\bar{X} = \mu_X) \approx 0$.

These exercises show we can view h as a ***function*** of p, i.e., $h(p)$.

Exercise 152. What are the values $h(1)$ and $h(0)$?

A: We showed in Exercises 150 and 151 that $h(1) = \infty$, $h(0) = 0$.

The function h inputs the level of confidence and outputs the corresponding margin of error.

Definition of a confidence interval

Definition 43. Let γ be a parameter and $\hat{\gamma}$ an estimator of γ. A confidence interval (CI) of level p (for γ) is a random interval $(\hat{\gamma} - h, \hat{\gamma} + h)$ such that

$$\mathrm{P}(\hat{\gamma} - h \leq \gamma \leq \hat{\gamma} + h) = p.$$

i.e., the CI covers the parameter with probability p.

$\hat{\gamma} - h$ is the lower bound, and $\hat{\gamma} + h$ is the upper bound of the CI.

- $\hat{\gamma} - h$ and $\hat{\gamma} + h$ are each random *variables*.
- $(\hat{\gamma} - h, \hat{\gamma} + h)$ is a random *interval*.

Example 32. Suppose $\gamma := \mu_X := \mathrm{E}(X_i)$. Then, $\hat{\gamma} := \bar{X}$ and our CI is $\left(\bar{X} - h, \bar{X} + h\right)$ with the following property:

$$\mathrm{P}(\bar{X} - h \leq \mu_X \leq \bar{X} + h) = p.$$

The frustration of statistics

Example 33. Suppose you explain all this to your boss, and you make some nice beamer slides about if we had 1000 samples.

- If she's the head of the Idescat, she probably understands econometrics better than all of us, but she wants to push you a bit. So she asks you the following:

  That's nice, but we only have one sample. Does the confidence interval for our one sample contain the population mean or not?

We don't know.

- That's frustrating, but that's statistics.
- In statistics, we almost never know the true answer.
- It's also confusing: we only have one sample, but to understand the concept that we are "80% sure our *CI* contains the parameter", we need to think about multiple samples.

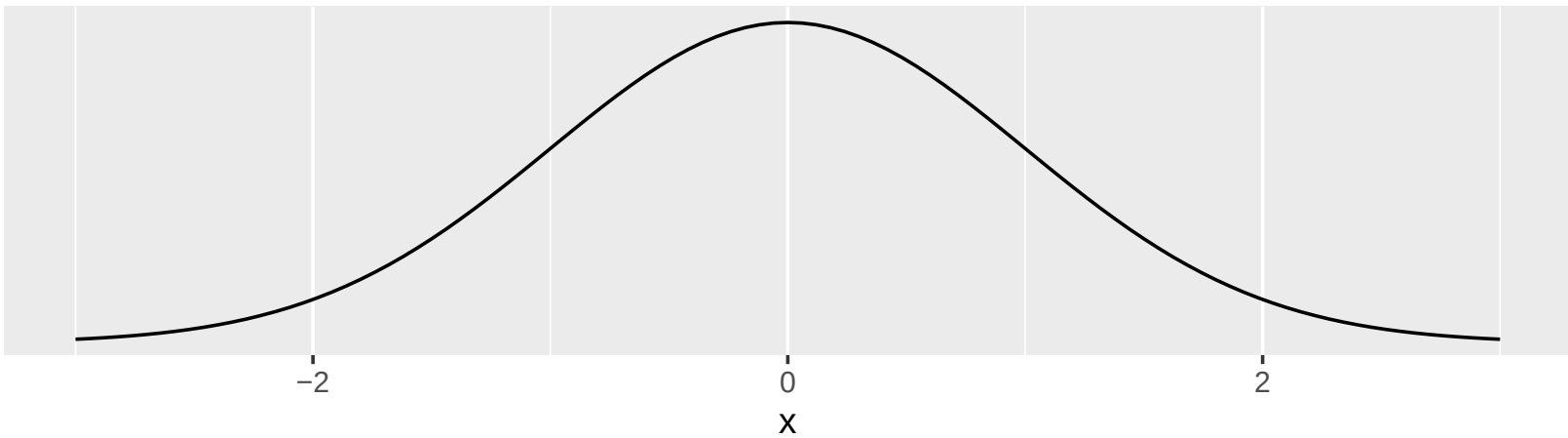
Simple example for understanding random interval (1/2)

Let's first get a feeling for the concepts of a random interval and coverage.

Suppose $M \sim N(0, 1)$.

Exercise 153. Find c so that $\mathrm{P}(-c \leq M \leq c) = 0.95$. $c = 1.96$.

Example 34. What is the distribution of $Q := -M$? $Q \sim N(0, 1)$ because of symmetry. This can be seen graphically:



Simple example for understanding random interval (2/2)

Reminder: $M \sim N(0, 1)$.

$$\mathrm{P}[-1.96 \leq M \leq 1.96] = 0.95. \tag{1}$$
$$\mathrm{P}[-1.96 \leq -M \leq 1.96] = 0.95. \tag{2}$$
$$\mathrm{P}[-1.96 + M \leq -M + M \leq 1.96 + M] = 0.95. \tag{3}$$
$$\mathrm{P}[M - 1.96 \leq 0 \leq M + 1.96] = 0.95. \tag{4}$$

Exercise 154. The random interval $(M - 1.96, M + 1.96)$ has 95% coverage of... what? $\mathrm{P}\left[0 \in (M - 1.96, M + 1.96)\right] = 0.95$.

Why 0? What's special about 0? It's the mean of M: $\mathrm{E}(M) = 0$.

- This means that if we take a draw from a $N(0, 1)$ and construct the observed interval by adding and subtracting 1.96, then 95% of the time, the interval will contain 0.

Constructing a CI for μ_X

Suppose $X_1, \ldots, X_n$ are iid and $X_i \sim N(\mu_X, \sigma_X^2)$. Then, $\bar{X} \sim N\left(\mu_X, \frac{\sigma_X^2}{n}\right)$. It follows that

$$\mathrm{P}\left(\bar{X} - 1.96 \cdot \sqrt{\frac{\sigma_X^2}{n}} \leq \mu_X \leq \bar{X} + 1.96 \cdot \sqrt{\frac{\sigma_X^2}{n}}\right) = 0.95.$$

That is, $\mathrm{P}\left[\mu_X \in \left(\bar{X} - 1.96 \cdot \sqrt{\frac{\sigma_X^2}{n}}, \bar{X} + 1.96 \cdot \sqrt{\frac{\sigma_X^2}{n}}\right)\right] = 0.95.$

Exercise 155. In the interval above, are any of the terms parameters?

σ_X^2 is the *population* variance. We can't have a parameter in the *CI* (otherwise we couldn't calculate a *CI* from the sample). What do we do? We estimate σ_X^2:

$$S_X^2 := \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2.$$

The importance of normality

Exercise 156. On the previous slide, where do we rely on $\bar{X}$ having a normal distribution? The 1.96.

Suppose $X_1, \ldots, X_n$ are iid and $X_i \sim \not{N}(\mu, \sigma^2)$.
Then, $\bar{X} \sim \not{N}\left(\mu_X, \frac{\sigma_X^2}{n}\right)$.

- If $\bar{X}$ is not normal we cannot use the 1.96 cut-off values. Those are from a normal distribution!
- We can still plug numbers into the formula, but it's not clear the CI will have correct coverage of μ_X.
- It seems we're in trouble.

There is hope

Suppose $X_1, \ldots, X_n$ are iid and the X_i are not normally distributed. Then $\bar{X}$ is not normally distributed, but...

- $\bar{X}$ is approximately normally distributed if n is large, and we write $\bar{X} \dot{\sim} N\left(\mu_X, \frac{\sigma_X^2}{n}\right)$.
  - This result is known as the Central Limit Theorem (CLT).
- The CI thus has approximately correct coverage for large n:

$$\mathrm{P}\left(\bar{X} - 1.96 \cdot \sqrt{\frac{S_X^2}{n}} \leq \mu_X \leq \bar{X} + 1.96 \cdot \sqrt{\frac{S_X^2}{n}}\right) \approx 0.95.$$

This is an approximate CI for 2 reasons:

1. We use S_X^2 instead of σ_X^2.
2. $\bar{X}$ is only *approximately* normally distributed.

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Learning objectives

1. Introduce hypothesis tests, emphasizing connections to previous topics (parameters, estimation, standardization).
   - Constructing test statistics.
   - Learn the unintuitive logic behind hypothesis testing.
   - Following the steps of a hypothesis test is easy. Really understanding how it works is hard.
2. Examples on income, fair coins, and approval polls.

Stating hypotheses

- A hypothesis is a logical expression (i.e., true or false) about parameters.
  - The expression can be about a parameter equalling a value.
  - It can be about one parameter equalling a different parameter.
- To test a hypothesis means to use data to explore whether there is evidence that the hypothesis is false.
- A null hypothesis (H_0) is the hypothesis that we will test.
- An alternative hypothesis (H_A) is a hypothesis that is necessarily true if the null hypothesis is false.

Hypothesis about mean income

Example 35. Let $X_1, \ldots, X_n$ be a random sample of incomes (in 1,000€'s) from CAT. Consider the following null and alternative hypotheses:

$$H_0 : \mathrm{E}(X_i) = 30.$$
$$H_A : \mathrm{E}(X_i) \neq 30.$$

Exercise 157. Recall that a hypothesis is about a parameter. What is the parameter in the above hypotheses? $\mathrm{E}(X_i)$.

The hypothesized value is the number that the parameter is theorized to equal.

Exercise 158. What's the hypothesized value in Example 35? 30.

- We're not stating that $\mathrm{E}(X_i)$ *does* equal 30. Rather, we are posing a *question*. Think of it more as: $\mathrm{E}(X_i) \overset{???}{=} 30$.

Example to understand equivalent expressions

Example 36. Suppose we have a random sample of size n from CAT where $F_i := 1\{i'\text{th person is female sex}\}$ and $M_i := 1\{i'\text{th person is male sex}\}$. Consider the following null and alternative hypotheses:

$$H_0 : \mathrm{E}(F_i) = \mathrm{E}(M_i).$$
$$H_A : \mathrm{E}(F_i) \neq \mathrm{E}(M_i).$$

Exercise 159. Suppose that sex can be only male or female. How can we restate the null and alternative hypotheses?

$$H_0 : \mathrm{E}(F_i) = 0.5.$$
$$H_A : \mathrm{E}(F_i) \neq 0.5.$$

These first pair of hypotheses is equivalent to the second pair.

Proof by contradiction

Before we study the steps of a hypothesis test, we introduce a proof technique that is related in terms of logic.

Consider the following proposition:

Proposition 1. Suppose $a, b \in \mathbb{Z}$. If $a + b \geq 19$, then $a \geq 10$ and/or $b \geq 10$. Is this proposition true? Yes.

Proof.

- Suppose the proposition is *not* true. What would that imply?
- There exists $a, b \in \mathbb{Z}$ with $a + b \geq 19$ and $a < 10$ and $b < 10$.
- But if $a < 10, b < 10$, then what is the most $a + b$ could be? $9 + 9 = 18$.
- This ***contradicts*** $a + b \geq 19$.
- Therefore, our initial supposition that the proposition is not true must be false. i.e., the proposition must be true.

Carrying out a hypothesis test

Definition 44. The phrase "under H_0, ...", or "under the null (hypothesis), ...", means "assuming the null hypothesis is true, ...".

We carry out a hypothesis test with the following steps:

1. We construct a test statistic, T, based on our estimator of the parameter inside the null hypothesis.
2. Under H_0, $T \dot{\sim} N(0, 1)$.
3. If the null is not true, then $T \dot{\not\sim} N(0, 1)$.
4. We consider whether the observed value from our sample, $t := T(\omega^*)$, is a surprising value from a $N(0, 1)$.

The result of a test is called a decision. A decision is to (a) reject the null hypothesis at a specified level of confidence, or to (b) fail to reject the null hypothesis at a specified level of confidence.

The murder trial of O.J. Simpson (1995)

In a murder trial in the US, O.J. Simpson was accused of murdering Nicole Brown Simpson (his ex-wife) and Ronald Goldman.

The case against O.J. Simpson can be framed as a hypothesis test:

$$H_0 : \text{OJ is innocent}$$
$$H_1 : \text{OJ is guilty}$$

Suppose H_0 is true.

- The probability of a match of O.J.'s blood to the blood at the crime scene was stated as 1 in 170 million.
- Since this prob is low, we might reject H_0 with high confidence.

(O.J. Simpson was found *not* guilty, partly because the evidence was not handled correctly by the police.)

Imagine the following: Scott has been accused of stealing Joan Miró's *Personatge davant el sol* from the Fundació Joan Miró.

The facts are the following:

1. Scott was seen teaching 25117 seminar but suspiciously ended seminar early at 17.30h.
2. The painting was stolen at 18.00h.
3. Scott was seen returning home (in Gràcia) at 18.20h.

The case can be framed as a hypothesis test:

$$H_0 : \text{Scott is innocent}$$
$$H_1 : \text{Scott is guilty}$$

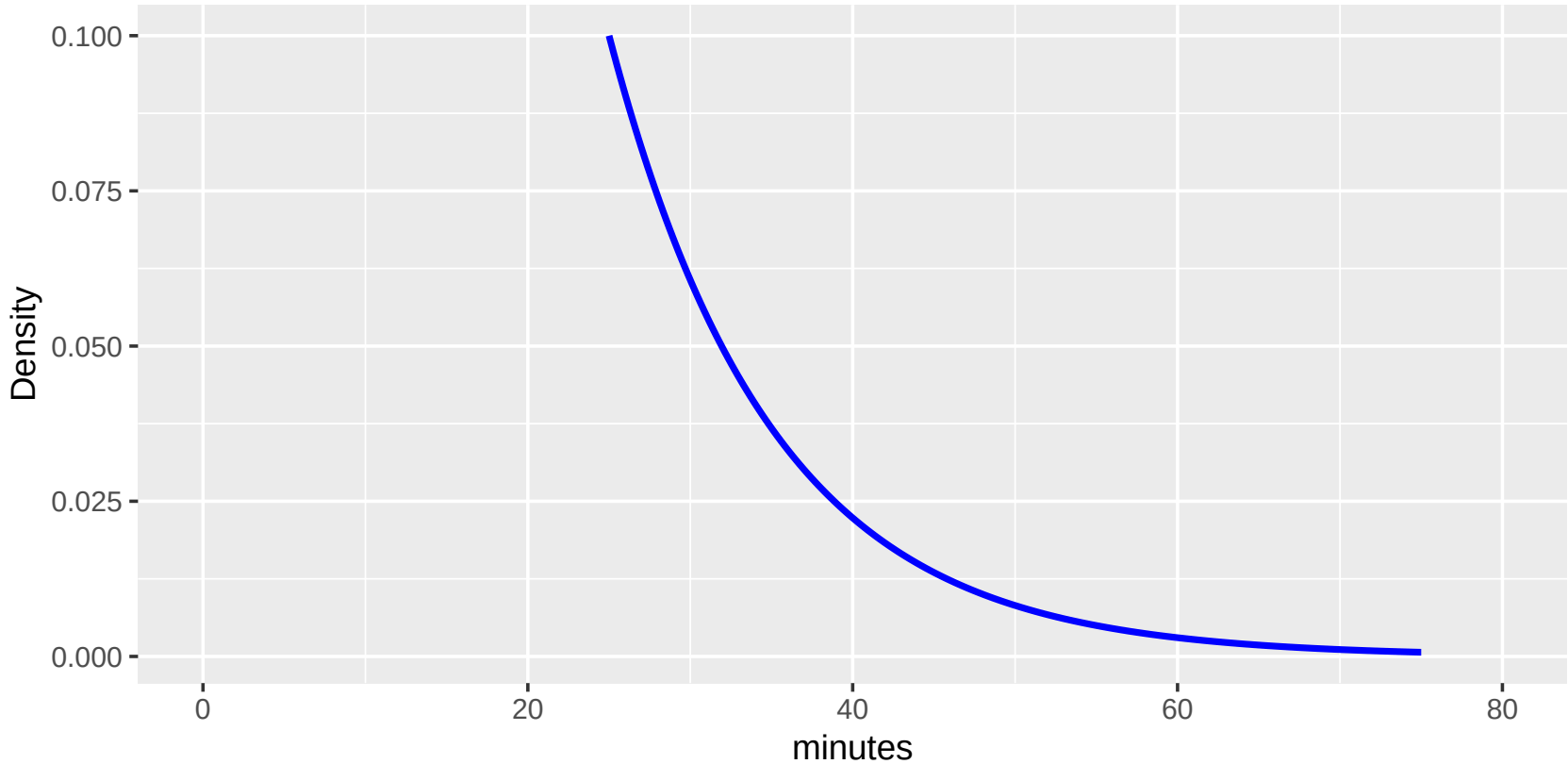
Exercise 160. What can we use as a test stat? (we want a number that, if Scott *were* innocent, a large value would be surprising.)

- $T :=$ The time it took Scott to commute home.
  $T(\omega^*) = 18.20h - 17.30h = 50m.$

How to use T to carry out the hypothesis test

- Under H_0, is $T(\omega^*) = 50$ a surprisingly long time for Scott to commute home? What must we know to answer this question?
- We need to know the dist of T under H_0.
- Note that we don't discuss the dist of T under H_1.
  - We don't know and we don't need to know for a hypothesis test.
- Scott's phone was suspiciously turned off the day of the theft.
- But we did get the dist of Scott's commuting time for the past year (during which there was no art theft in Barcelona) from the phone's GPS data.
  - It has an approximate exponential dist with param $\lambda = 1/10$ shifted by 25.

Distribution of Scott's commute home when no reported thefts in Barcelona $= 25 + Z$, where $Z \sim \exp(1/10)$.



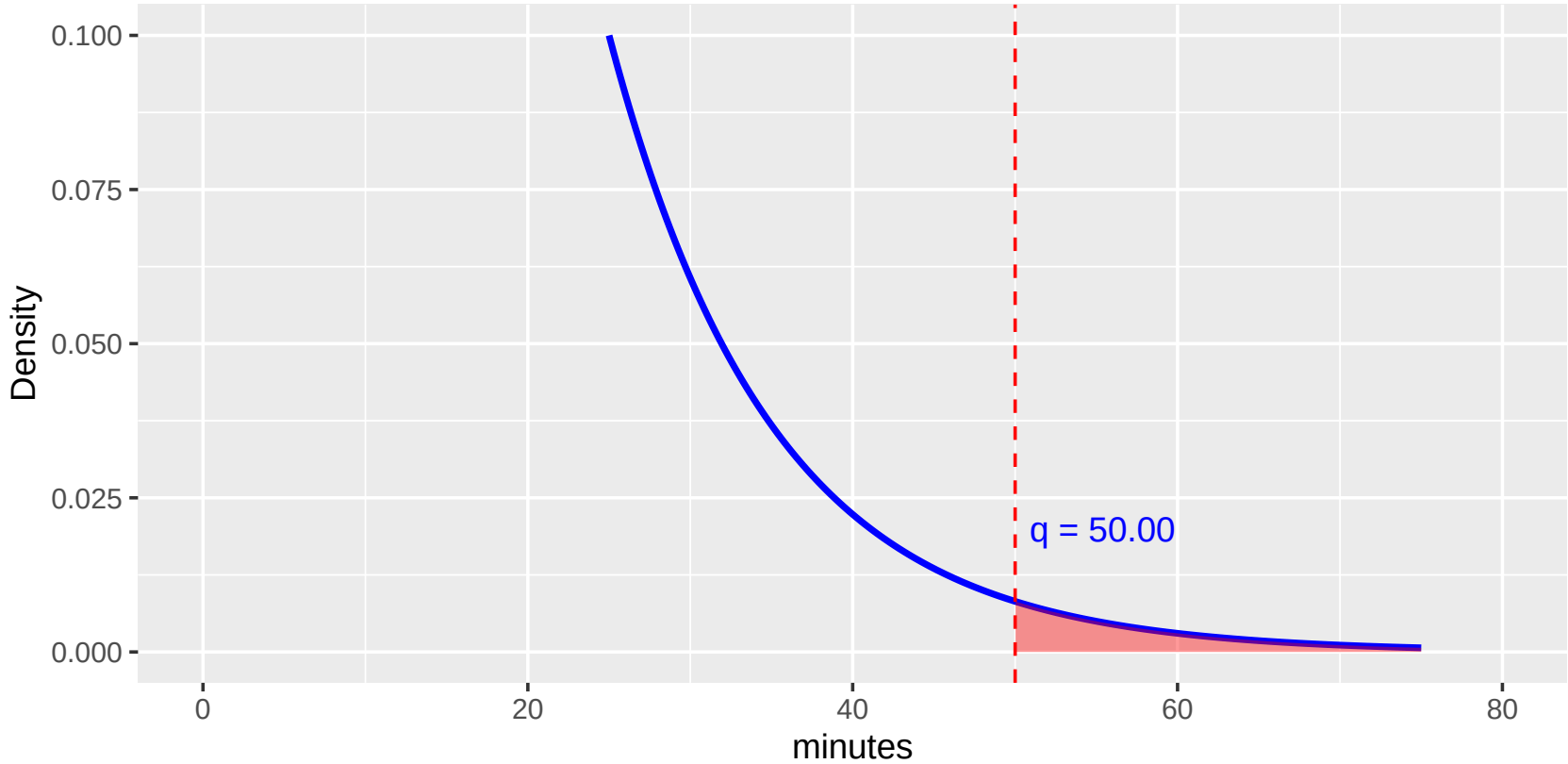
Using T

The cdf is given by $F(x) = 1 - e^{-\frac{1}{10}(x-25)}\mathrm{I}\{x \geq 25\}$.

Exercise 161. What is $\mathrm{P}(T \geq 50)$? We don't know the dist of T. We don't know if $T(\omega^*)$ was drawn from the same dist as Scott's *innocent* commutes home.

Exercise 162. What is $\mathrm{P}(T \geq 50)$ *under* H_0?

$1 - F(50) = 1 - \left(1 - e^{-\frac{1}{10}(50-25)}\right) = e^{-\frac{25}{10}} = e^{-\frac{5}{2}} \approx 0.082.$



Under H_0, $\mathrm{P}(T \geq 50) \approx 1 - F(50) \approx 0.082$. That is the area of the shaded region.

Exercise 163. How do we interpret 0.082?
Only 8.2% of Scott's innocent commutes home are longer than the commute home on the night of the theft.

Exercise 164. If the percentage were *smaller*, would we reject H_0 with more or less confidence? More.

The test statistic

Definition 45. Let $X_1, \ldots, X_n$ be a random sample. A test statistic is a statistic we use to decide whether to reject or fail to reject the given null hypothesis.

Exercise 165. Is a test statistic a rv?

- Yes because it's a statistic.
  - Using some samples we might reject the null, and using other samples (from the *same* pop) we might fail to reject it.
- However, in practice we often use "test statistic" to refer to both the rv, T, and its observed value, $T(\omega^*)$.
- To differentiate, sometimes we say *observed* test statistic to refer to $T(\omega^*)$.

Intuition of the test statistic

Recall the hypothesis about mean income: Let $X_1, \ldots, X_n$ be a random sample of incomes (in 1,000€'s) from CAT. Consider the following null and alternative hypotheses:

$$H_0 : \mathrm{E}(X_i) = 30.$$
$$H_A : \mathrm{E}(X_i) \neq 30.$$

Key intuition: under the null, $\bar{X}(\omega^*)$ should be close to 30.

- How do we say this in math? Under the null, we expect $\left|\bar{X}(\omega^*) - 30\right|$ to be small.
- But even under H_0, $\left|\bar{X}(\omega^*) - 30\right|$ won't likely equal exactly 0.
- Hypothesis testing is about figuring out whether the value $\left|\bar{X}(\omega^*) - 30\right|$ is unlikely large under H_0.

How large is *too* large?

Suppose $\bar{X}(\omega^*) = 55$. Is the following *unlikely* large under H_0?
$\left|\bar{X}(\omega^*) - 30\right| = |55 - 30| = |25| = 25$.

- Recall that the variance in the pop is $\mathrm{V}(X_i)$.
- If $\mathrm{V}(X_i)$ is large, then $\mathrm{V}(\bar{X}) = \frac{\mathrm{V}(X_i)}{n}$ could be large.
- Similarly, if n is small, then $\mathrm{V}(\bar{X})$ could be large.
- Thus, if (1) $\mathrm{V}(X_i)$ is large and (2) n is small, it would not be surprising, under H_0, if $\left|\bar{X} - 30\right|$ is sometimes small and sometimes large.
- The distribution of $\bar{X}$ under the null is what tells us reasonable distances $\left|\bar{X} - 30\right|$ when the null is true.
- Under H_0, $\bar{X} \dot{\sim} N\left(30, \frac{\mathrm{V}(X_i)}{n}\right)$ so we can create a $\dot{\sim} N(0, 1)$ by standardizing $\bar{X}$ by subtracting 30 and dividing by $\sqrt{\frac{\mathrm{V}(X_i)}{n}}$.

Constructing the test statistic (1/2)

Let $X_1, \ldots, X_n$ be a random sample. Let $\mu := \mathrm{E}(X_i)$. Suppose we want to test $H_0 : \mu = 30$ vs. $H_A : \mu \neq 30$. Consider the following:

$$Z := \frac{\bar{X} - 30}{\sqrt{\mathrm{V}(\bar{X})}}.$$

Exercise 166. But wait, is 30 a parameter? (we can't have a parameter in our test *statistic*)

- 30 is the hypothesized value of the parameter. It's not calculated from the pop so it's not a parameter.

Exercise 167. Then is Z a stat? No, $\mathrm{V}(\bar{X}) = \frac{\mathrm{V}(X_i)}{n}$ is a param.!

- So what do we do? We estimate it.
- $S^2 := \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2$ is our estimator of $\mathrm{V}(X_i)$.
- So what is our estimator of $\mathrm{V}(\bar{X})$? S^2/n.

Constructing the test statistic (2/2)

Let $X_1, \ldots, X_n$ be a random sample. Let $\mu := \mathrm{E}(X_i)$. Suppose we want to test $H_0 : \mu = 30$ vs. $H_A : \mu \neq 30$. Consider the following:

$$T := \frac{\bar{X} - 30}{\sqrt{S^2/n}}.$$

Exercise 168. Is $T \dot{\sim} N(0,1)$? We don't know.

- Knowing that $T \dot{\sim} N(0,1)$ would mean we know $\mathrm{E}(X_i) = 30$, which would mean we know H_0 is true!
- Specifically, we don't know the distribution of $\bar{X}$. If we did, then we would know $\mathrm{E}(\bar{X})$ and thus $\mathrm{E}(X_i)$.
- *Under the null,* $T \dot{\sim} N(0,1)$ by standardization.
- Even under H_0, T isn't *exactly* distributed $N(0,1)$. Why not?

1. Instead of $\mathrm{V}(\bar{X})$ we use an estimator.
2. The X_i aren't necessarily $\sim N$. Why then can we claim that $\bar{X} \dot{\sim} N(0,1)$ under the null? The CLT.

Recall that $Z := \frac{\bar{X}-30}{\sqrt{\mathrm{V}(\bar{X})}}$ is not a valid test statistic.

Instead, we use $T := \frac{\bar{X}-30}{\sqrt{S^2/n}}$.

Exercise 169. For those who have seen hypothesis testing before, what do we call $\sqrt{S^2/n}$? It is the standard error of $\bar{X}$.

Definition 46. Let $\hat{\gamma}$ be an estimator (of some parameter γ). The standard error of $\hat{\gamma}$ is the estimator of $\mathrm{SD}(\hat{\gamma})$, and is written as $\mathrm{SE}(\hat{\gamma})$, or as $\widehat{\mathrm{SD}}(\hat{\gamma})$, where the $\hat{}$ means the *estimator* of $\mathrm{SD}(\hat{\gamma})$

- The standard error is not specific to hypothesis testing. It's a component of confidence intervals as well.

Terminology summary

Let $\bar{X}$ be our estimator of $\mathrm{E}(X_i)$.

Exercise 170. What term do we use for $\sqrt{\mathrm{V}(\bar{X})}$?
The standard deviation of $\bar{X}$. Sometimes we denote this as $\mathrm{SD}(\bar{X})$.

Exercise 171. Is $\mathrm{SD}(\bar{X})$ a number or a rv? A number.

Exercise 172. What term do we use for $\sqrt{S^2/n}$?
The standard error of $\bar{X}$. We denote this as $\mathrm{SE}(\bar{X})$ or $\hat{\mathrm{SD}}(\bar{X})$.

Exercise 173. $\mathrm{SE}(\bar{X})$ a number or a rv? A rv.

Exercise 174. What term do we use for $\sqrt{S^2(\omega^*)/n}$?
The *observed* standard error of $\bar{X}$. Alternatively, we could call this the estimate of $\mathrm{SD}(\bar{X})$.

Exercise 175. Is the observed SE of $\bar{X}$ a number or a rv? A number.

Summary

- The intuition of test statistic.
- The distribution of the test statistic under the null, versus the distribution of the test statistic not under the null.
- The "proof by contradiction" logic behind a hypothesis test.
  - Optional: at end of class, we will practice applying this logic to a game of Sudoku.

Exercise 176. What's the next step on this Sudoku? (Hint: the "easiest" step is placing a 2 in box 4.)

	4	6				2		8
		9				3	5	6
5		3			6	7		4
9	5	7	6	2	3	8	4	1
6	3	8	4	7	1	5	2	9
	1		5	9	8	6	3	7
		5	1		9	4		3
3	9					1		
		1				9	8	

Exercise 176. What's the next step on this Sudoku? (Hint: the "easiest" step is placing a 2 in box 4.)

	4	6				2		8
		9				3	5	6
5		3			6	7		4
9	5	7	6	2	3	8	4	1
6	3	8	4	7	1	5	2	9
	1	2	5	9	8	6	3	7
		5	1		9	4		3
3	9					1		
		1				9	8	

Proof.

A 2 must go in r6c1 or r6c3. *Suppose* the 2 goes in r6c1. Then, the 2 in c3 goes in r8c3.

But in r7 there is nowhere a 2 can go,

which *contradicts* the rules of Sudoku. Thus, the initial supposition must be false. This proves the 2 goes in r6c3. □

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Learning objectives

1. Continue hypothesis tests.
2. Introduce p-values.
3. Hypothesis tests vs. confidence intervals.
4. Generalize steps for hypothesis testing and confidence intervals beyond the mean.

Reframing a hypothesis test

Recall $T := \frac{\bar{X}-30}{\sqrt{S^2/n}}$ is a test statistic for $H_0 : \mathrm{E}(X_i) = 30$.

Exercise 177. What is the distribution of T?

- Under the null, $T \dot{\sim} N(0,1)$.
- Under the alternative, we are not subtracting off $\mathrm{E}(\bar{X})$ so we are not standardizing $\bar{X}$. So $T \dot{\not\sim} N(0,1)$.
- There is thus an equivalence: $\mathrm{E}(X_i) = 30 \Leftrightarrow T \dot{\sim} N(0,1)$.

Thus, we can reframe the above H_0 to $H_0 : T \dot{\sim} N(0,1)$.

Exercise 178. But we don't observe the distribution of T. What do we observe? For the observed sample, ω^*, we observe $T(\omega^*)$, i.e., one outcome of T.

The logic behind a hypothesis test (1/2)

- It might seem intuitive to use that outcome, $T(\omega^*)$, to decide whether it's likely that $T \dot{\sim} N(0,1)$, i.e., $\mathrm{E}(X_i) = 30$.
- But this is actually not how hypothesis testing works.

Example 37. Can we use $T(\omega^*)$ to say something like "There's a 95% chance H_0 is true" ?

- i.e., can we say something like "$\mathrm{P}\left(\mathrm{E}(X_i) = 30\right) = 0.95$"?
- No. $\mathrm{P}\left(\mathrm{E}(X_i) = 30\right) \in \{0, 1\}$.
- Or can we say $\mathrm{P}\left(T \dot{\sim} N(0,1)\right) = 0.95$? No. It is not the distribution of T that is random. It is fixed, but we don't know what it is.
- The above attempts are closer to what is done in Bayesian statistics, which is not covered in this course.

The logic behind a hypothesis test (2/2)

- Instead of exploring "How likely is it that $\mathrm{E}(X_i) = 30$, given $T(\omega^*)$?", we flip the logic... and instead ask: "how likely is observing $T(\omega^*)$ given that $\mathrm{E}(X_i) = 30$".
- The flipped logic is less intuitive, but is the only way we can proceed. We proceed using logic similar to a proof by contradiction:

1. Suppose H_0 is true.
2. Then, $T \dot{\sim} N(0, 1)$.
3. Assess whether the value $T(\omega^*)$ is consistent with coming from a $N(0, 1)$. If not, then maybe our initial proposition was false.

We'll now discuss how we perform the assessment of step 3?

Critical region

- For $Q \dot{\sim} N(0,1)$, $\mathrm{P}(-1.96 < Q < 1.96) \approx 0.95$.
- i.e., $\mathrm{P}\left[Q \in (-1.96, 1.96)\right] \approx 0.95$.
- Then, $\mathrm{P}(|Q| > 1.96) = \mathrm{P}\left[Q \notin (-1.96, 1.96)\right] = 1 - \mathrm{P}\left[Q \in (-1.96, 1.96)\right] \approx 0.05$.
- Since $\mathrm{P}(|Q| > 1.96)$ is small, if $T(\omega^*) \notin (-1.96, 1.96)$ we claim there is evidence that $T \dot{\not\sim} N(0,1)$ at the 95% level.

Definition 47. The critical region is the set A for which we reject H_0 if $T \in A$.

- In our example, for a test at the 5% level, the critical region is all numbers in $(-\infty, -1.96)$ and $(1.96, \infty)$.
- We often want to know for which levels of confidence a hypothesis test would reject.
  - We could find the corresponding critical regions for several different levels, but there's an easier way.

Testing a fair coin (1/2)

Let $X_1, \ldots, X_{1000}$ be an iid sample of 1,000 coin flips (i.e., $X_i := \mathrm{I}\{i'\text{th coin flip is } H\}$). Suppose we are interested in the following hypotheses:

$$H_0 : \mathrm{P}(X_i = 1) = 0.5$$
$$H_A : \mathrm{P}(X_i = 1) \neq 0.5.$$

What is our estimator of $\mathrm{P}(X_i = 1)$? $\bar{X}$.

- Suppose for the sample we collect, ω^*, we observe 997 heads. That is, $\bar{X}(\omega^*) = 0.997$.
- How can we argue that under H_0 observing 0.997 is surprising?
- We could say that under H_0, $\mathrm{P}(\bar{X} = 0.997)$ is small.
- But $\mathrm{P}(\bar{X} = 0.500)$ is also small!
- Key intuition: the *distance* from the observed statistic to the hypothesized value, $|0.997 - 0.500|$, is large.

Testing a fair coin (2/2)

Let's consider the logic of the previous slide:

1. Suppose the coin is fair.
2. If the coin is fair, it's shocking to get 997 heads out of 1,000 flips.
    - The reason it's shocking is because 997 is so far from 500.
3. Therefore, we reject the initial supposition.

Intuition of the p-value

Exercise 179. What is $\mathrm{P}(|T| > 1.96)$?
We don't know, because we don't know the distribution of T.

- *Under the null*, $\mathrm{P}(|T| > 1.96)$ is ≈ 0.05.

We can generalize the intuition of a hypothesis test. Let $Q \sim N(0,1)$. Suppose that $T(\omega^*) = 1.428$.

- To think about whether $T(\omega^*)$ is a *likely* value from a $N(0,1)$, we might think about whether $\mathrm{P}[Q = 1.428]$ is large.
- But $\mathrm{P}[Q = 1.428] = 0$ since Q is a continuous rv.
- So instead, we consider the probability of observing an outcome as extreme, or more extreme, than $T(\omega^*)$.
- i.e., we focus on whether $\mathrm{P}[|Q| \geq 1.428]$ is large.
- We use magnitude (i.e., abs. value) because, e.g., -4.0 is more extreme (departs further from the null) than 1.428.

Definition of the p-value

Definition 48. The p-value is the probability, *under the null*, of obtaining test results at least as extreme as the test statistic.

Example 38. The p-value for a 2-sided test of the pop mean is $2 \cdot \Phi(-|T|)$.

Exercise 180. Is the p-value a parameter, estimator, or statistic? Statistic because it's a function of T.

- The *observed* p-value, $2 \cdot \Phi\left(-|T(\omega^*)|\right)$, is a number.
- The p-value is also the smallest level at which a hypothesis test would reject the null.

Approval example

- CNN published an article on 3 April 2023 with the following title "Majority of Americans approve of Trump indictment".
- Opening sentence of the article:

  Sixty percent of Americans approve of the indictment of former President Donald Trump, according to a new CNN Poll...

Exercise 181. Why might I claim that this sentence is deceptive?
It uses the estimate as the population value!

- Only at the end of the article it mentions that 1,048 participants were surveyed.

Exercise 182. How could we check whether the claim in the title is deceptive also? We could do a hypothesis test.

Carrying out the test based on 60% approval of *sample*

Let $X_1, \ldots, X_{1048}$ be a random sample, where $X_i := 1\{i'\text{th person approves of the indictment}\}$. $\bar{X}(\omega^*) = 0.60$.

Exercise 183. How can we state the hypothesis test that the majority of Americans approve of the Trump indictment?

$H_0 : \mathrm{P}(X_i = 1) = 0.5$ vs. $H_A : \mathrm{P}(X_i = 1) \neq 0.5$.

- What is our estimator of $\mathrm{P}(X_i = 1)$? $\bar{X}$.
- Recall also that $\mathrm{V}(X_i) = \mathrm{P}(X_i = 1)\left[1 - \mathrm{P}(X_i = 1)\right]$.
- Then, $\mathrm{V}(\bar{X}) = \frac{\mathrm{V}(X_i)}{n} = \frac{\mathrm{P}(X_i=1)[1-\mathrm{P}(X_i=1)]}{n}$ so a natural estimator of $\mathrm{V}(\bar{X})$ is $\hat{\mathrm{V}}(\bar{X}) := \frac{\bar{x}[1-\bar{x}]}{n}$. i.e., $\mathrm{SE}(\bar{X}) = \sqrt{\hat{\mathrm{V}}(\bar{X})} = \sqrt{\frac{\bar{x}[1-\bar{x}]}{n}}$.

The test statistic is then:

$$T = \frac{\bar{X} - 0.50}{\mathrm{SE}(\bar{X})} = \frac{\bar{X} - 0.50}{\sqrt{\frac{\bar{X}[1-\bar{X}]}{n}}}.$$

For our observed sample, ω^*, the observed value $|T(\omega^*)|$ is:

$\left|\frac{0.60-0.50}{\sqrt{\frac{0.60(1-0.60)}{1048}}}\right| \approx 6.61 > 1.96$ so we reject H_0 at the 95% level.

Exercise 184. If we used any hypothesized value less than 0.50 (e.g., 0.45), can we say we would also reject that number at the 95% level? Yes, $|T(\omega^*)|$ would be even larger.

- It's amazing we can use 1,048 people to say something about millions of people.
- The bedrock underlying this magic is *random sampling*.
  - Were the 1,048 text messages indeed a random sample?
  - Was everyone in the pop equally likely to be selected?
  - Was there a 100% response rate?

Relationship between hypothesis tests and CIs

- There is an important relationship between hypothesis tests and CIs: if a hypothesized value is rejected, that value is also not in the ci, and vice-versa.
- The CI is the set of theorized values for which the hypothesis test does not reject (at the same confidence level).

Exercise 185. Recall, in the poll example, that we rejected $H_0 : \mathrm{P}(X_i = 1) = 0.5$ at the 95% level. What can we say about the 95% ci for $\mathrm{P}(X_i = 1)$? $0.5 \notin ci$.

Theorem 7. Let γ be a parameter, let c be a theorized value, and let ci be the observed CI for γ at level p. Then,

$$\text{Reject } H_0 : \gamma = c \text{ at level } p \iff c \notin ci.$$

Trends away from hypothesis tests towards CIs

- There has been a gradual movement away from focusing policy decisions on hypothesis tests.
- Suppose we reject $H_0 : \mathrm{E}(X_i) = 30$.
  - It could be that $\mathrm{E}(X_i) = 30.2$ or that $\mathrm{E}(X_i) = 300$.
- We could do a bunch of different hypothesis tests to get a feeling for which hypothesized values we would reject. Or...
- We could focus on a CI, which provides a range of hypothesized values that γ might be.

Confidence intervals

- A CI is often more informative than a hypothesis test in some senses.
- Because $\bar{X}$ is approximately normal (by the CLT), we can use our standard formula for a 95% CI:

$$\bar{X} \pm 1.96 \cdot \mathrm{SE}(\bar{X}) = \bar{X} \pm 1.96 \cdot \sqrt{\frac{\bar{X}\left[1-\bar{X}\right]}{n}}.$$

- The above CI gives the following observed ci: $(0.57, 0.63)$.

 The previous hypothesis test and ci can be carried out in R with the following command:

prop.test(x = 0.60 * 1048, n = 1048, p = 0.50, conf. level = 0.95)

Exercise 186. Recall we rejected $H_0 : \mathrm{P}(X_i = 1) = 0.5$ at the 95% level. What if we test 0.55 at the 95% level as the theorized value?

- By Theorem 7, we would reject 0.55, since $0.55 \notin (0.57, 0.63)$.

Hypothesis tests and CIs for other parameters

- So far we've mostly focused on hypothesis tests about $\mathrm{E}(X_i)$.
- The following are common steps to take for constructing hypothesis tests and CIs for other parameters.
- Suppose γ is a parameter with estimator $\hat{\gamma}$.

1. Argue that $\hat{\gamma}$ is approximately normally distributed.
   a. For $\bar{X}$, we did this using the CLT.
2. Construct $\mathrm{SE}(\hat{\gamma})$:
   a. Find an expression for $\mathrm{V}(\hat{\gamma})$.
   b. Estimate the parameters in the expression for $\mathrm{V}(\hat{\gamma})$ to get $\hat{\mathrm{V}}(\hat{\gamma})$.
   c. We then have $\mathrm{SE}(\hat{\gamma}) = \sqrt{\hat{\mathrm{V}}(\hat{\gamma})}$.
3. Construct the 95% CI as $\hat{\gamma} \pm 1.96 \cdot \mathrm{SE}(\hat{\gamma})$.

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Learning objectives

Suppose we're interested in the relationship between education (educ), measured in years, and wage, measured in dollars per hour.

- This topic is studied a lot in economics (in the subfields of "labor economics", "economics of education", and "economics of discrimination").

In this lesson, we explore data on educ and wage from a simulated (i.e., made up) data set. We have 2 goals:

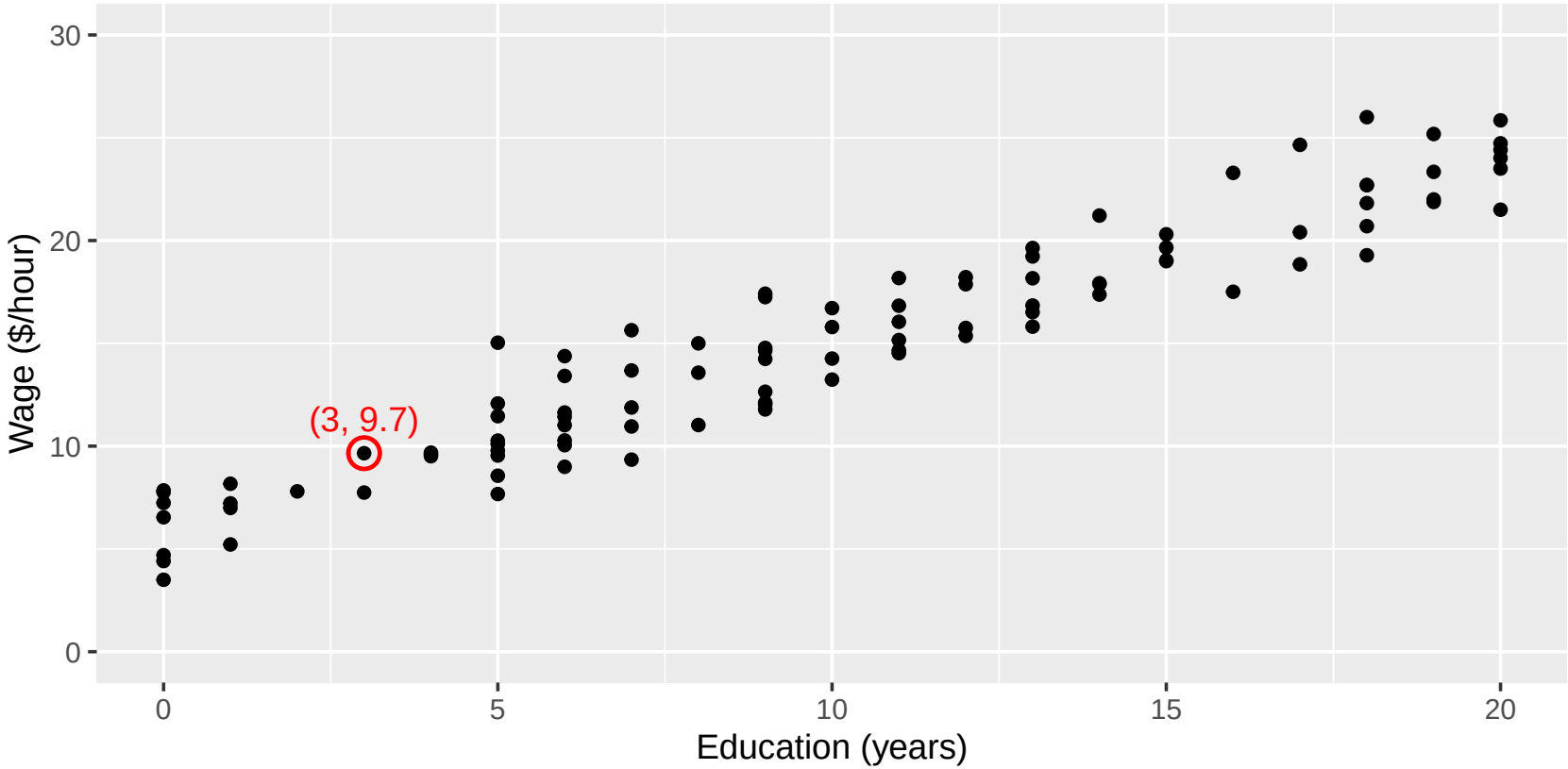
1. Understand the methods (i.e., scatter plots and lines) to describe the relationship of these 2 vars. in the ***population***.
    - We accomplish this by first working with data on the entire (hypothetical) population.
2. Understand how to ***estimate*** the relationship of education and wage in the population using only a ***sample***.

Motivation: why is a scatter plot useful?

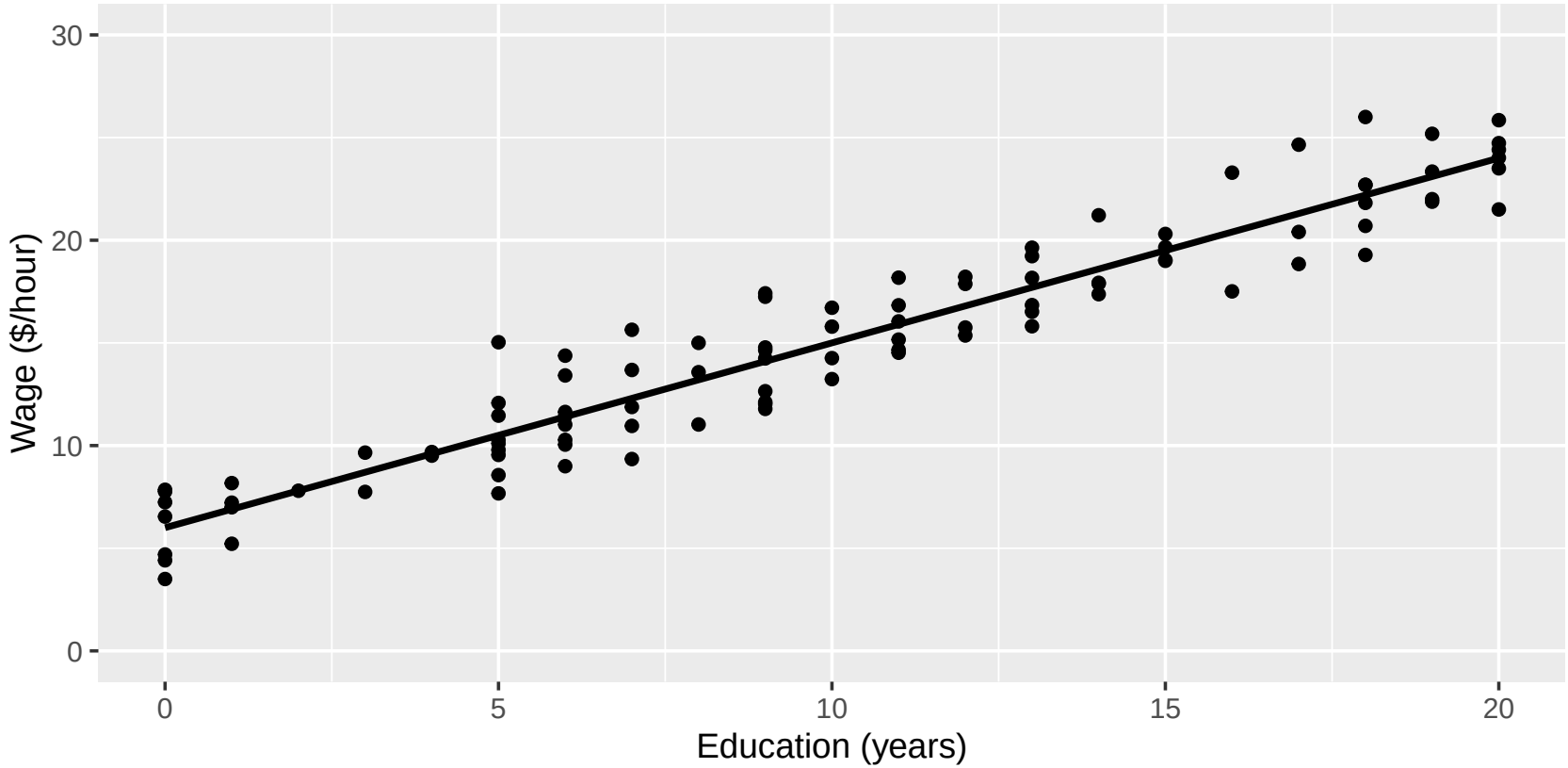
Without a scatter plot, it's difficult to see patterns, or trends, in the data by just studying the numeric values directly.

Below are 30 observations, sorted by education:

educ	wage	educ	wage	educ	wage
0	4.4	1	7.2	5	12.1
0	7.2	2	7.8	5	9.8
0	4.7	3	9.7	5	8.6
0	7.8	3	7.7	5	11.5
0	6.5	4	9.5	5	10.3
0	7.8	4	9.7	5	9.5
0	3.5	4	9.6	6	11.6
1	7.0	5	7.7	6	10.0
1	5.2	5	10.1	6	9.0
1	8.2	5	15.0	6	10.3



- Here, a point (the unit of observation) could be the wage and education of a **person**, or it could be the *average* wage and *average* education of workers in a **state**, **country**, or **industry**.
- For our population, the unit of observation is a ***person***. For example, the circled point represents a person with 3 years of education earning 9.7 dollars per hour.



Example 39. What contains more information, the line or the dots?

The dots.

1. For one, the line is drawn *based on* the dots.
   - i.e., the line is a *function* of the dots.
2. Further, how many numbers define a line? 2 (slope, intercept).
   - *Each* point in the scatter plot has two numbers in itself!

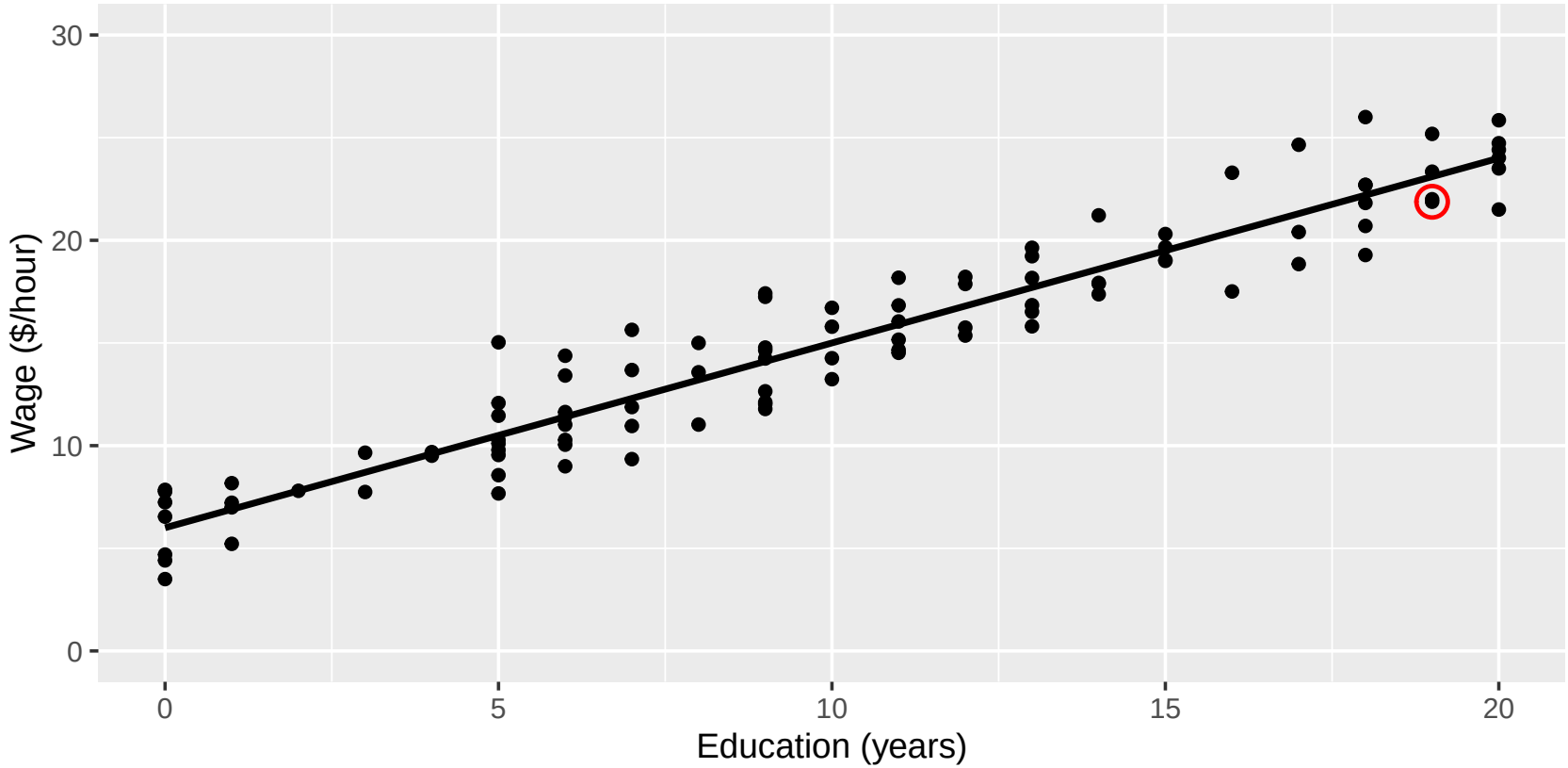
Then why draw a line? (1/2)

If the dots contain more information, then why would we spend time studying the line instead of the dots?

- We draw a line because we *want* to **summarize** the joint distribution of two variables.
  - Just like we use, e.g., the mean to summarize the distribution of a variable.
  - The disadvantage of a line (less info) is also its advantage.
- We can define parameters of interest (the slope and intercept).

  - After defining parameters, we can then discuss inference on them (i.e., estimation, confidence intervals, and hypothesis tests).

Then why draw a line? (2/2)

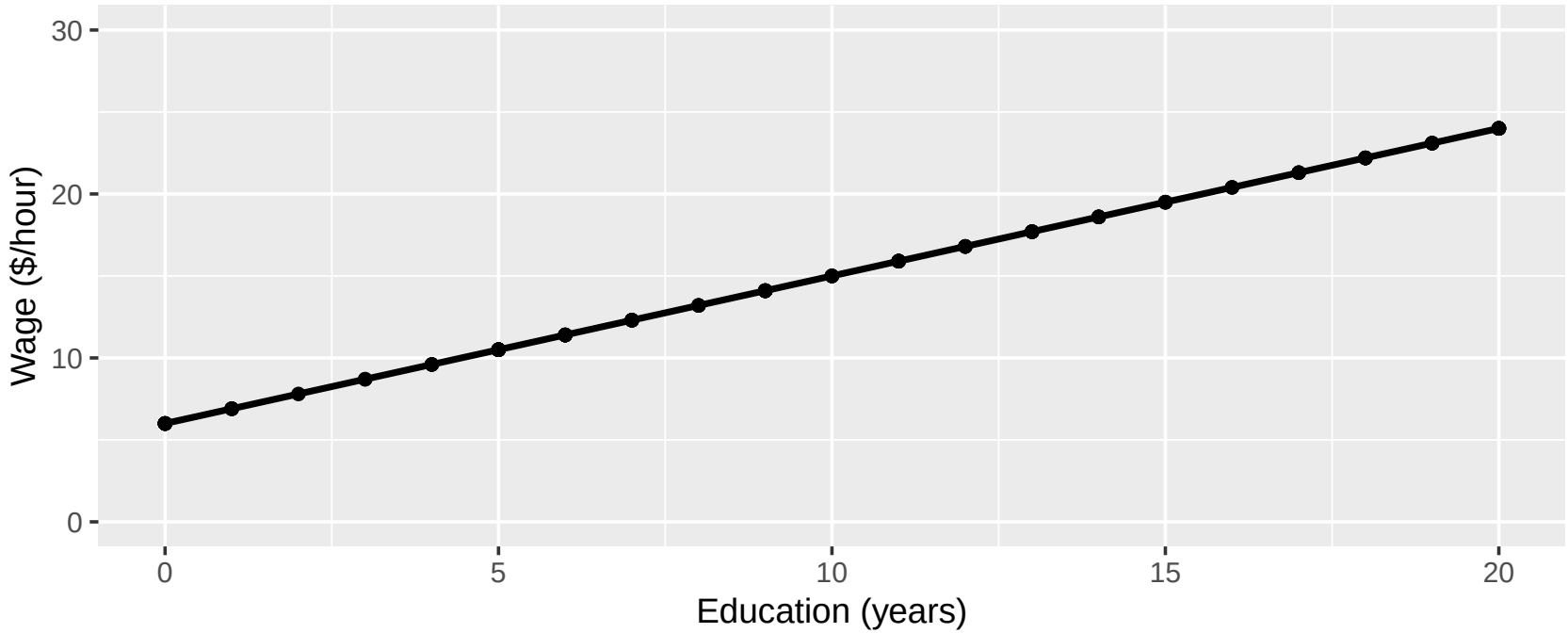


Exercise 187. What can we say about the circled person in the pop?

- Since the point is below the line, they have low wage *relative* to the wages of people with similar education.

What if the dots are all on the line?

Exercise 188. What would it mean if the dots were exactly on the line?



If you knew someone's education, you would know their *exact* wage.

- The math answer is: $\text{corr}(\text{educ}, \text{wage}) \in \{-1, 1\}$.
- The relationship would satisfy the following 2 properties:
  1. wage is completely determined by educ.
  2. The relationship between wage and educ is linear.

Why aren't the dots on the line?

In our case, the relationship of educ and wage in the data does appear to have a linear trend, so it must be that there are more determinants of wage than education.

Exercise 189. Think of 3 variables that determine a person's wage besides educ.

1. Ability (e.g., IQ) of the individual
2. Experience of the individual
3. Living expenses of the city where the individual lives

These are (some of) the reasons why the dots are not on the line.

The regression function

Definition 49 (Regression function). Let X, Y be two rvs. The regression function (of Y wrt X) is the function $f : \mathrm{supp}(X) \mapsto \mathbb{R}$ defined by:

$$f(x) = \mathrm{E}(Y|X = x).$$

- The notation $\mathrm{E}(Y|x)$ is equivalent to $\mathrm{E}(Y|X = x)$.

Example 40. Consider the regression function of wage with respect to educ: $f(\mathrm{educ}) := \mathrm{E}(\mathrm{wage}|\mathrm{educ})$.

- f **inputs** a value of educ and **outputs** the *average* wage of the subset of people in the pop with that value of educ.

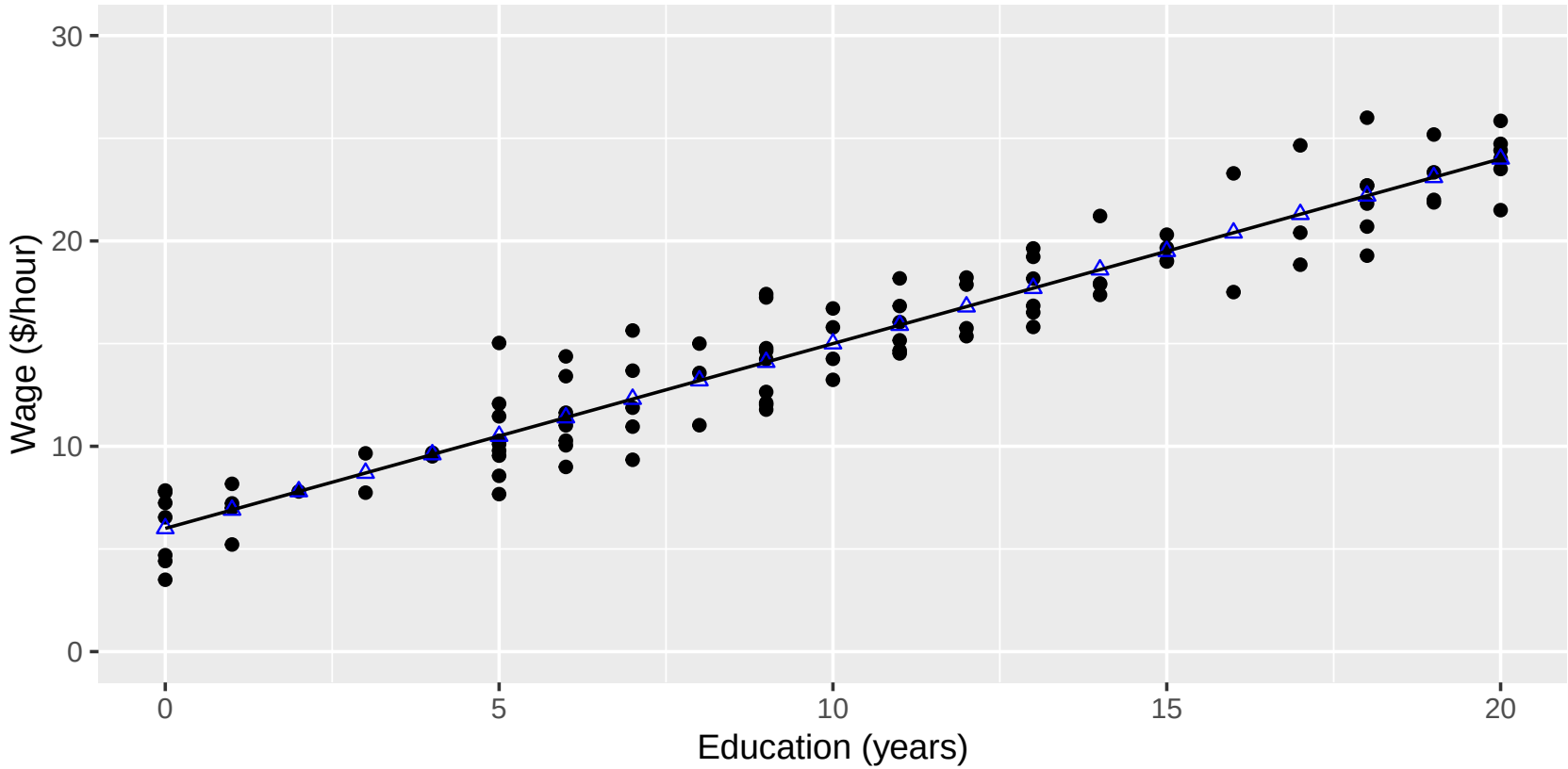
Linearity assumption

Assumption 1 (Linearity of the regression function).
$\mathrm{E}(Y|X = x) = \beta_0 + \beta_1 x$ for some $\beta_0, \beta_1 \in \mathbb{R}$.

- β_0 is the intercept. It's pronounced "beta-naught" or "beta-zero".
- β_1 is the slope. It's pronounced "beta-one".
- We call β_0 and β_1 regression coefficients.
  - It's strange to think of β_0 as a coefficient, but it can be viewed as the coefficient on 1. i.e., $\beta_0 = \beta_0 \cdot 1$.
- When Assumption 1 holds, we can refer to the regression function as the regression *line*.

Typically we do not have data on the population, so we cannot directly check whether Assumption 1 holds.

However, in this lecture we have started with the (unrealistic) case where we do observe the population.



Since we have the pop, we can check the assumption that $f(\text{educ}) := \text{E}(\text{wage}|\text{educ})$ is linear.

For $f(\text{educ})$ to be linear, we must be able to draw a line through the avg. wages (the blue $\triangle$s above). Indeed, the assumption holds.

For our population, $\beta_0 = 6,\ \beta_1 = 0.9$. These are the parameters.

Interpretation of β_0 and β_1

Substituting these values into the regression function, we get:

$$\mathrm{E}(\mathrm{wage}|\mathrm{educ}) = 6 + 0.9 \cdot \mathrm{educ}.$$

An important task in econometrics is ***interpreting*** the values of β_0 and β_1 in the ***context*** of the particular variables.

- Interpretation depends on the ***assumptions*** that we argue are valid for the context.
- This is the line for the *average* wage given a specific amount of education.

Example 41. Interpret the values $\beta_0 = 6$ and $\beta_1 = 0.9$.

First, let's walk through some exercises to help us think about what we can and cannot say based on the values for β_0 and β_1.

What can we use β_0 (=6) and β_1 (=0.9) to say? (1/5)

Exercise 190. Suppose a person in the pop has 7 years of educ. Can we use β_0 and β_1 to say what their wage is?

No. The regression function only tells us the *average* wage across *all* people in the population with 7 years of education.

What can we use β_0 (=6) and β_1 (=0.9) to say? (2/5)

Exercise 191. Suppose we take all the people in the population with 7 years of education. Can we use β_0 and β_1 to say what their *average* wage is?

Yes. This is exactly what the regression function tells us:

$$\overline{\text{wage}} = 6 + 0.9 \cdot 7 = 6 + 6.3 = 12.3.$$

Their average wage is 12.3 dollars per hour.

What can we use β_0 (=6) and β_1 (=0.9) to say? (3/5)

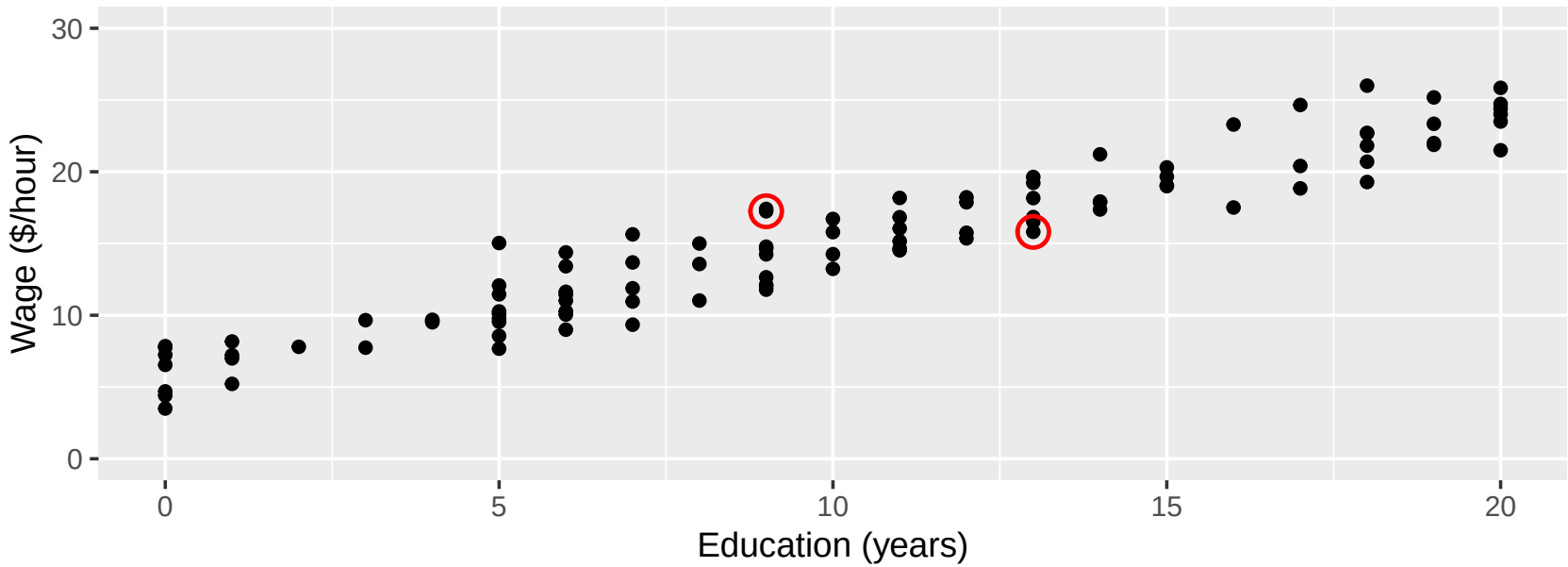
Exercise 192. Suppose *everyone* in the pop who had 7 years of educ stays in school for one more year. Can we use β_0 and β_1 to say by how much the *average* wage will increase for these people when they have 8 years of educ compared to when they had 7 years of educ? 

No.

- The regression function is based on a snapshot ("cross-section") of the population.
- Even though the question is about an average, it is not about the *current* population's average.
- It's about the *future* of people in the population.
- Without further assumptions, we can't "extrapolate".

What can we use β_0 (=6) and β_1 (=0.9) to say? (4/5)

Exercise 193. Suppose we take one person at random from the population who has 9 years of education and one person at random who has 13 years of education. Is the wage of the person with 13 years of education larger than the person with 9 years? Not necessarily.



Why is the wage of the person with 9 years higher? Because of other determinants of wage (maybe more experience or more IQ).

What can we use β_0 (=6) and β_1 (=0.9) to say? (5/5)

Exercise 194. Suppose we take *many* people at random with 9 years of education and many people at random with 13 years. Is the average wage higher for those with 13 than those with 9?

Yes, because (1) we're talking about the *average*; (2) we're talking about the *current* population; and (3) $\beta_1 > 0$.

Back to our example on interpretation

After working through these exercises, let's come back to our motivating example:

Example 2. Interpret the values $\beta_0 = 6$ and $\beta_1 = 0.9$.

Recall our equation: $\mathrm{E}(\mathrm{wage}|\mathrm{educ}) = 6 + 0.9 \cdot \mathrm{educ}$.

- If we take a large iid sample from the pop who have a certain value of educ, say e, and we take a large iid sample from the pop who have educ $e + 1$, the difference in the *average* wages of these two groups will be approximately 0.9 dollars.
- How do we interpret the value we have for β_0, which is 6? If we plug in 0 for educ, we get $\mathrm{E}(\mathrm{wage}|\mathrm{educ} = 0) = 6$, i.e., the *average* wage of people in the pop with 0 years of education is 6 dollars.

Why squared?

Let $x_1, \ldots, x_n$ be a sequence with $x_i \in \mathbb{R}$. Consider the following function:

$$f(a) = \sum_{i=1}^{n} (x_i - a)^2$$

We will now show that $\bar{x} = \operatorname{argmin}_{a \in \mathbb{R}} \sum_{i=1}^{n} (x_i - a)^2$.

$$\frac{d}{da} f(a) = \frac{d}{da} \sum_{i=1}^{n} (x_i - a)^2 \tag{1}$$

$$= \sum_{i=1}^{n} \frac{d}{da} (x_i - a)^2 \tag{2}$$

$$= \sum_{i=1}^{n} 2(x_i - a)(-1). \tag{3}$$

$$0 = \sum_{i=1}^{n} (x_i - a^*) \tag{4}$$

$$na^* = \sum_{i=1}^{n} x_i \tag{5}$$

$$a^* = \frac{1}{n} \sum_{i=1}^{n} x_i. \tag{6}$$

Formulas

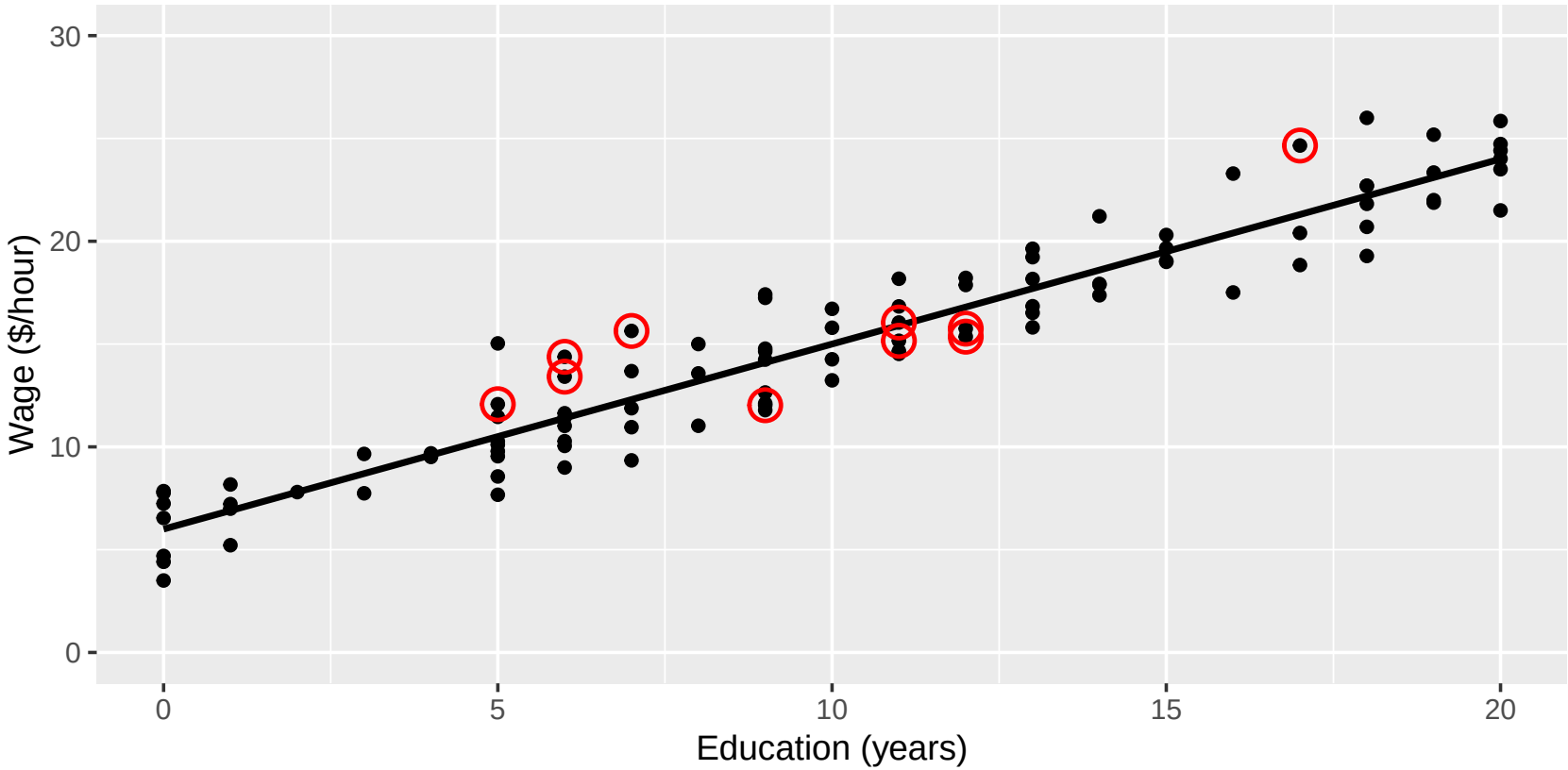
The solution to the minimization problem gives the following formulas:

$$\beta_1 = \frac{\operatorname{cov}(x, y)}{\operatorname{V}(x)}, \tag{1}$$

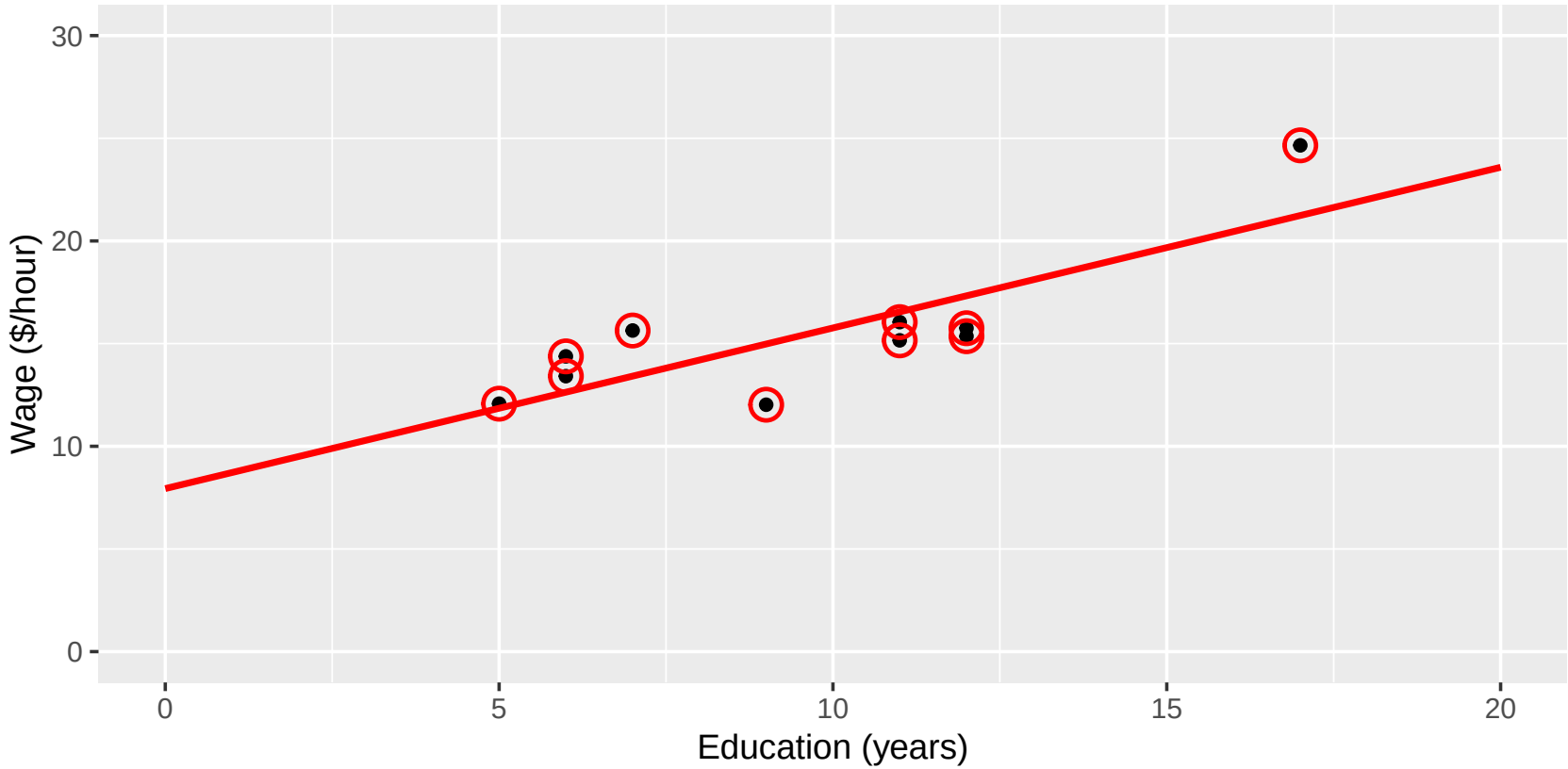
$$\beta_0 = \overline{y} - \beta_1 \overline{x}. \tag{2}$$

Big picture of the regression line

- The most important thing we did was to define ***parameters*** and motivate why they are interesting, i.e., why we would ideally like to know their values.
- We now consider a more realistic situation: suppose we only have data on a sample from the population.
- Our goal is clear: we would like to know, β_0 and β_1.
  - But because we no longer have the population, we cannot just calculate the parameters.
  - Even if we cannot observe β_0 and β_1, they are still numbers.
  - We have to ***estimate*** β_0 and β_1 using a sample.



- We consider an iid sample of size 10, $(X_1, Y_1), ..., (X_{10}, Y_{10})$.
- For our particular sample, ω^*, we circle in red the pairs $(X_1(\omega^*), Y_1(\omega^*)), \ldots, (X_{10}(\omega^*), Y_{10}(\omega^*))$.

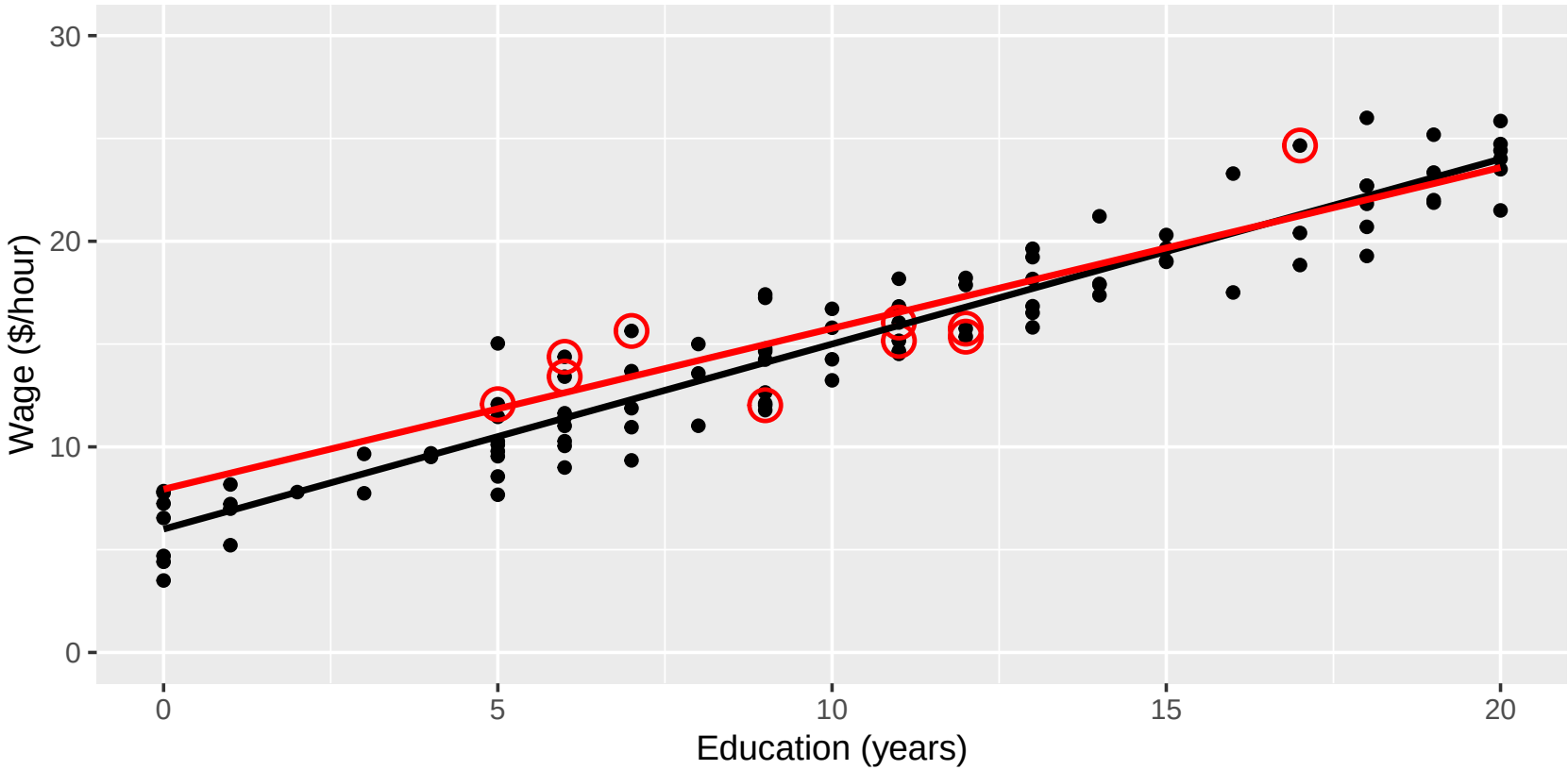


The black line still *exists*. We just don't know what it is.

Exercise 195. How would we estimate the black line (i.e., the intercept, β_0, and slope, β_1) using only the sample?

By using the intercept and slope of a line through the *sample*.

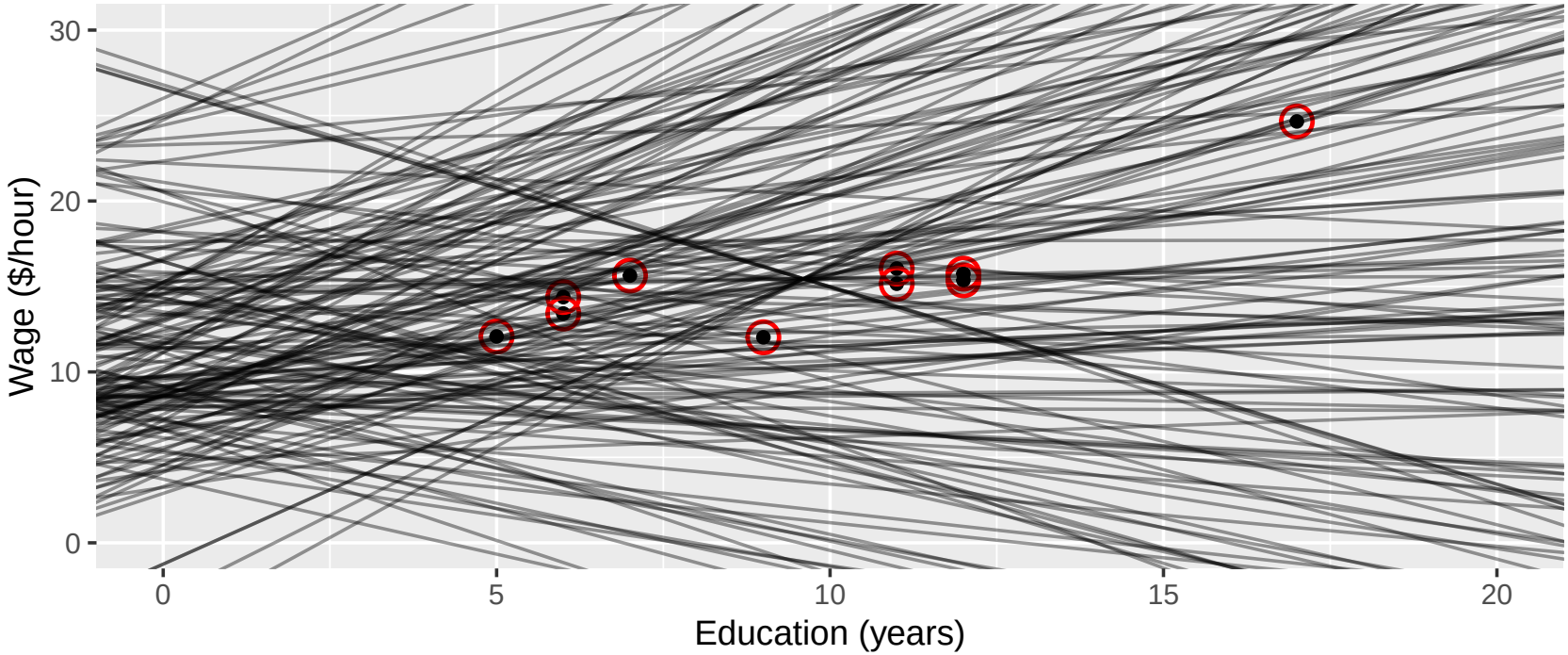
- The intuition is the same as, e.g., estimation of the mean: we do to the sample what we would like to do to the pop.



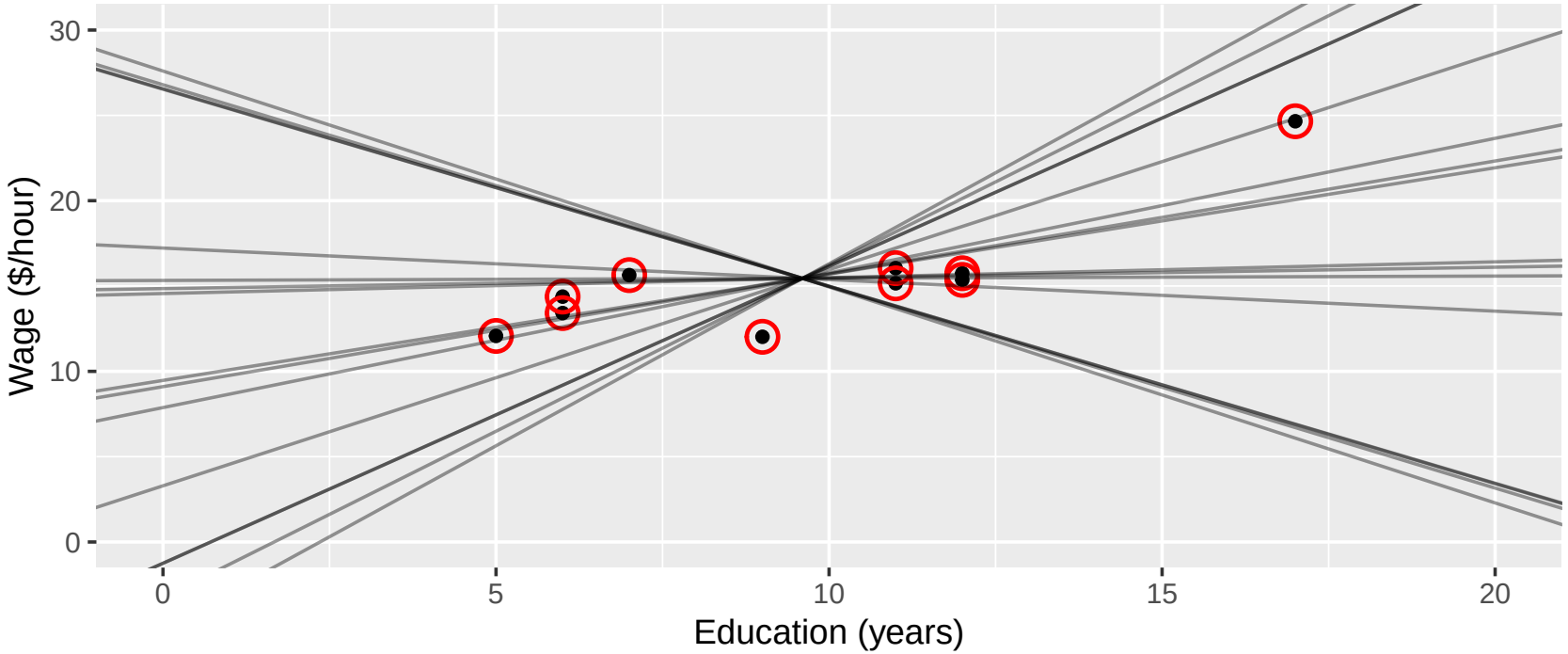
- We define the equation for the black line (the line through the points in the ***population***) as: $\overline{\text{wage}} = \beta_0 + \beta_1 \text{educ}$.
- We define the equation for the red line (the line through the points in the ***sample***) as: $\widehat{\text{wage}} := \widehat{\overline{\text{wage}}} = \hat{\beta}_0 + \hat{\beta}_1 \text{educ}$.
  - $\widehat{\overline{\text{wage}}}$ is complex notation, so it's standard to use $\widehat{\text{wage}}$ instead.
  - $\widehat{\text{wage}}$ estimates the avg wage in pop for a given year of educ.

The estimators

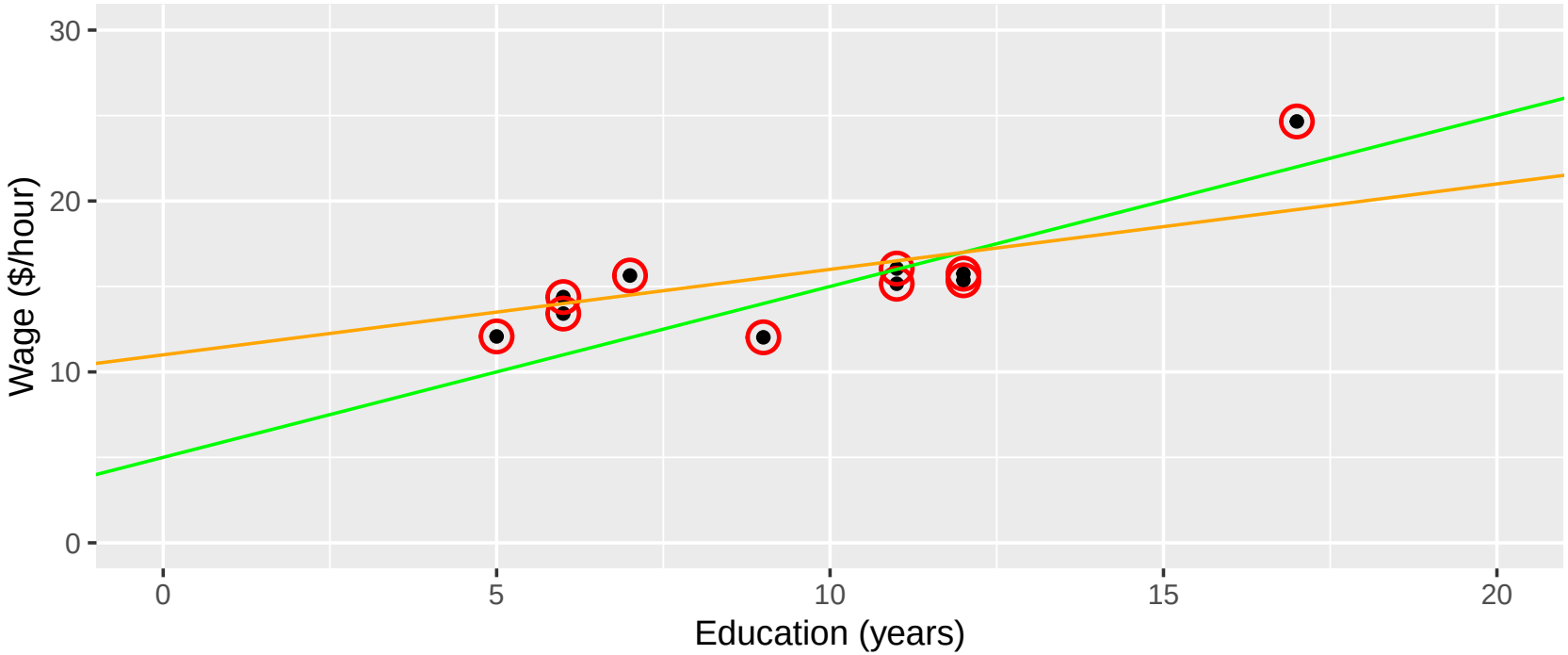
- But how do we draw a line through the sample?
- Even if the population satisfies Assumption 1 (linearity of the regression function), we do not expect the sample to have mean linearity.
- The answer is Ordinary Least Squares (OLS).



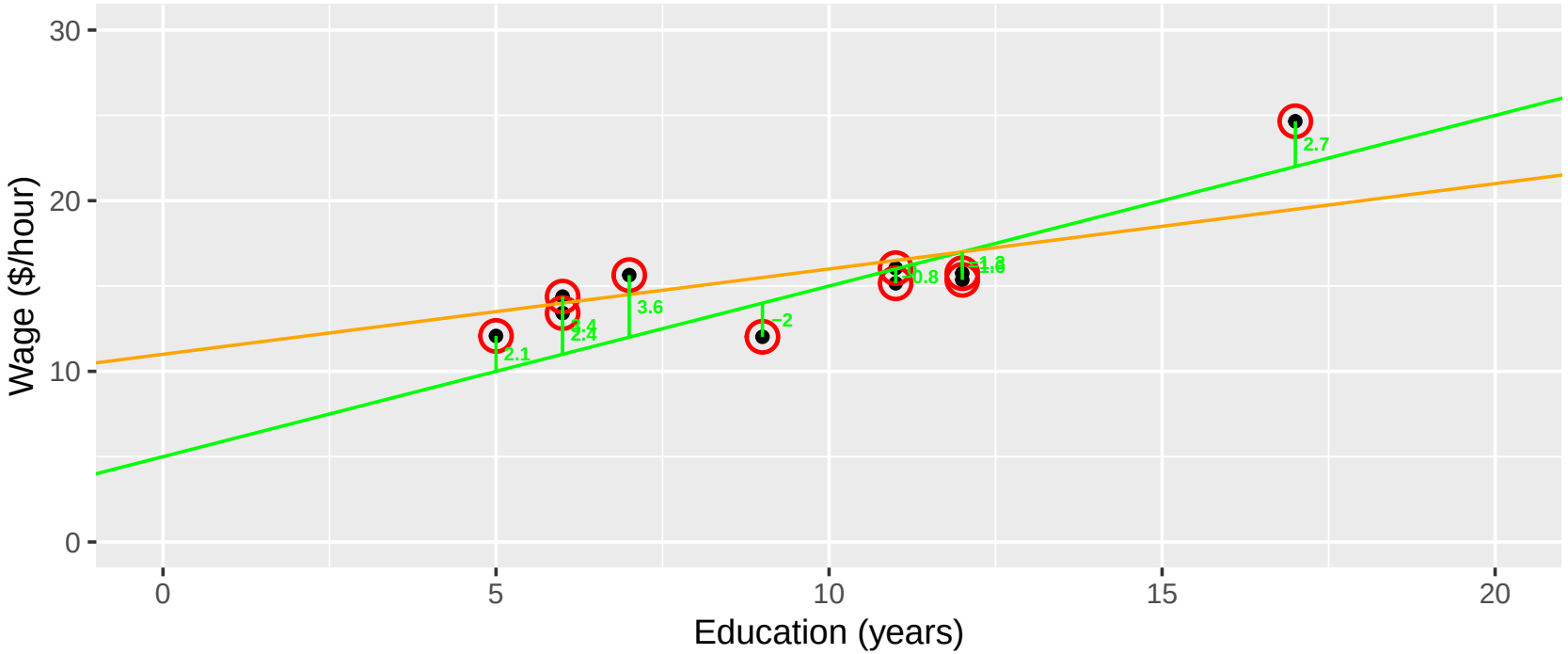
- Imagine a million lines drawn on the above plot. They all take the form $\text{wage} = b_0 + b_1 \text{educ}$, for different values of b_0 and b_1.
- Which line does OLS choose?



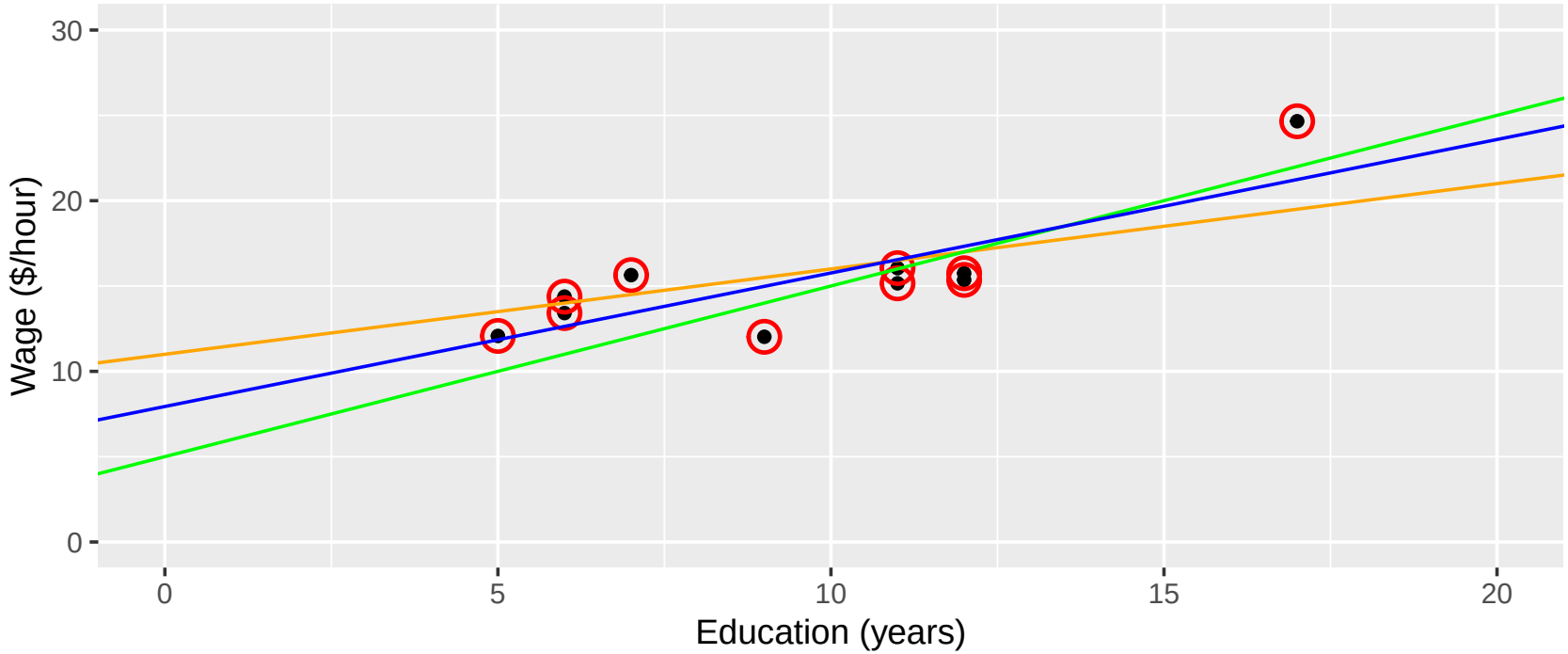
- Some of the lines are clearly not ideal because they could be moved up or down and would get closer to the dots. We can get rid of those ones right away.
- But the rest cannot be easily improved on by shifting up/down.
- We'll start by considering two lines, and we'll work through which one OLS prefers.



- Consider 2 potential lines of the form $\text{wage} = b_0 + b_1 \text{educ}$.
- Green line: $b_0 = 5, b_1 = 1$. Orange line: $b_0 = 11, b_1 = 0.5$.
- How does OLS decide which line is closer to the dots?
- Define $u_i(b_0, b_1) := \text{wage}_i - (b_0 + b_1 \text{educ}_i)$ as the difference between the y-value (wage) of a point and the line with coefficients b_0, b_1. These are called the residuals wrt a line.
- Intuitively, OLS chooses b_0, b_1 so the $u_i(b_0, b_1)$ are "small".



- Recall: $u_i(b_0, b_1) := \text{wage}_i - (b_0 + b_1 \text{educ}_i)$.
- The vertical lines from the dots show the residuals of each point *wrt the green line*.
- We sum the *squared* (to turn negatives to positives) residuals:
  For the green line: $\sum_{i=1}^{10} [u_i(b_0 = 5, b_1 = 1)]^2 = 50.7$.
  For the orange line: $\sum_{i=1}^{10} [u_i(b_0 = 11, b_1 = 0.5)]^2 = 48.7$.
- $48.7 < 50.7$ so OLS prefers the orange line over the green line.



- If we tried all pairs $(b_0, b_1) \in \mathbb{R}^2$, the *smallest* we can get the sum is $\sum_{i=1}^{10} [u_i(b_0 = 7.9, b_1 = 0.8)]^2 = 37.6$.
- The OLS coefficients $\hat{\beta}_0$, $\hat{\beta}_1$ are the *values* of b_0 and b_1 that minimize $\sum_{i=1}^{10} [u_i(b_0, b_1)]^2$.
- The formal def. is: $(\hat{\beta}_0, \hat{\beta}_1) := \operatorname{argmin}_{(b_0,b_1)\in\mathbb{R}^2} \sum_{i=1}^{n} [u_i(b_0, b_1)]^2$.
- For our sample, $\hat{\beta}_0(\omega^*) = 7.9, \hat{\beta}_1(\omega^*) = 0.8$.

Technical details

- $(\hat{\beta}_0, \hat{\beta}_1) := \operatorname{argmin}_{(b_0,b_1)\in\mathbb{R}^2} \sum_{i=1}^{n} [u_i(b_0, b_1)]^2$
- We can analytically solve for the values of $\hat{\beta}_0$, $\hat{\beta}_1$ as follows:
  1. Differentiate $\sum_{i=1}^{10} [u_i(b_0, b_1)]^2$ with respect to b_0 and b_1.
  2. Solve for the b_0, b_1 such that the expressions from (1) equal 0.
  3. Show that $\sum_{i=1}^{10} [u_i(b_0, b_1)]^2$ is convex in b_0, b_1.
- For *calculation*, the analytical results are useful:

$$\hat{\beta}_1 := \frac{\hat{\mathrm{cov}}(x, y)}{\hat{\mathrm{V}}(x)} = \frac{\hat{\mathrm{cov}}(\mathrm{educ}, \mathrm{wage})}{\hat{\mathrm{V}}(\mathrm{educ})}$$
$$\hat{\beta}_0 := \overline{y} - \hat{\beta}_1 \overline{x} = \overline{\mathrm{wage}} - \hat{\beta}_1 \overline{\mathrm{educ}},$$

  where $\hat{\mathrm{cov}}$ and $\hat{\mathrm{V}}$ are covariance and variance in the *sample*.
- For *interpretation*, understanding the minimization problem carries the most intuition.

But why *squared* distance?

The OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are intuitive, but why should we minimize the sum of *squared* residuals?

- If we run OLS on the *population* data, the OLS coefficents would exactly equal the regression coefficients β_0, β_1.
  - Intuition: the solution to $\min_a \sum_{i=1}^n (x_i - a)^2$ is $a^* = \bar{x}$.
- If we minimize the sum of *absolute* residuals, $\sum_{i=1}^n |u_i(b_0, b_1)|$, in the pop data, we would not obtain β_0, β_1.
  - Intuition: the solution to $\min_a \sum_{i=1}^n |x_i - a|$ is $a^* =$ median of x_i.
- So using OLS on the sample to estimate β_0, β_1 satisfies the "plug-in principle": we do to the sample what we wish we could do to the population.

Properties of estimators

Exercise 196. What properties do we want $\hat{\beta}_0$ and $\hat{\beta}_1$ to have?

1. $\mathrm{E}(\hat{\beta}_j) = \beta_j$ for $j = 0, 1$.
   - OLS estimators do satisfy this property.
2. What if we want to use $\hat{\beta}_j$ for a CI of β_j? We want $\hat{\beta}_j \dot{\sim} N$.
   - Then, we can make a 95% CI for β_j with $\hat{\beta}_j \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_j)$.
   - OLS estimators do satisfy this property for large n.

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Confidence intervals for $\hat{\beta}_j$

For this class, assume that $\hat{\beta}_0$, $\hat{\beta}_1$ are the ***OLS*** estimators of the population OLS line.

- We will not show it, but theories prove that $\hat{\beta}_j \dot{\sim} N$ for "large" sample sizes.
- Then, a 95% CI can be constructed as $\hat{\beta}_j \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_j)$.

Summarizing the error term

Recall our model

$$y_i = \beta_0 + \beta_1 x_i + u_i \, .$$

- If education were the *only* thing that affected wage, then all the dots would be on the line.
- The u_i term represents all the *other* factors that affect wage.
  - We make many assumptions in economics, but here we actually aren't assuming that education is the only thing that affects wage. There can be other factors and that is still OK in our model.
  - The u_i term allows us to think about the relationship between x_i and y_i but not require that x_i *completely* determines y_i.
  - The u_i term likely "contains" a lot of other variables. e.g.,

$$u_i = \beta_2 a_i + \beta_3 b_i + \beta_4 c_i + \ldots \, .$$

Terminology review

In econometrics, we see equations like the following a lot:

$$y_i = \beta_0 + \beta_1 x_i + u_i \, .$$

So let's make sure we know the terminology.

- This model corresponds to the specification $y_i \sim x_i$.
- β_0 is "beta naught" or "beta-zero" or the "intercept".
- β_1 is the "slope coefficient" or "beta-one".
- We call u_i the "error term", or the "unobserved variables term".

Getting a handle on the error term

Let's consider the following simple model:

$$y_i = \beta_0 + \beta_1 x_i + u_i \, .$$

The 1st step to approach a model is "get a feel for the error term".

- Although we cannot know *all* of the variables inside u_i, it helps to just list a few.

Exercise 197. Think of three variables that are likely in the u_i term.

- job characteristics
  - long hours
  - stressful
- individual characteristics
  - intelligence
  - GPA
  - quality of education

Test score exercise

Let y_i be the test score of student i in a class, and x_i the number of hours they studied. Consider the specification $y_i \sim x_i$ and the model $y = \beta_0 + \beta_1 x_i + u_i$.

Exercise 198. What are some variables inside u_i?

1. $IQ_i :=$ IQ of student i.
2. $S_i :=$ Stress at time of test of student i.

Then, $u_i = \beta_{IQ} IQ_i + \beta_S S_i + \cdots$.

Exercise 199. What are the signs of β_{IQ} and β_S?

1. $\beta_{IQ} > 0$.
2. $\beta_S < 0$.

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Learning objectives

1. Understand $\hat{\beta}_1$ as a rv.

$\hat{\beta}_1$ has a distribution

It should be clear that $\hat{\beta}_1$ is an estimator of β_1.

- $\hat{\beta}_1$ is a function of the sample, ω.
- $\hat{\beta}_1$ is a r.v., i.e., a function from Ω to $\mathbb{R}$.
- $\hat{\beta}_1(\omega)$ is the slope of the OLS line through the sample of data from ω.
- If we take a different sample (i.e., if ω changes), then we will (very likely) have a different OLS line and thus $\hat{\beta}_1(\omega)$ will change.

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Learning objectives

1. Storytelling.
2. The concept of a "counterfactual" (cf).
3. Causal versus descriptive/predictive statements.
4. How policy recommendations depend on the underlying mechanism.

- We will first work with a couple of simple, but made up, examples.
- We then discuss some real examples in research and the press.

Storytelling

Definition 50 (Consistent with the data). A story is consistent with the data if the following statement holds: if the story *were* true, a clear chain of logic would lead to the patterns in the data.

- The word "were" in Definition 50 is important: Whether a story is "consistent with the data" has nothing to do with the likelihood the story is true.
- Specifically, we want to separate the following two questions:

1. Is the story consistent with the patterns in the data?
   - This is the brainstorming phase.
   - This phase is more objective: does the logic of the story imply the patterns in the data?
2. Is the story true?
   - This is more subjective.
   - It may require further investigation (e.g., data collection, or study of the subject) to figure out.

Example: calçots

Suppose we see the following pattern:

> *People in Barcelona who eat calçots are on average healthier than people in Barcelona who do not.*



Example: calçots

Example 3 (calçots). Suppose we see the following pattern:

People in Barcelona who eat calçots are on average healthier than people in Barcelona who do not.

Exercise 200. What are stories consistent with this finding?

1. Calçots are healthy. ▸ Aren't they just onions?
2. Salvitxada is healthy.
3. Calçots consumption is positively associated with income.
   - Those with high income might have better health for other reasons (can pay for extra medical care, the gym, etc.)
4. Calçotadas are social events, and it has been shown that socializing is good for health.

Example: petanca

Imagine a survey was done of 10,000 residents of Barcelona in 2024, and a follow-up was done in 2025. It found that people who played petanca in 2024 were more likely to die over the next year.



Example: petanca

Example 4. Imagine a survey was done of 10,000 residents of Barcelona in 2024, and a follow-up was done in 2025. It found that people who played petanca in 2024 were more likely to die over the next year.

Exercise 201. What stories are consistent with this finding?

1. Petanca is a dangerous sport.
2. Petanca is played especially by men, and men have lower life expectancy than women.
    - For 2019-2023 Catalunya, life expectancy in Catalunya at birth was ~86 years for women and ~81 for men. (idescat)
3. Older people play petanca, and older people are also more likely to die due to health problems at old age.

Definition 51. A counterfactual (cf) is an alternative to what happened.

Example 5. Example of every day use: If I had 1M€ last year, I would have bought a house.

- This is a cf since I didn't have 1M€ last year.

Example 6. If you didn't go to university, what would your wage be right now?

- You'd be earning a wage, we just don't know what that number would have been.

Example 7. If you had studied an extra hour for your math exam, you would have received a (potentially) different exam score.

- That cf exam score is a number, and differs by individual.
- Think of these numbers as existing (and they are not random), we just don't know what they are.
- cfs are especially important for policy making.

Causal statements vs descriptive statements

Definition 52. A statement about x and y is causal if it implies that a cf value for x would lead to a change in y. Otherwise, it is descriptive.

Definition 53. We say that X is a predictor of Y if $\operatorname{corr}(X, Y) \neq 0$.

Exercise 202. Consider the statement "The proportion of people in the metro with umbrellas is a predictor of whether it will rain." Is it causal or descriptive?

- On a given day when many people had umbrellas and it rained, consider a cf: *if no one had brought an umbrella...*
  - Does the statement imply that it wouldn't have rained? No.
- The statement is descriptive. It's just mentioning the correlation between the presence of umbrellas and it raining.

Policy recommendations

Depending on whether there is a causal link regarding the patterns in the data, we might consider policy recommendations.

- Should we prohibit/tax petanca? Require helmets?
- Should we subsidize calçots?
- A policy rec is not just relevant to *government* policies.
- We will see an example on advertising and sales in companies.

For the *same* observed pattern in the data, the policy recommendation depends on which story we think is true.

- Suppose we think a story is the true one and implement a policy accordingly.
- If in fact a different story were true, the policy could have no effect (and wasted resources) or even a negative effect.

Example 8 (calçots). Suppose we see the following pattern:

People in Barcelona who eat calçots are on average healthier than people who do not.

Exercise 203. What are stories consistent with this finding?

1. Calçots are healthy (they have vitamins C, A, and K).
2. Salvitxada is healthy.
3. Calçots consumption is positively associated with income.
   - Those with high income might have better health for other reasons (can pay for extra medical care, the gym, etc.)
4. Calçotadas are social events, and it has been shown that socializing is good for health.

Exercise 204. Is the initial statement descriptive/causal? Descriptive. It's describing an observation.

- While reading, we might think the author is trying to convince people calçots are the reason for better health. But...
- There's no *logic* that implies the difference in health is *because* of calçots consumption.

Example 8 (calçots). Suppose we see the following pattern:

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4. Calçotadas are social events, and it has been shown that socializing is good for health.

Exercise 205. Which stories are causal?

- (1), (2), and (4): they suggest that if people who did not eat calçots had eaten calçots, their cf health would be better.
- (3) is descriptive: their cf health would be the same (because their cf income would be the same).

Example 8 (calçots). Suppose we see the following pattern:

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Exercise 203. What are stories consistent with this finding?

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Exercise 206. What is the policy rec?

- If we think (1), (2), or (4) is true, we could consider subsidizing calçots.
- If (3), then nothing.

Imagine a survey was done of 10,000 residents of Barcelona in 2024, and a follow-up was done in 2025. It found that people who played petanca in 2024 were more likely to die over the next year.

Exercise 207. What stories are consistent with this finding?

1. Petanca is a dangerous sport.
2. Petanca is played especially by men, and men have lower life expectancy than women.
    - For 2019-2023 Catalunya, life expectancy in Catalunya at birth was ~86 years for women and ~81 for men. (idescat)
3. Older people play petanca, and older people are also more likely to die due to health problems at old age.

Exercise 208. Is the initial statement descriptive/causal?

- Descriptive. If it said "...*because* they play petanca" it would be causal.

Imagine a survey was done of 10,000 residents of Barcelona in 2024, and a follow-up was done in 2025. It found that people who played petanca in 2024 were more likely to die over the next year.

Exercise 207. What stories are consistent with this finding?

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3. Older people play petanca, and older people are also more likely to die due to health problems at old age.

Exercise 209. Which stories are causal?

- Only (1) implies a cf: if people who played petanca had not played petanca, their health would have increased.

Imagine a survey was done of 10,000 residents of Barcelona in 2024, and a follow-up was done in 2025. It found that people who played petanca in 2024 were more likely to die over the next year.

Exercise 207. What stories are consistent with this finding?

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    - For 2019-2023 Catalunya, life expectancy in Catalunya at birth was ~86 years for women and ~81 for men. (idescat)
3. Older people play petanca, and older people are also more likely to die due to health problems at old age.

Exercise 210. What is the policy rec?

- If (1) were true: people should play less petanca.
- If (2)–(3): no rec.

Bald men and Covid

The petanca example might seem silly, but this type of mistake is not rare in research or the press.

A June 2020 Telegraph article references research that finds "Bald men at higher risk of severe case of Covid-19".



From telegraph.co.uk

Bald men and Covid

Article: Forbes article referenced the same research.

Quotes from Forbes article:

> *Researchers at Brown University point to androgens, the group of hormones which causes hair loss in men.*
>
> *Lead author Dr. Carlos Wambier said, "We think androgens or male hormones are definitely the gateway for the virus to enter our cells. We really think baldness is a perfect predictor of severity."*

Exercise 211. Is the 1st statement causal or descriptive? Causal.

Exercise 212. Is the 2nd statement causal or descriptive? Descriptive.

They go on to suggest a policy rec:

> *"A vaccine might ultimately be found for SARS-CoV-2; however, if a vaccine is not found or found to be ineffective, androgen suppression as a prophylactic treatment could reduce Covid-19 disease burden."*

An alternative story

Exercise 213. What are other (besides androgens) stories that could explain the results?



Forbes updated the article, and changed the title from "Bald Men at Higher Risk of Severe Coronavirus Symptoms" to "Study Suggests Bald Men Could Be More At Risk For Severe Coronavirus, But More Research Is Needed".

And they added some disclaimers:

> *Updated (6/8/20): This piece has been clarified to note that the study did not control for age, which is a risk factor for hair loss and severe Covid-19.*
>
> *Updated (6/9/20): The headline of this story has been updated to reflect that the research is preliminary.*
>
> New research is showing why a larger percentage of men—particularly bald men—are having worse Covid-19 outcomes than women.
>
> Researchers at Brown University point to androgens, the group of hormones which causes hair loss in men. They've determined androgens are linked to severe cases of

However, even in the first sentence of the revised article the use of "why" is a causal statement!

OK OK but...

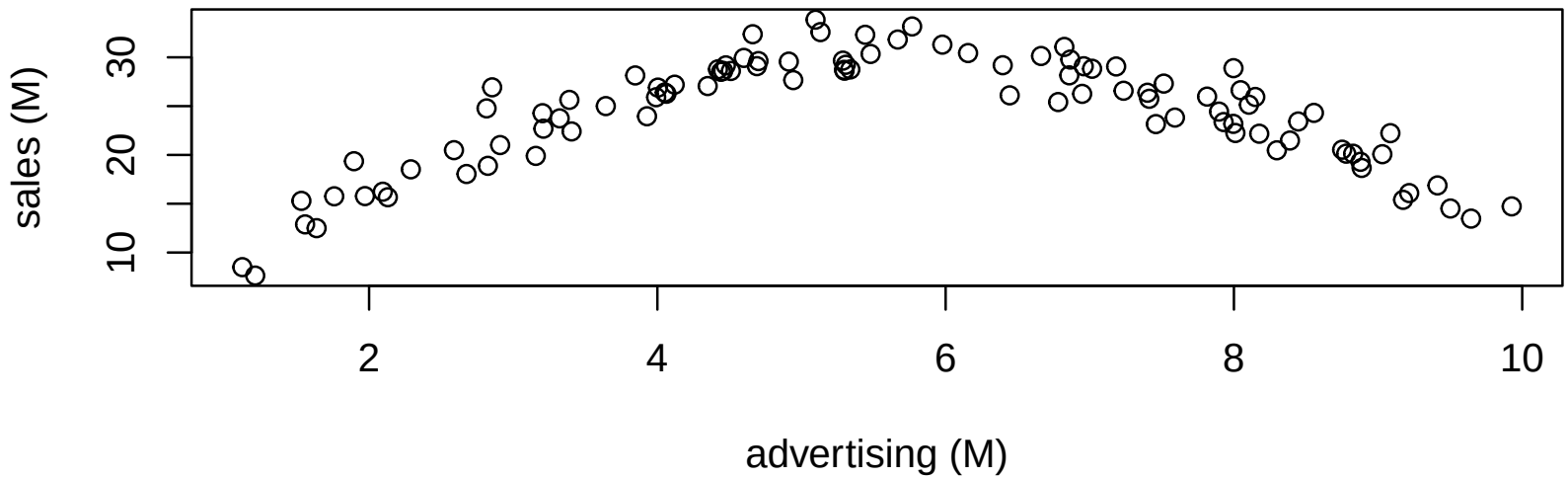
But those news outlets (Forbes and Telegraph) aren't the New York Times. They wouldn't make a mistake like this, right? Right?!?!?



Exercise 214. Which statement is descriptive? "...people who reported going to a museum or concert even once a year lived longer than those who didn't".

Exercise 215. Which statement is causal? "Want to live longer? Try going to the opera." This suggests the cf that if people who didn't go to the opera *had* gone, they would have lived longer.

Suppose we have data on advertising and sales (both in M€) for the year 2025.



Exercise 216. How can we "safely" describe the joint dist? For the year 2025, the mean of sales across companies, conditional on advertising, at first increases and then decreases.

Exercise 217. Causal/descriptive?: "A medium amount of advertising is good for sales, but too much is bad."

- Causal: it implies the following cf: For a company spending a medium amount, if it were to have spent a large amount it would have been worse (i.e., a decline in sales).

Advertising

Exercise 218. What are stories consistent with the data?

1. Too much advertising can annoy consumers into not buying.
2. Poor quality products need lots of advertising to compensate for poor quality.

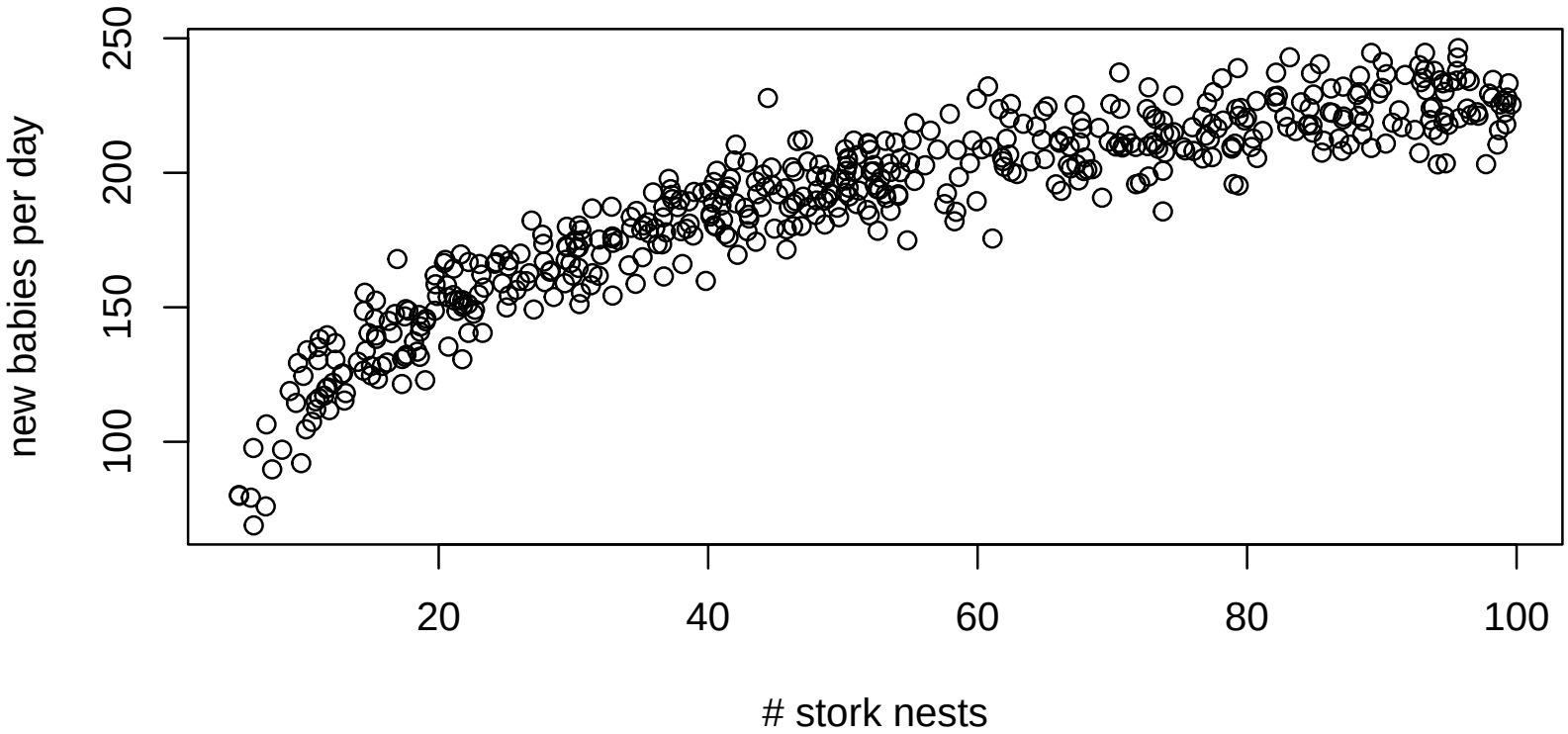
Exercise 219. You are working at a company that's currently spending 9M€ on advertising. Your boss asks you "What would happen to our sales if our company instead spent 5M€?"

1. If Story 1 were true, we would predict that the company's sales would increase.
2. If Story 2 were true, we would predict that the company's sales would decrease.

Both of the above are the answer. That is, we would work through the cf for each story.

Storks

Suppose we observe the following data in towns across the world:



Exercise 220. Describe the data.

- More stork nests is associated with more new babies born.

Storks

Exercise 221. What stories are consistent with the data?

1. Storks really do bring babies.
2. There are more storks in bigger cities, and bigger cities also have a higher number of births (since there are more people).

Exercise 222. Which stories are causal?

1. (1) is causal: under (1) it suggests a cf: if we somehow increased the stork population of a city, they would bring more babies.
2. (2) is not causal: under (2), if we somehow increased the stork population, we would not see a change in new babies.

A general approach

Although each case is different, here are some general steps that are good to follow in many cases:

1. Describe the data descriptively.
2. Brainstorm stories consistent with patterns in the data.
3. State a relevant cf.
4. For each story from step (2), step through the logic of what *would have been* the cf outcome *if* the story were true.
5. We can then give a policy rec for each story.

What we've learned

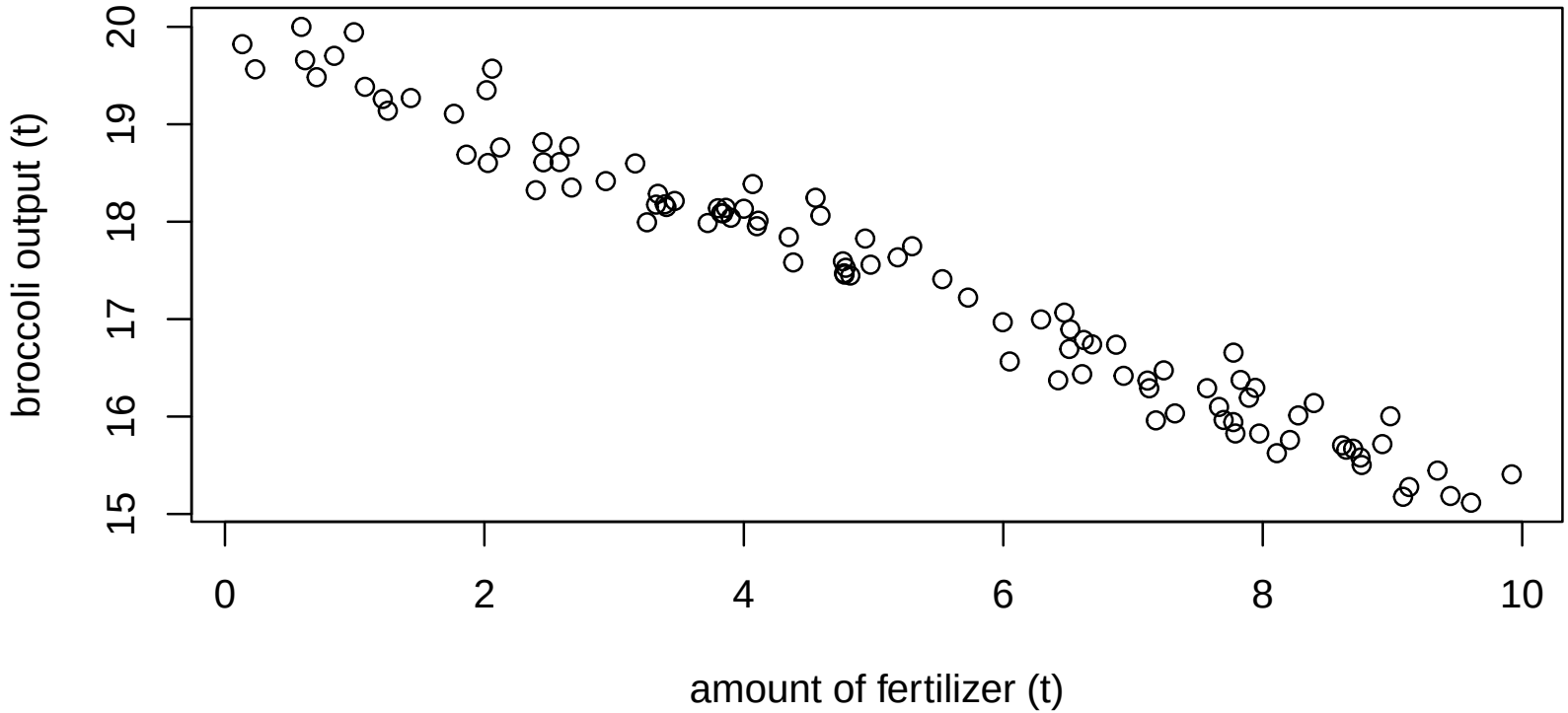
1. If you learn one thing in this class, it's to be skeptical of any causal implications drawn from statistics.
2. If we think a story is true when it is not, that would lead to potentially dangerous policy recs (e.g., the case of baldness and Covid-19).

Next steps: we investigate

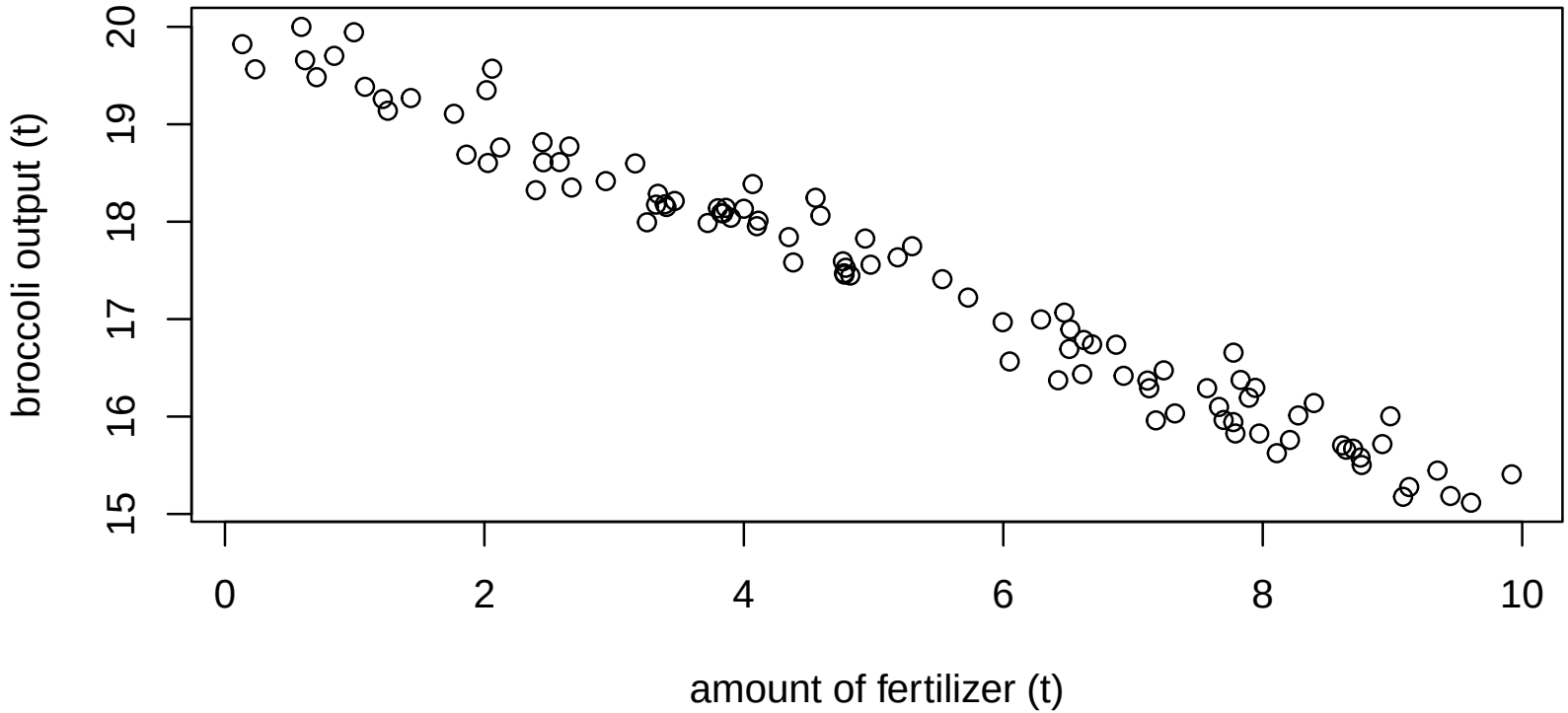
1. If we have the underlying individual data for a pattern, we can dig into which stories are true.
    a. For the storks, if we have data on city size (population or area), what could we do?
    b. For the finding of bald men and Covid, what data would you want and what could we do?
2. Even if we have the data, since it's observational data it might be difficult to disentangle causality.
    a. In some cases, research can use a powerful tool called Randomized Controlled Trials.

Broccoli

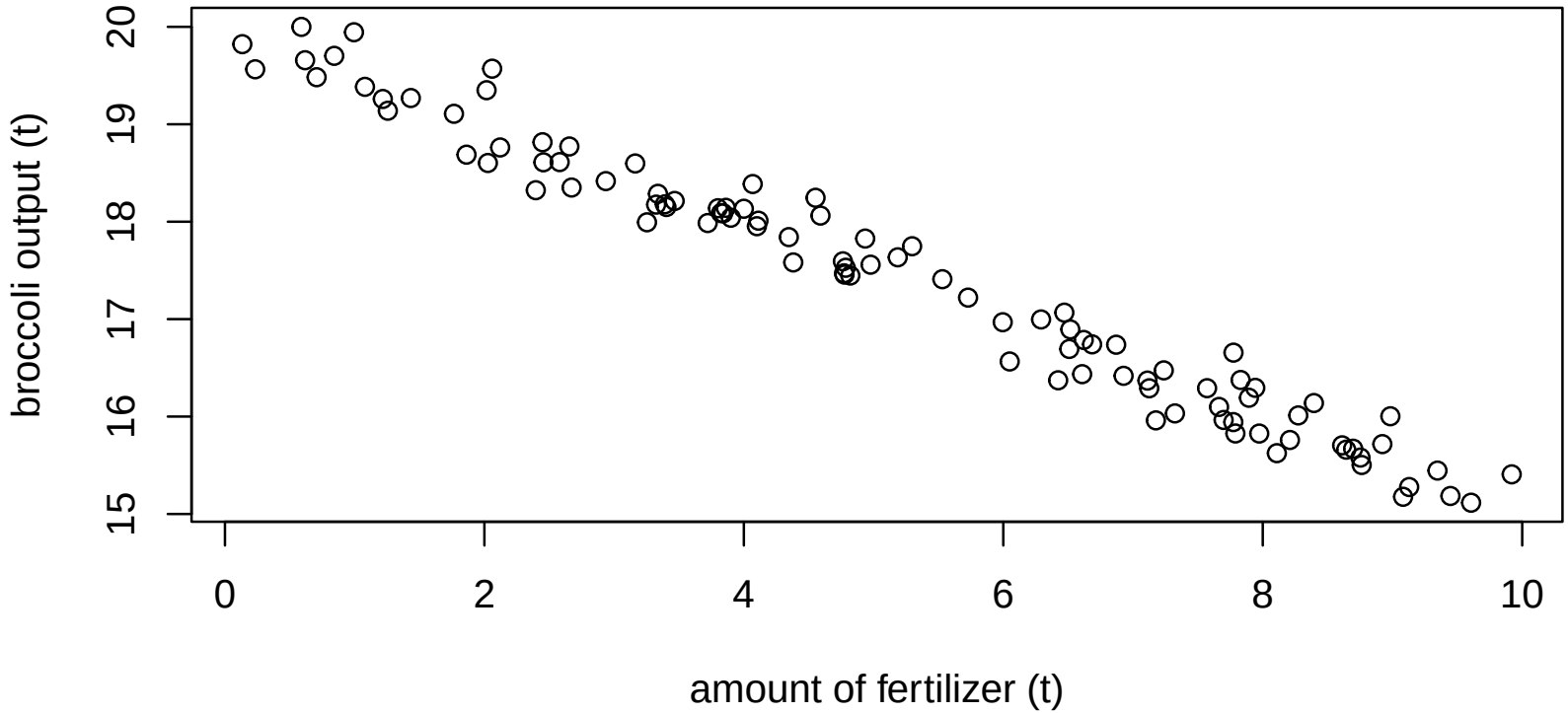
- Suppose the government of Kenya is considering a subsidy for fertilizer to broccoli farmers.
- To decide whether to give the subsidy, the government wants to know the benefit of using fertilizer for growing broccoli.
- Kenya brought us in as consultants and has given us data on fertilizer usage and yield for each plot of land.



- The joint dist of the 100 plots is shown in the figure.
- Each point on the scatter plot represents a plot of land.
- On the x-axis is the # tons of fertilizer used on a plot.
- On the y-axis is the # tons of broccoli yielded by the plot.



Exercise 223. How can we summarize the joint distribution between broccoli and fertilizer using statistics terminology? The covariance of output and fertilizer is negative.



Exercise 224. How can we interpret the joint dist in non-technical terms? Plots with more fertilizer are producing less broccoli than plots with less fertilizer.

Exercise 225. Is this statement causal? No. We are simply describing the plots. We do not say *why* plots with more fertilizer produce less.

- We don't suggest a cf like what *would have* happened to plots that received a lot of fertilizer *if they had* received less.

Exercise 226. What are stories consistent with the data on broccoli output in Kenya?

1. The fertilizer was expired, or the farmers used the wrong type of fertilizer (e.g., the type used might have had too much nitrogen for broccoli).
2. Fertilizer has beneficial nutrients for broccoli; however the fertilizer attracts rodents which eat the broccoli.
3. Farmers are compensating for a negative aspect of the plots.
    - e.g., the plots that receive ↑fertilizer have bad soil quality.

Exercise 227. What is a relevant cf to decide if a story is causal?

- If the plots of land that received a lot of fertilizer had received less fertilizer, would their cf broccoli output have changed?

Story (2) is interesting: even though the fertilizer itself is beneficial, its "causal effect" is negative because it causes rodents to come.

Randomized controlled trials (RCTs)

- Randomized controlled trials (RCTs) are a convincing way to learn about cfs and thus measure causal effects.
- We can't always use an RCT, but even then the concept of an RCT helps us to define what the causal effect is.
- J-PAL is a research institute focused on running RCTs in developing countries to learn how to improve lives.

Definition 54. Experimental data is obtained when the treatment variable is assigned by a researcher.

- Data from RCTs are experimental data.

Definition 55. Observational data is obtained without intervening/manipulating the subjects.

What we learned

1. How to think about counterfactuals.
2. How to categorize stories as consistent with patterns, and as causal or descriptive.
3. How to identify causal statements and observational statements.
4. How to state assumptions that our conclusions rely on.



Calçots: the next superfood?

Back to lecture

Comparison of vitamin content in calçots vs regular onions:

Nutrient	**Calçots** (per 100 g)	**Regular Onions** (per 100 g)
Vitamin C	18–20 mg	7–9 mg
Vitamin A (β-carotene)	80–100 µg	∼0 µg
Vitamin K	40–45 µg	0.4 µg
Folate (B9)	60–65 µg	18–20 µg
Vitamin B6	0.15–0.20 mg	0.12–0.13 mg
Other B vitamins	small amounts	small amounts

Table of Contents

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Learning objectives

- Learn how to use OLS to explore causal effects.
- Learn the key assumption for when the population OLS line has a causal interpretation.
- Learn how to get a handle on the error term in order to argue whether the assumption is true or not for a specific context.
- Understand what it means for a parameter to be "identified".
- Reminder: cf = "counterfactual".

The causal line

Assumption 2. The causal effect of x on y is the same for everyone in the population.

Assumption 3. The causal effect is linear.

Let β_1^{causal} be the causal effect on y of increasing x by 1 unit.

Exercise 228. Is β_1^{causal} a number or a rv? A number.

Definition 56. A number γ is identified if it could be calculated from the population data.

- γ can be "identified" even if we don't have data on the pop.
- Whether γ is identified or not depends on the following: ***if*** we had data on the population, then could we calculate it from the population?

The essence of econometrics

How is it possible to have data on the pop but still not be able to calculate something?

- Causal effects depend on cfs!
- We do not have cf outcomes in the pop data.

How is it possible for *any* causal parameter to be identified if we don't have data on cfs?

- Under certain assumptions, we can use a control group to know (average) cf outcomes.
- The question of whether we can *convincingly* perform causal analysis then boils down to the following:
  - Can we convincingly argue that our assumptions are (at least approximately) true?
- Answering this question is the obsession of modern econometrics.

Causal identification in practice

- Causal identification is the goal of many companies.
- A common goal at companies is: how will demand change if we change price?
  - Two categories of relevant cf outcomes:
    1. How will short-term demand change?
    2. How will long-term demand change?
- In this class we've developed the intuition that we can apply generally.
- An important skill is training your critical eye.
  - Now you have the intuition to raise red flags when you see causal statements without stating the required assumptions to interpret findings as causal.

Exercise 229. Is the causal effect of fertilizer on broccoli identified from the observational data from the Kenyan government?

- Not without assumptions. Even if we had data on all plots of land, because soil quality could be different across the plots and we think the farmer allocated fertilizer based on soil quality, we cannot calculate the causal effect.

Exercise 230. What assumption would allow us to use plots that received low fertilizer as a valid control group for plots that received high fertilizer?

- That the plots that received high fertilizer do not differ in any way *that affects broccoli output* from the plots that received low fertilizer, except forthe amount of fertilizer.
- Let $H := 1$ if a plot received high fertilizer, and 0 if low.
- What do we want the relationship to be between H and soilq to yield a valid control group? $H \perp\!\!\!\perp \mathrm{soilq}$.
- And we would need to assume independence of H with *any* variable that affects broccoli output (besides fert).

Generalizing the assumption

- Suppose we have population data x_i and y_i.
- Let u_i be all components (variables and coefficients) that affect y_i besides x_i.

Exercise 231. What assumption do we need in order to identify the causal effect of x_i on y_i?

- $x_i \perp\!\!\!\perp u_i$.
- It turns out that we can make a weaker assumption that still yields identification: $\operatorname{cov}(x_i, u_i) = 0$.

Same individuals, different datasets

We may have different datasets about the *same* individuals.

Example 9. For the plots of land producing broccoli, consider two different datasets:

1. Observational data provided by the Kenyan government.
2. Experimental data by running an RCT on the same plots.

Exercise 232. Would we expect the joint dist of broccoli and fertilizer to be the same in (1) and (2)?

- Not necessarily. If soil quality is a confounder, we would expect $\text{cov}(\text{fert}_\text{i}, \text{broccoli}_\text{i}) < 0$ in (1) and $\text{cov}(\text{fert}_\text{i}, \text{broccoli}_\text{i}) > 0$ in (2).

- Whether we can identify a causal effect depends on the relationship of the variables.
  - which in turn depends on *how* the data were collected.

Example: education and wage

Let β_1^{causal} be the causal effect on wage of increasing education by 1 year. Consider the following equation:

$$\text{wage}_i = \beta_0 + \beta_1^{causal}\text{educ}_i + \text{u}_i.$$

Exercise 233. Is β_1^{causal} identified? Hint: think of some vars. inside u_i.

No, because $\text{cov}(\text{educ}_i, u_i) \neq 0$. To make this claim (of "no"), we need to think of a variable and argue two things about it:

1. We think that IQ_i is inside u_i. Why do we think this is true, in this context? For two individuals with the same number of years of education, the one with higher IQ likely earns more because they are more productive.
2. $\text{cov}(\text{educ}_i, \text{u}_i) \neq 0$. Why do we think this is true, in this context? Individuals who have higher IQ are more likely to stay in school because performing well is easier, get accepted to college (because they perform better), and choose to attend college.